RTL Design Sherpa

RTL Math Library Module Reference — Rev 0.90

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1 RTL Math Library

Location: `rtl/math/*.sv` **Generators:** `bin/math_generate.py`, `bin/math_generate.sh`, `bin/rtl_generators/` **Status:** Production Ready

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1.1 Overview

`rtl/math/` is a ~170-module arithmetic library: integer add/subtract/multiply, IEEE-754-style floating-point (bf16, fp16, fp32, fp8) arithmetic, comparison, conversion, and machine-learning activation functions. Almost all of it is **code-generated** by the Python framework under `bin/rtl_generators/` — and that should tell you how to read it. Don't approach this as a flat list of 170 files; you'll drown. Read it by **operation** and, within each operation, by the **methodology** (algorithm) that implements it.

This document is the organizing map. Each operation lists its methodologies, the research each is based on, the module name pattern, and a link to the detailed per-methodology doc. The floating-point cores are themselves built from the integer methodologies below — Dadda trees for the mantissa multiplies — so the two halves of the library share a foundation.

Naming convention: width-parameterized generated instances carry the width suffix (`_008`, `_016`, `_032`, …) and/or the format tag (`bf16`, `fp16`, `fp32`, `fp8_e4m3`, `fp8_e5m2`, `ieee754_2008_*`). One methodology doc covers all of its width/format instances.

1.2 Integer Arithmetic

1.2.1 Addition

Methodology	Modules	Research	Detail
Ripple-carry / full / half	`math_adder_full`, `math_adder_half`, `math_adder_full_nbit`	Classic ripple carry	math_adder_full.md, math_adder_basic.md
Carry-save (CSA)	`math_adder_carry_save_nbit`	Redundant carry-save form (used in multiplier trees)	math_adder_carry_save.md
Carry chain (serial P/G – NOT Weinberger-Smith O(log N) lookahead; the	(see subtractor / adder-basic)	Weinberger & Smith, "A Logic for High-Speed	math_adder_pg_chain.md

Methodology	Modules	Research	Detail
linked page explains)		Addition," NBS Circular 591 (1958)	
Brent-Kung (parallel prefix)	`math_adder_brent_kung_{008,016,032,064}` + prefix cells (`_black`, `_gray`, `_pg`, `_bitwisepg`, `_grouppg_*`, `_sum`)	Brent, R.P. & Kung, H.T. (1982), "A Regular Layout for Parallel Adders," *IEEE Trans. Computers* C-31(3):260-264	math_adder_brent_kung.md, math_prefix_cell.md
Han-Carlson (parallel prefix)	`math_adder_han_carlson_{016,022,032,044,048,072}`	Han, T. & Carlson, D.A. (1987), "Fast area-efficient VLSI adders," *Proc. 8th IEEE Symp. Computer Arithmetic (ARITH)*:49-56	math_adder_han_carlson.md
Add/subtract	`math_addsub_full_nbit`	Two's-complement add/sub select	math_addsub.md

Parallel-prefix trade-off: all three prefix adders compute the carry network in O(log n) depth — what differs is where each spends its area, wiring, and fan-out budget. Brent-Kung minimizes cells/wiring (more depth), Kogge-Stone minimizes depth (most wiring), and Han-Carlson is the middle ground (a Kogge-Stone/Brent-Kung hybrid). See the prefix-cell and gray-cell docs for the shared generate/propagate building blocks.

1.2.2 Subtraction

Methodology	Modules	Detail
Ripple-carry / carry-lookahead / full / half	`math_subtractor_{full,half,full_nbit,ripple_carry,carry_lookahead}`	math_subtractor.md

1.2.3 Multiplication

Methodology	Modules	Research	Detail
Basic cell / carry-save	`math_multiplier_basic_cell`, `math_multiplier_carry_save`	Shift-add partial-product cells	math_multiplier_basic.md
Dadda tree	`math_multiplier_dadda_tree_{008,016,032}`, `math_multiplier_dadda_4to2_{008,011,024}`	Dadda, L. (1965), "Some schemes for parallel multipliers," *Alta Frequenza* 34:349-356	math_multiplier_dadda_tree.md, math_multiplier_dadda_4to2.md
Wallace tree	`math_multiplier_wallace_tree_{008,016,032}`, `math_multiplier_wallace_tree_csa_{008,016,032}`	Wallace, C.S. (1964), "A Suggestion for a Fast Multiplier," *IEEE Trans. Electronic Computers* EC-13(1):14-17	math_multiplier_wallace_tree.md

Dadda vs Wallace: both reduce the partial-product matrix to two rows in O(log n) depth before a final carry-propagate add. The difference is the reduction schedule: Dadda uses the *minimum* number of (3:2) counters that still meets the height schedule (fewer gates, slightly more wiring), while Wallace reduces as early and greedily as possible. The `_csa` Wallace variant reduces with explicit carry-save adders. The `4to2` variants use 4:2 compressors (see math_compressor_4to2.md) and are the building block the floating-point mantissa multipliers reuse.

1.3 Floating-Point Arithmetic

IEEE-754-2008 single (fp32) and half (fp16), plus the ML-oriented narrow formats bfloat16 (bf16) and 8-bit (fp8 `e4m3` / `e5m2`). Each format's core operators are assembled from the integer methodologies above: the **exponent path** uses a behavioral 10-bit adder (not a prefix-adder instance) and the **mantissa multiply** uses a Dadda 4:2 tree.

1.3.1 Formats

Format	Modules tag	Notes
bfloat16	`math_bf16_*`	1-8-7; truncated fp32 range
IEEE fp16	`math_fp16_*`, `math_ieee754_2008_fp16_*`	1-5-10
IEEE fp32	`math_fp32_*`, `math_ieee754_2008_fp32_*`	1-8-23
fp8 E4M3	`math_fp8_e4m3_*`	1-4-3
fp8 E5M2	`math_fp8_e5m2_*`	1-5-2

1.3.2 Core operators (per format)

Operation	Modules	Method	Detail
Exponent add	`math_*_exponent_adder`	Behavioral 10-bit add (`assign`, no prefix instance)	math_bf16_exponent_adder.md
Mantissa multiply	`math_*_mantissa_mult`	Dadda 4:2 tree	math_bf16_mantissa_mult.md
Add	`math_*_adder`	Align → add → normalize → round (IEEE-754 §5)	math_bf16_adder.md
Multiply	`math_*_multiplier`	Exponent add + mantissa mult + normalize/round	math_bf16_multiplier.md
Fused multiply-add	`math_*_fma`	Single-rounding a·b + c	math_bf16_fma.md

Detailed per-format coverage: math_fp16_modules.md, math_fp32_modules.md, math_fp8_modules.md, math_ieee754_modules.md, and the bf16 set (math_bf16_extended.md).

1.3.3 Division & reciprocal

Methodology	Modules	Research
Goldschmidt (multiplicative convergence)	`math_bf16_goldschmidt_div`, `math_bf16_divider`	Goldschmidt, R.E. (1964), “Applications of Division by Convergence,” M.Sc. thesis, MIT
Newton-Raphson reciprocal	`math_bf16_newton_raphson_recip`, `math_bf16_reciprocal`, `math_bf16_fast_reciprocal`	Iterative `x_{n+1} = x_n(2 - a·x_n)` refinement of a seed

1.3.4 Conversion

`math_{src}_to_{dst}` — all bidirectional conversions among bf16 / fp16 / fp32 / fp8_e4m3 / fp8_e5m2, plus the integer bridges (`math_bf16_to_int`, `math_int_to_bf16`, `math_bf16_scale_to_int8`). Each does exponent re-bias plus mantissa round/truncate with correct saturation and subnormal handling for the destination format.

1.3.5 Comparison & range

`math_*_comparator`, `math_*_max`, `math_*_min`, `math_*_max_tree_8`, `math_*_min_tree_8`, `math_*_clamp` — magnitude compare (sign/exp/mantissa ordering), pairwise and 8-wide reduction trees, and clamp-to-range.

1.3.6 Activation functions (ML)

`math_*_{relu, leaky_relu, gelu, silu, sigmoid, tanh, softmax_8}` plus the transcendental helpers `math_bf16_{exp2, log2, log2_scale}`. These implement the standard neural-network activations directly in each low-precision format (piecewise / polynomial / LUT approximations sized to the format’s mantissa), so an accelerator can keep activations in bf16/fp8 without promoting to fp32.

1.4 Generation Automation (bin/)

Almost every module above is emitted by a Python code-generator — the RTL is regenerated, not hand-edited. Two entry points:

1.4.1 Integer arithmetic — `bin/math_generate.py`

```
# one methodology + width per invocation
python bin/math_generate.py --type <brent_kung|dadda|wallace_fa|
wallace_csa> \
                            --path <out_dir> --buswidth <N>
```

`bin/math_generate.sh` is the batch driver that sweeps the standard widths (e.g. Brent-Kung at 8/16/32) into `math_outputs/`. The `--type` values map to the methodologies: `brent_kung` (prefix adder), `dadda` / `wallace_fa` / `wallace_csa` (the three multiplier reduction schemes). The emitters live in `bin/rtl_generators/utils` (`write_bk` / `write_dadda` / `write_wallace`) and `bin/rtl_generators/{adders,multipliers}/`.

1.4.2 Floating-point — `bin/rtl_generators/ieee754/generate_all.py`

```
python bin/rtl_generators/ieee754/generate_all.py [output_directory]
```

Generates the full FP library: the shared integer cores (`han_carlson_adder`, `dadda_4to2_multiplier`) followed by each format's `{mantissa_mult, exponent_adder, multiplier, adder, fma}` and all cross-format conversions. The bf16 set has its own `bin/rtl_generators/bf16/generate_all.py`, and the activation / comparison / conversion families are emitted by `fp_activations.py`, `fp_comparisons.py`, and `fp_conversions.py`.

1.4.3 The emitter framework — `bin/rtl_generators/`

Path	Role
`verilog/module.py`, `param.py`, `signal.py`, `verilog_parser.py`	Structured Verilog emission (ports, params, signals) — the backend every generator writes through
`utils/utils.py`	Shared helpers (`write_bk`, `write_dadda`, `write_wallace`, formatting)
`adders/`, `multipliers/`	Integer methodology generators (Brent-Kung, Dadda, Wallace)
`ieee754/`, `bf16/`	Per-format FP operator generators + `generate_all.py`
`ecc/`, `amba/`	Other generated families (not part of `math_*`)
`unittests/`	Generator self-tests

1.4.4 Regeneration rule

These are generated files, and the repo's Critical Rule #0 is not negotiable. When a **generator** changes, you delete and regenerate **all** affected outputs — not a single width or format — and re-run the tests. Widths and cells share interfaces, so a partial regen buys you silent interface mismatches that surface three modules downstream. And never hand-edit a `math_*.sv` that a generator owns. Change the generator, regenerate, move on.

1.5 Related Modules

- Prefix cells: math_prefix_cell.md · math_prefix_cell_gray.md
- Multipliers: dadda_tree · dadda_4to2 · wallace_tree · basic · compressor_4to2
- Subtraction: math_subtractor.md
- Floating-point: bf16 extended · fp16 · fp32 · fp8 · ieee754

1.6 Testing

This is the library catalogue – each module page above carries its own Testing section with the suite that covers it. The whole area runs with `make -C val/math clean-all && make -C val/math run-all-{gate,func,full}-parallel` (119 wrappers, REG_LEVEL x TEST_LEVEL grid), and the formal suite lives in `formal/common/math_*/`.

1.7 References

- Brent, R.P. & Kung, H.T. (1982). "A Regular Layout for Parallel Adders." *IEEE Transactions on Computers*, C-31(3), 260-264.
- Han, T. & Carlson, D.A. (1987). "Fast area-efficient VLSI adders." *Proc. 8th IEEE Symposium on Computer Arithmetic (ARITH-8)*, 49-56.
- Kogge, P.M. & Stone, H.S. (1973). "A Parallel Algorithm for the Efficient Solution of a General Class of Recurrence Equations." *IEEE Transactions on Computers*, C-22(8), 786-793.
- Dadda, L. (1965). "Some schemes for parallel multipliers." *Alta Frequenza*, 34, 349-356.
- Wallace, C.S. (1964). "A Suggestion for a Fast Multiplier." *IEEE Transactions on Electronic Computers*, EC-13(1), 14-17.
- Weinberger, A. & Smith, J.L. (1958). "A Logic for High-Speed Addition." *National Bureau of Standards Circular 591*, 3-12.
- Goldschmidt, R.E. (1964). "Applications of Division by Convergence." M.Sc. thesis, MIT.
- IEEE Std 754-2008, *IEEE Standard for Floating-Point Arithmetic*.

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1.8 Navigation

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2 Basic Adder Building Blocks

Everything in this library eventually bottoms out here: the half adder, the full adder, and an N-bit array of full adders.

2.1 Overview

Three modules form the foundation of every adder architecture in the library: - **math_adder_half** - Single-bit half adder (2 inputs: A, B) - **math_adder_full** - Single-bit full adder (3 inputs: A, B, Cin) - **math_adder_full_nbit** - N-bit array of full adders (identical to ripple carry adder)

You'll find them inside ripple carry adders, carry-save adders, multipliers, and just about every other arithmetic circuit here.

2.1.1 Module Declarations

Half Adder

```
module math_adder_half #(
    parameter int N = 1      // Unused parameter (kept for consistency)
) (
    input  logic i_a,        // Input A
    input  logic i_b,        // Input B
    output logic ow_sum,     // Sum output
    output logic ow_carry    // Carry output
);
```

Full Adder

```
module math_adder_full #(
    parameter int N = 1      // Unused parameter (kept for consistency)
) (
    input  logic i_a,        // Input A
    input  logic i_b,        // Input B
    input  logic i_c,        // Carry input
    output logic ow_sum,     // Sum output
    output logic ow_carry    // Carry output
);
```

N-bit Full Adder Array

```
module math_adder_full_nbit #(
    parameter int N = 4      // Number of bits
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic         i_c,       // Initial carry input
    output logic [N-1:0] ow_sum,    // Sum output
    output logic         ow_carry   // Final carry output
);
```

2.2 Parameters

2.2.1 Half Adder & Full Adder

Parameter	Type	Default	Description
N	int	1	Unused legacy parameter (single-bit modules)

Note: The `N` parameter exists for consistency but is not used. These are single-bit modules.

2.2.2 N-bit Full Adder Array

Parameter	Type	Default	Description
N	int	4	Number of bits in the adder (range: 1-64)

2.3 Ports

2.3.1 Half Adder

Port	Direction	Width	Description
i_a	Input	1	Input A
i_b	Input	1	Input B
ow_sum	Output	1	Sum output (A ^ B)
ow_carry	Output	1	Carry output (A & B)

2.3.2 Full Adder

Port	Direction	Width	Description
i_a	Input	1	Input A
i_b	Input	1	Input B
i_c	Input	1	Carry input
ow_sum	Output	1	Sum output (A ^ B ^ Cin)
ow_carry	Output	1	Carry output

2.3.3 N-bit Full Adder Array

Port	Direction	Width	Description
i_a	Input	N	Operand A
i_b	Input	N	Operand B
i_c	Input	1	Initial carry input
ow_sum	Output	N	Sum output (A + B + Cin)
ow_carry	Output	1	Final carry output

2.4 Functional Description

2.4.1 Half Adder

The half adder adds two single-bit inputs without a carry input:

Logic Equations:

```
Sum = A ⊕ B      (XOR)
Carry = A • B    (AND)
```

Truth Table: | A | B | Sum | Carry | |—|—|—–|——-| | 0 | 0 | 0 | 0 | | 0 | 1 | 1 | 0 | | 1 | 0 | 1 | 0 | | 1 | 1 | 0 | 1 |

Implementation:

```
assign ow_sum   = i_a ^ i_b;    // XOR for sum
assign ow_carry = i_a & i_b;    // AND for carry
```

Use Cases: - LSB addition when no carry input exists - Building block for full adders - Parity generation (sum output) - Detecting when both inputs are 1 (carry output)

2.4.2 Full Adder

The full adder adds three single-bit inputs (A, B, Cin):

Logic Equations:

```
Sum = A ⊕ B ⊕ Cin
Carry = (A • B) | (Cin • (A ⊕ B))
      = Majority(A, B, Cin)
```

Truth Table: | A | B | Cin | Sum | Cout | |—|—|—–|—–|——| | 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 1 | 1 | 0 | | 0 | 1 | 0 | 1 | 0 | | 0 | 1 | 1 | 0 | 1 | | 1 | 0 | 0 | 1 | 0 | | 1 | 0 | 1 | 0 | 1 | | 1 | 1 | 0 | 0 | 1 | | 1 | 1 | 1 | 1 | 1 |

Implementation:

```
assign ow_sum   = i_a ^ i_b ^ i_c;              // 3-input XOR
assign ow_carry = (i_a & i_b) | (i_c & (i_a ^ i_b));  // Majority
function
```

Alternative Carry Implementation (equivalent):

```
assign ow_carry = (i_a & i_b) | (i_a & i_c) | (i_b & i_c);  // Full
majority
```

Use Cases: - Building block for ripple carry adders - Building block for carry-save adders - Multi-bit addition (chained together) - 3:2 compression in multiplier trees

2.4.3 N-bit Full Adder Array

Chains N full adders to create a multi-bit adder:

Structure:

```
Bit 0:   FA(A[0], B[0], Cin)     → Sum[0], C[0]
Bit 1:   FA(A[1], B[1], C[0])    → Sum[1], C[1]
Bit 2:   FA(A[2], B[2], C[1])    → Sum[2], C[2]
...
Bit N-1: FA(A[N-1], B[N-1], C[N-2]) → Sum[N-1], Cout
```

Implementation:

```
logic [N:0] w_c;  // Intermediate carries
assign w_c[0] = i_c;  // Initial carry

genvar i;
generate
    for (i = 0; i < N; i++) begin : gen_full_adders
        math_adder_full fa (
            .i_a(i_a[i]),
            .i_b(i_b[i]),
```

```
            .i_c(w_c[i]),
            .ow_sum(ow_sum[i]),
            .ow_carry(w_c[i+1])
        );
    end
endgenerate

assign ow_carry = w_c[N];  // Final carry out
```

Note: This module is functionally identical to `math_adder_ripple_carry`. The only difference is internal structure: - `math_adder_ripple_carry`: Instantiates first FA separately, then generates remaining - `math_adder_full_nbit`: Uniform generate loop for all FAs

2.5 Timing Characteristics

2.5.1 Propagation Delays

Module	Logic Levels	Typical Delay (ns)
Half Adder	1 (XOR/AND parallel)	~0.2
Full Adder	2 (XOR then carry)	~0.4
N-bit Array	2N (ripple chain)	~0.4N

Critical Paths:

Half Adder:

```
i_a/i_b → ow_sum    (1 XOR gate)
i_a/i_b → ow_carry  (1 AND gate)
```

Full Adder:

```
i_a/i_b/i_c → ow_sum    (2 XOR gates)
i_a/i_b/i_c → ow_carry  (2 gates: XOR + OR/AND)
```

N-bit Array:

```
i_c → FA[0].Cout → FA[1].Cout → ... → FA[N-1].Cout → ow_carry
     (2N gate delays)
```

2.6 Usage Examples

2.6.1 Half Adder: Parity Generator

```
logic [7:0] data;
logic parity;

// Generate even parity bit
```

```
logic [7:0] parity_chain;
assign parity_chain[0] = data[0];
genvar i;
generate
    for (i = 1; i < 8; i++) begin : gen_parity
        math_adder_half u_parity (
            .i_a(data[i]),
            .i_b(parity_chain[i-1]),
            .ow_sum(parity_chain[i]),  // XOR accumulation
            .ow_carry()                 // Unused
        );
    end
endgenerate

assign parity = parity_chain[7];
```

2.6.2 Full Adder: Carry-Save Adder Stage

```
// Add 3 numbers using carry-save adder (1 level)
logic [7:0] a, b, c;
logic [7:0] sum, carry;

genvar i;
generate
    for (i = 0; i < 8; i++) begin : gen_csa
        math_adder_full fa (
            .i_a(a[i]),
            .i_b(b[i]),
            .i_c(c[i]),
            .ow_sum(sum[i]),      // Sum bits
            .ow_carry(carry[i])   // Carry bits (shifted left)
        );
    end
endgenerate

// Final result: sum + (carry << 1) using conventional adder
logic [9:0] result;   // 3 x 255 = 765 needs 10 bits (see
math_adder_carry_save's rule)
assign result = {2'b0, sum} + {1'b0, carry, 1'b0};
```

2.6.3 N-bit Full Adder: BCD Digit Adder

```
// Add two BCD digits (0-9)
logic [3:0] bcd_a, bcd_b, bcd_sum_raw;
logic carry_raw;

math_adder_full_nbit #(.N(4)) u_bcd_add (
    .i_a(bcd_a),
    .i_b(bcd_b),
```

```
    .i_c(1'b0),
    .ow_sum(bcd_sum_raw),
    .ow_carry(carry_raw)
);

// BCD correction
logic bcd_overflow;
assign bcd_overflow = carry_raw | (bcd_sum_raw > 4'd9);

logic [3:0] bcd_sum_corrected;
math_adder_full_nbit #(.N(4)) u_bcd_correct (
    .i_a(bcd_sum_raw),
    .i_b(bcd_overflow ? 4'd6 : 4'd0),  // Add 6 for correction
    .i_c(1'b0),
    .ow_sum(bcd_sum_corrected),
    .ow_carry()
);
```

2.6.4 Building a 2-bit Adder from Scratch

```
logic [1:0] a, b, sum;
logic cin, cout;
logic c_intermediate;

// Bit 0: Half adder (if no carry input) or full adder
math_adder_full fa0 (
    .i_a(a[0]),
    .i_b(b[0]),
    .i_c(cin),
    .ow_sum(sum[0]),
    .ow_carry(c_intermediate)
);

// Bit 1: Full adder
math_adder_full fa1 (
    .i_a(a[1]),
    .i_b(b[1]),
    .i_c(c_intermediate),
    .ow_sum(sum[1]),
    .ow_carry(cout)
);
```

2.7 Design Notes

2.7.1 Why Use These Modules

Advantages: - **Minimal area**: Absolute smallest implementation - **Simple**: Easy to understand and verify - **Building blocks**: Compose into larger structures - **FPGA-friendly**: Maps well to carry chain primitives

Disadvantages: - **Slow for wide widths**: O(N) delay - **Not optimal**: Synthesis tools can often do better - **Manual instantiation**: More code than behavioral + operator

2.7.2 Behavioral vs Structural

Behavioral (Let synthesis infer):

```
// Simple, synthesis tool optimizes
assign sum = a + b + cin;
```

Structural (Explicit full adders):

```
// Manual control, educational
genvar i;
generate
    for (i = 0; i < N; i++) begin
        math_adder_full fa (...);
    end
endgenerate
```

Recommendation: Use behavioral unless you need explicit control (e.g., carry-save adders, multipliers, custom timing control).

2.7.3 FPGA-Specific Considerations

Modern FPGAs have dedicated adder resources: - **Xilinx**: CARRY4 primitives - **Intel**: Carry chain logic - **Result**: Behavioral + often as fast as manual instantiation

Exception: Carry-save adders and multiplier trees benefit from explicit full adder instantiation.

2.7.4 Performance

Module	Structure
Half Adder	1 XOR + 1 AND
Full Adder	2 XOR + 2 AND + 1 OR
N-bit Array	N full adders, carry rippling

FPGA Note: Modern FPGAs (Xilinx, Intel) have dedicated carry chain logic that implements full adders very efficiently — often faster than generic LUT logic.

When to Use Each Module

Half Adder: - First bit of addition chain (no carry input) - Parity generation - Educational/demonstration purposes - Rarely used standalone in production designs

Full Adder: - Building block for custom arithmetic circuits - Carry-save adder stages - Multiplier partial product reduction - When instantiating adder arrays manually

N-bit Array: - Quick multi-bit adder for small widths (N ≤ 8) - When ripple carry is acceptable - For N > 8, consider carry lookahead or parallel prefix

2.7.5 Common Pitfalls

Anti-Pattern 1: Using half adder where full adder needed

```
// WRONG: Half adder can't accept carry input
math_adder_half fa0 (
    .i_a(a[0]), .i_b(b[0]),
    .i_c(cin),  // ERROR: No i_c port!
    ...
);

// RIGHT: Use full adder
math_adder_full fa0 (
    .i_a(a[0]), .i_b(b[0]), .i_c(cin), ...
);
```

Anti-Pattern 2: Over-engineering simple additions

```
// WRONG: Manual full adder chain for simple addition
// (More code, no benefit)
genvar i;
generate
    for (i = 0; i < 16; i++) begin
        math_adder_full fa (...);
    end
endgenerate

// RIGHT: Use behavioral code (simpler, tool optimizes)
assign sum = a + b;
```

Anti-Pattern 3: Ignoring synthesis capabilities

```
// Behavioral code synthesizes to optimal adder anyway:
assign sum = a + b;

// No need for explicit full adders unless:
// - Building carry-save adder
```

```
// - Building multiplier tree
// - Educational/demonstration purpose
```

Anti-Pattern 4: Incorrect carry chain connection

```
// WRONG: both stages fed the SAME carry-in -- fa1 must consume fa0's
// carry-OUT, not the chain's original carry-in
math_adder_full fa0
(.i_a(a[0]), .i_b(b[0]), .i_c(cin),  .ow_sum(s[0]), .ow_carry(c[0]));
math_adder_full fa1
(.i_a(a[1]), .i_b(b[1]), .i_c(cin),  .ow_sum(s[1]), .ow_carry(c[1]));

// RIGHT: carry ripples -- each stage's i_c is the previous stage's
ow_carry
math_adder_full fa0
(.i_a(a[0]), .i_b(b[0]), .i_c(cin),  .ow_sum(s[0]), .ow_carry(c[0]));
math_adder_full fa1
(.i_a(a[1]), .i_b(b[1]), .i_c(c[0]), .ow_sum(s[1]), .ow_carry(c[1]));
```

2.8 Related Modules

- **math_adder_ripple_carry.sv** - Uses full adders for N-bit addition
- **math_adder_carry_save.sv** - Parallel full adders (no carry chain)
- **math_multiplier_*.sv** - Uses full adders for partial product reduction
- **math_adder_full_nbit.sv** - Identical functionality to ripple carry adder

2.9 Testing

Covered by 3 test suites:

- `val/math/test_math_adder_half.py`
- `val/math/test_math_adder_full.py`
- `val/math/test_math_adder_full_nbit.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

2.10 References

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3 Brent-Kung Parallel Prefix Adder

A family of parallel prefix adders built on the Brent-Kung algorithm — O(log N) depth without the Kogge-Stone area bill, thanks to an asymmetric tree and area-efficient gray cells.

3.1 Overview

The `math_adder_brent_kung` module family implements the Brent-Kung parallel prefix addition algorithm, which achieves O(log N) critical path depth while minimizing hardware area compared to other parallel prefix adders like Kogge-Stone. The trick is the asymmetric tree: a forward tree computes intermediate results, then a reverse tree fills in the carries using area-efficient "gray" cells that only propagate generate signals.

Available widths: **8-bit**, **16-bit**, **32-bit**, **64-bit**

3.1.1 Module Hierarchy

Top-Level Modules (User-Facing): - `math_adder_brent_kung_008.sv` - 8-bit adder - `math_adder_brent_kung_016.sv` - 16-bit adder - `math_adder_brent_kung_032.sv` - 32-bit adder - `math_adder_brent_kung_064.sv` - 64-bit adder (used as the final CPA of the 32-bit Dadda and Wallace multipliers)

Internal Building Blocks: - `math_adder_brent_kung_pg.sv` - Single-bit P/G generator - `math_adder_brent_kung_black.sv` - Black cell (outputs P and G) - `math_adder_brent_kung_gray.sv` - Gray cell (outputs only G, area-efficient) - `math_adder_brent_kung_bitwisepg.sv` - Bitwise P/G logic stage - `math_adder_brent_kung_grouppg_008/016/032/064.sv` - Width-specific prefix networks - `math_adder_brent_kung_sum.sv` - Final sum calculation

3.1.2 Module Declaration

```
module math_adder_brent_kung_032 #(
    parameter int N = 32     // Adder width in bits
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic         i_c,       // Carry input
    output logic [N-1:0] ow_sum,    // Sum output
    output logic         ow_carry   // Carry output
);
```

Note: Replace `_032` with `_008`, `_016` or `_064` for the 8-, 16- or 64-bit version. All four share this port list; only `N` and the internal prefix network change.

3.2 Parameters

Parameter	Type	Default	Description
N	int	32	Adder width in bits (8, 16, 32 or 64)

Width Options: - **8-bit**: Set N=8 in `math_adder_brent_kung_008` - **16-bit**: Set N=16 in `math_adder_brent_kung_016` - **32-bit**: Set N=32 in `math_adder_brent_kung_032` - **64-bit**: Set N=64 in `math_adder_brent_kung_064`

Why Fixed Widths? The parallel prefix network structure is width-specific and optimized for each size during design. Generic parameterization would require runtime generation of the tree structure — not something a parameter can express.

3.3 Ports

Port	Direction	Width	Description
i_a	Input	N	Operand A (addend)
i_b	Input	N	Operand B (addend)
i_c	Input	1	Carry input (0 for addition, 1 for +1 increment)
ow_sum	Output	N	Sum output (A + B + Cin)
ow_carry	Output	1	Carry output (overflow)

3.4 Functional Description

3.4.1 Parallel Prefix Addition Theory

Parallel prefix adders compute all carries in parallel using **propagate (P)** and **generate (G)** signals:

Bit-Level Definitions: - **Generate**: G_i = A_i & B_i (carry generated at position i) - **Propagate**: P_i = A_i ^ B_i (carry propagated through position i) - **Sum**: S_i = P_i ^ C_i (sum bit given incoming carry)

Group-Level Recurrence: - **G[i:j]** = G[i:k] | (P[i:k] & G[k-1:j]) (group generate) - **P[i:j]** = P[i:k] & P[k-1:j] (group propagate)

3.4.2 Brent-Kung Algorithm Stages

The Brent-Kung adder executes in three stages:

3.4.2.1 Stage 1: Bitwise P/G Generation

```
For each bit i:
  P[i] = A[i] ^ B[i]
  G[i] = A[i] & B[i]
```

Hardware: Array of AND/XOR gates (1 level of logic)

3.4.2.2 Stage 2: Parallel Prefix Network (Forward + Reverse Trees)

```
Forward Tree (log2(N) - 1 levels, black cells):
  Build binary tree upward using black cells
  Compute group P/G for increasing spans: [1:0], [3:2], [7:4], etc.

Reverse Tree (log2(N)-1 levels):
  Propagate group G downward using gray cells
  Fill in missing intermediate positions
```

Hardware: Network of black cells (forward) and gray cells (reverse)

Key Optimization: Gray cells only compute G (not P), saving ~30% area vs Kogge-Stone

3.4.2.3 Stage 3: Final Sum Calculation

```
For each bit i:
  Sum[i] = P[i] ^ G[i-1]  (XOR with incoming carry)

Carry_out = G[N-1]  (final group generate)
```

Hardware: Array of XOR gates (1 level of logic)

3.4.3 Three-Stage Pipeline

The top-level module instantiates three sub-modules:

```
// Stage 1: Generate bitwise P and G signals
math_adder_brent_kung_bitwisepg #(.N(N)) BitwisePGLogic_inst (
    .i_a (i_a),
    .i_b (i_b),
    .i_c (i_c),
    .ow_g(ow_g),  // Generate signals [N:0]
```

```
    .ow_p(ow_p)   // Propagate signals [N:0]
);

// Stage 2: Parallel prefix network (forward + reverse trees)
math_adder_brent_kung_grouppg_032 #(.N(N)) GroupPGLogic_inst (
    .i_g  (ow_g),
    .i_p  (ow_p),
    .ow_gg(ow_gg),  // Group generate [N:0]
    .ow_pp()        // Unused (gray cells only compute G)
);

// Stage 3: Compute final sum
math_adder_brent_kung_sum #(.N(N)) SumLogic_inst (
    .i_p      (ow_p),
    .i_gg     (ow_gg),
    .ow_sum   (ow_sum),
    .ow_carry (ow_carry)
);
```

3.4.4 Internal Building Blocks

3.4.4.1 PG Cell (math_adder_brent_kung_pg)

```
// Single-bit P/G generator
assign ow_g = i_a & i_b;  // Generate
assign ow_p = i_a ^ i_b;  // Propagate
```

Usage: Initial stage to create bit-level P/G signals

3.4.4.2 Black Cell (math_adder_brent_kung_black)

```
// Combines two P/G pairs, outputs both
assign ow_g = i_g | (i_p & i_g_km1);  // Group generate
assign ow_p = i_p & i_p_km1;          // Group propagate
```

Usage: Forward tree (needs both P and G for further propagation)

Logic Depth: 2 gates (1 AND + 1 OR/AND)

3.4.4.3 Gray Cell (math_adder_brent_kung_gray)

```
// Combines two P/G pairs, outputs only G (area-efficient)
assign ow_g = i_g | (i_p & i_g_km1);  // Group generate
// No P output (saves one AND gate per cell)
```

Usage: Reverse tree (only needs G, not P)

Area Savings: ~33% fewer gates vs black cell

Logic Depth: 2 gates (1 AND + 1 OR)

3.4.5 Prefix Network Structure (32-bit Example)

```
Depth 0 (Bitwise):  P0 G0  P1 G1  P2 G2  P3 G3  ...  P31 G31

Depth 1 (Forward):  [1:0]g  [3:2]  [5:4]  [7:6] ... [31:30]      (15
black + 1 gray;

[1:0] is GRAY -- bit 1's

final prefix needs no P)

Depth 2 (Forward):  [3:0]g  [7:4]  [11:8]  [15:12] ... [31:28]   (7
black + 1 gray)

Depth 3 (Forward):  [7:0]g  [15:8]  [23:16]  [31:24]             (3
black + 1 gray)

Depth 4 (Forward):  [15:0]g  [31:16]                             (1
black + 1 gray)

Depth 5 (Root):     [31:0]g                                      (gray
-- the root

needs no P output)

Depth 6+ (Reverse fill): gray cells complete the remaining
prefixes ([2:0], [4:0], [5:0], [6:0], ...) by combining each
bit's own (g,p) with the nearest completed prefix below it

Depth 8 (Reverse):  deepest gap-fill (e.g. [30:29])            (Gray
cells)

Depth 9 (Sum):      S0    S1    S2    S3    ...    S31         (XOR
gates)
```

Key Features: - **Forward tree**: log2(N) - 1 levels; these are the levels that contain black cells - **Reverse tree**: log2(N) - 1 further levels, gray cells only - **Prefix-network depth**: 2 x log2(N) - 2 cell levels - **Total depth**: 2 x log2(N) cell levels, adding the bitwise PG stage and the sum XOR stage - **Black cells**: Used in the forward tree (outputs both P and G) - **Gray cells**: Used where only G is needed further (outputs only G, saves area); the reverse tree — including the [31:0] root — is gray-only. The RTL confirms: the root instantiates `math_adder_brent_kung_gray`, not a black cell

Measured from the generated prefix networks (`math_adder_brent_kung_grouppg_*.sv`):

Brent-Kung prefix network depth and cell counts, counted from the RTL

Width	Prefix levels	Total levels	Black cells	Gray cells
8-bit	4	6	4	8
16-bit	6	8	11	16
32-bit	8	10	26	32
64-bit	10	12	57	64

Black cells follow `N - log2(N) - 1` and gray cells `N`. Don't take my word for it — count them yourself with `grep -c math_adder_brent_kung_black rtl/math/math_adder_brent_kung_grouppg_032.sv`.

3.5 Timing Characteristics

3.5.1 Combinational Delay Analysis

Logic levels below are cell levels counted from the RTL. The delay and frequency columns are rough estimates only — no synthesis run backs them, and they will move substantially with technology, constraints and effort level.

Brent-Kung timing, logic levels measured and delays estimated

Width	Logic Levels	Est. Delay (ns)	Est. Max Frequency
8-bit	6	~2.0	~500 MHz
16-bit	8	~2.5	~400 MHz
32-bit	10	~3.0	~333 MHz
64-bit	12	~3.5	~285 MHz

Logic Level Breakdown (32-bit): 1. Bitwise PG: 1 level (AND/XOR) 2. Forward tree: 4 levels (levels containing black cells) 3. Reverse tree: 4 levels (gray cells only) 4. Sum calculation: 1 level (XOR) 5. **Total**: 10 levels

Comparison to Other Adders:

Adder Type	Logic Depth	Area	Best Use Case
Ripple Carry	O(N)	Minimal	Low-speed, area-critical
Carry Lookahead	O(log N)	Medium	Moderate speed
Brent-Kung	**O(log N)**	**Medium**	**Balanced**

Adder Type	Logic Depth	Area	Best Use Case
			speed/area
Kogge-Stone	O(log N)	Maximum	Maximum speed (datapath)

3.5.2 Critical Paths

1. **Forward Tree Path**: i_a/i_b -> PG -> black cells (prefix levels 1-4) -> Group G
2. **Reverse Tree Path**: Group G -> gray cells (prefix levels 5-8) -> Final G
3. **Sum Path**: Final G → XOR with P → ow_sum

Optimization Tip: Pipeline between stages for >1 GHz operation:

```
// Pipeline registers (example for 2-stage pipeline)
logic [N-1:0] r_p_stage1, r_g_stage1;
logic [N-1:0] r_sum;

always_ff @(posedge clk) begin
    // Stage 1: PG generation + first half of prefix tree
    r_p_stage1 <= ow_p;
    r_g_stage1 <= partial_gg;  // Intermediate prefix result

    // Stage 2: Second half of prefix tree + sum
    r_sum <= ow_sum;
end
```

3.6 Usage Examples

3.6.1 Basic 32-bit Addition

```
logic [31:0] a, b, sum;
logic carry_out;

math_adder_brent_kung_032 #(
    .N(32)
) u_adder (
    .i_a      (a),
    .i_b      (b),
    .i_c      (1'b0),        // No carry input
    .ow_sum   (sum),
    .ow_carry (carry_out)
);

// Example: 123 + 456
```

```
initial begin
    a = 32'd123;
    b = 32'd456;
    #1;  // Wait for combinational propagation
    assert(sum == 32'd579);
end
```

3.6.2 8-bit Incrementer

```
logic [7:0] count, count_plus_1;

math_adder_brent_kung_008 #(
    .N(8)
) u_incrementer (
    .i_a      (count),
    .i_b      (8'b0),        // B = 0
    .i_c      (1'b1),        // Carry in = 1 → +1 increment
    .ow_sum   (count_plus_1),
    .ow_carry ()             // Unused
);
```

3.6.3 16-bit Subtraction (A - B)

```
logic [15:0] a, b, difference;
logic borrow;

// Subtraction: A - B = A + (~B) + 1 (two's complement)
math_adder_brent_kung_016 #(
    .N(16)
) u_subtractor (
    .i_a      (a),
    .i_b      (~b),          // Invert B
    .i_c      (1'b1),        // Carry in = 1 (for two's complement)
    .ow_sum   (difference),
    .ow_carry (borrow)       // Borrow = ~carry_out
);

// Example: 1000 - 300
initial begin
    a = 16'd1000;
    b = 16'd300;
    #1;
    assert(difference == 16'd700);
end
```

3.6.4 Multi-Precision Addition (64-bit via 2×32-bit)

```
logic [31:0] a_low, a_high, b_low, b_high;
logic [31:0] sum_low, sum_high;
```

```
logic carry_low, carry_high;

// Low 32 bits
math_adder_brent_kung_032 u_add_low (
    .i_a      (a_low),
    .i_b      (b_low),
    .i_c      (1'b0),
    .ow_sum   (sum_low),
    .ow_carry (carry_low)
);

// High 32 bits (chain carry from low)
math_adder_brent_kung_032 u_add_high (
    .i_a      (a_high),
    .i_b      (b_high),
    .i_c      (carry_low),     // Carry from low adder
    .ow_sum   (sum_high),
    .ow_carry (carry_high)
);

// Concatenate results
logic [63:0] sum_64 = {sum_high, sum_low};
```

3.6.5 Overflow Detection (Signed Addition)

```
logic [31:0] a_signed, b_signed, sum_signed;
logic overflow;

math_adder_brent_kung_032 u_signed_add (
    .i_a      (a_signed),
    .i_b      (b_signed),
    .i_c      (1'b0),
    .ow_sum   (sum_signed),
    .ow_carry ()
);

// Overflow detection for signed addition
// Overflow = (A[31] == B[31]) && (Sum[31] != A[31])
assign overflow = (a_signed[31] == b_signed[31]) &&
                  (sum_signed[31] != a_signed[31]);

// Example: Detect overflow
initial begin
    // Max positive + positive = overflow
    a_signed = 32'h7FFFFFFF;  // +2,147,483,647
    b_signed = 32'h00000001;  // +1
    #1;
```

```
    assert(overflow == 1'b1);  // Result wraps to negative
end
```

3.7 Design Notes

3.7.1 Width Selection

Available Widths: - **8-bit**: Suitable for byte operations, small arithmetic units - **16-bit**: Common for ALU slices, address arithmetic - **32-bit**: Standard integer width, general-purpose ALU - **64-bit**: Wide datapaths and product-width CPAs; this is the final adder inside the 32-bit Dadda and Wallace multipliers

For Non-Standard Widths: - Extend inputs with zeros: `{24'b0, a[7:0]}` for 8-bit in 32-bit adder - Truncate outputs: `sum_32[15:0]` for 16-bit result - Or use ripple carry / carry lookahead for custom widths

3.7.2 Synthesis Considerations

Optimization Tips: 1. **Flatten hierarchy** for best optimization: `tcl     set_flatten true -effort high`

2. **Pipeline for high frequency**:
   - Break at forward/reverse tree boundary
   - Break at prefix network / sum calculation boundary
3. **Critical path optimization**:

   ```
   set_max_delay -from [get_ports i_a] -to [get_ports ow_sum] 3.0
   ```

4. **Area optimization** (if needed):
   - Consider carry lookahead instead (smaller but still fast)
   - Share adders using time-division multiplexing

3.7.3 Performance

Cell counts below are exact, counted from the RTL.

Brent-Kung prefix-cell counts, measured from the RTL

Width	Black cells	Gray cells	Pipeline FFs
8-bit	4	8	0 (combinational)
16-bit	11	16	0

Width	Black cells	Gray cells	Pipeline FFs
			(combinational)
32-bit	26	32	0 (combinational)
64-bit	57	64	0 (combinational)

Speed vs Area Trade-offs:

Adder Architecture	Relative Speed	Relative Area	Logic Depth
Ripple Carry	1.0× (slowest)	1.0× (smallest)	O(N)
Carry Lookahead	4.0×	3.0×	O(log N)
Brent-Kung	**6.0×**	**4.5×**	**O(log N)**
Kogge-Stone	8.0× (fastest)	7.0× (largest)	O(log N)

When to Use Brent-Kung: - **Balanced designs**: Need good speed without excessive area - **Mid-range frequency**: 200-500 MHz targets - **FPGA implementations**: LUT utilization matters - **Multiple adders**: Area budget shared across many units

When to Use Alternatives: - Use **Ripple Carry** for: Low-speed, area-critical (e.g., control logic) - Use **Kogge-Stone** for: Critical datapath, maximum performance (e.g., FPU)

3.7.4 Common Pitfalls

Anti-Pattern 1: Using wrong width variant

```
// WRONG: Trying to parameterize to 24-bit
math_adder_brent_kung_032 #(.N(24)) u_add (...);
// Result: Mismatch with grouppg_032 network (designed for 32-bit)

// RIGHT: Use 32-bit, ignore upper bits
math_adder_brent_kung_032 #(.N(32)) u_add (
    .i_a({8'b0, a[23:0]}),  // Zero-extend
    .i_b({8'b0, b[23:0]}),
    .ow_sum(sum_32),
    ...
);
logic [23:0] sum = sum_32[23:0];  // Truncate
```

Anti-Pattern 2: Ignoring carry output for signed arithmetic

```
// WRONG: Using carry for signed overflow
logic [31:0] a_signed, b_signed, sum;
logic overflow;

math_adder_brent_kung_032 u_add (...);
assign overflow = ow_carry;  // INCORRECT for signed!

// RIGHT: Check sign bits
assign overflow = (a_signed[31] == b_signed[31]) &&
                  (sum[31] != a_signed[31]);
```

Anti-Pattern 3: Expecting registered outputs

```
// WRONG: Assuming outputs are registered
always_ff @(posedge clk) begin
    result <= ow_sum;  // ow_sum is combinational!
end

// RIGHT: Add explicit output register
logic [31:0] r_sum;
always_ff @(posedge clk) begin
    r_sum <= ow_sum;
end
```

Anti-Pattern 4: Chaining without considering timing

```
// WRONG: Long chain of adders in one cycle
logic [31:0] a, b, c, d, result;
logic [31:0] ab_sum, abc_sum;

math_adder_brent_kung_032 u_add1
(.i_a(a), .i_b(b), .ow_sum(ab_sum), ...);
math_adder_brent_kung_032 u_add2
(.i_a(ab_sum), .i_b(c), .ow_sum(abc_sum), ...);
math_adder_brent_kung_032 u_add3
(.i_a(abc_sum), .i_b(d), .ow_sum(result), ...);

// 3 × 3ns = 9ns path!  May not close timing at high frequency

// RIGHT: Pipeline the chain
logic [31:0] r_ab_sum, r_abc_sum;
always_ff @(posedge clk) begin
    r_ab_sum <= ab_sum;
    r_abc_sum <= abc_sum;
end
```

3.8 Related Modules

- **math_adder_pg_chain.sv** - Medium-speed adder (less area)
- **math_adder_ripple_carry.sv** - Minimal area adder (slow)
- **math_adder_carry_save_nbit.sv** - For multi-operand addition (multipliers)
- **math_subtractor_*.sv** - Subtraction variants (uses two's complement)

3.9 Testing

From the test suite (`val/math/test_math_adder_brent_kung.py`) — one parameterized test covering every width, not one file per width.

Key Test Scenarios: - Exhaustive testing (8-bit only) - Random stimulus (16/32-bit) - Corner cases: 0+0, 0+MAX, MAX+MAX - Carry propagation: All 1's + 1 - Signed overflow detection - Multi-precision chaining

Test Command:

```
# Test all widths the test sweeps (REG_LEVEL selects the width list)
pytest val/math/test_math_adder_brent_kung.py -v

# A single width
pytest
"val/math/test_math_adder_brent_kung.py::test_math_adder_brent_kung[32
]" -v
```

The width sweep is driven by `REG_LEVEL`: `FUNC` (the default) covers 8 and 16 bits, `FULL` covers 8, 16 and 32. The 64-bit variant is exercised indirectly, as the final CPA of the 32-bit Dadda and Wallace multiplier tests, rather than by a direct adder test.

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

3.10 References

- Brent, R.P., Kung, H.T. "A Regular Layout for Parallel Adders." IEEE Transactions on Computers, 1982.
- Harris, D. "A Taxonomy of Parallel Prefix Networks." Asilomar Conference, 2003.
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4 Carry-Save Adder

Single-bit and N-bit carry-save adders for high-speed multi-operand addition — 3:2 compression with constant-depth parallel operation.

4.1 Overview

Carry-save adders (CSAs) add three operands in a single level of logic by keeping the sum and carry outputs separate — no carry chaining between bits. That buys you: - **Constant depth**: O(1) logic depth regardless of bit width - **3:2 compression**: Reduce 3 numbers to 2 in one stage - **Multiplier trees**: Essential for Wallace and Dadda tree multipliers - **Multi-operand addition**: Add many numbers efficiently

These are the fundamental building blocks of fast multipliers and multi-operand addition trees.

Modules Covered: - **math_adder_carry_save** - Single-bit CSA (3:2 compressor) - **math_adder_carry_save_nbit** - N-bit parallel CSA array

4.1.1 Module Declarations

Single-bit Carry-Save Adder

```
module math_adder_carry_save (
    input  logic i_a,         // Input A
    input  logic i_b,         // Input B
    input  logic i_c,         // Input C
    output logic ow_sum,      // Sum output
    output logic ow_carry     // Carry output
);
```

N-bit Carry-Save Adder

```
module math_adder_carry_save_nbit #(
    parameter int N = 4      // Bit width
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic [N-1:0] i_c,       // Operand C (or carry from
previous CSA)
    output logic [N-1:0] ow_sum,    // Sum outputs (parallel)
    output logic [N-1:0] ow_carry   // Carry outputs (parallel, NOT
```

```
chained!)
);
```

4.2 Parameters

4.2.1 Single-bit CSA

No parameters (fixed single-bit operation).

4.2.2 N-bit CSA

Parameter	Type	Default	Description
N	int	4	Bit width (range: 1-64, typical: 8-64)

Width Guidelines: - **Typical**: 8-64 bits (matches datapath width) - **Multipliers**: Match partial product width - **No practical limit**: CSA depth is independent of N

4.3 Ports

4.3.1 Single-bit CSA

Port	Direction	Width	Description
i_a	Input	1	Input A
i_b	Input	1	Input B
i_c	Input	1	Input C
ow_sum	Output	1	Sum output bit
ow_carry	Output	1	Carry output bit

4.3.2 N-bit CSA

Port	Direction	Width	Description
i_a	Input	N	Operand A
i_b	Input	N	Operand B
i_c	Input	N	Operand C
ow_sum	Output	N	Sum vector (parallel)
ow_carry	Output	N	Carry vector. Weight 2^(i+1): shift left 1 in the

Port	Direction	Width	Description
			final reduction

Important: The N-bit CSA outputs TWO N-bit vectors. To get the final result, you must add them using a conventional adder (ripple carry, CLA, or parallel prefix).

4.4 Functional Description

4.4.1 Single-bit Carry-Save Adder (3:2 Compressor)

The CSA compresses 3 bits into 2 bits (sum and carry):

Logic Equations:

```
Sum = A ⊕ B ⊕ C                                         // 3-input XOR
Carry = (A • B) | (A • C) | (B • C)                     // Majority
function
      = Majority(A, B, C)
```

Truth Table: | A | B | C | Sum | Carry | A+B+C (decimal) | |—|—|—|—–|——-|—————–| | 0 | 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 1 | 1 | 0 | 1 | | 0 | 1 | 0 | 1 | 0 | 1 | | 0 | 1 | 1 | 0 | 1 | 2 | | 1 | 0 | 0 | 1 | 0 | 1 | | 1 | 0 | 1 | 0 | 1 | 2 | | 1 | 1 | 0 | 0 | 1 | 2 | | 1 | 1 | 1 | 1 | 1 | 3 |

Key Property: Sum + 2×Carry = A + B + C (mathematically exact)

Implementation:

```
assign ow_sum   = i_a ^ i_b ^ i_c;                         // XOR
assign ow_carry = (i_a & i_b) | (i_a & i_c) | (i_b & i_c); // Majority
```

Note: This is identical to a full adder, but used differently: - **Full adder**: Carry chains to next bit - **CSA**: Carry kept separate, processed in next stage

4.4.2 N-bit Carry-Save Adder

The N-bit CSA applies 3:2 compression to each bit position in parallel:

```
Bit 0:  CSA(A[0], B[0], C[0]) → Sum[0], Carry[0]
Bit 1:  CSA(A[1], B[1], C[1]) → Sum[1], Carry[1]
...
Bit N-1: CSA(A[N-1], B[N-1], C[N-1]) → Sum[N-1], Carry[N-1]
```

Implementation:

```
genvar i;
generate
    for (i = 0; i < N; i++) begin : gen_carry_save
        assign ow_sum[i]   = i_a[i] ^ i_b[i] ^ i_c[i];
        assign ow_carry[i] = (i_a[i] & i_b[i]) | (i_a[i] & i_c[i]) |
```

```
(i_b[i] & i_c[i]);
    end
endgenerate
```

Critical Difference from Ripple Carry: - **Carry NOT chained**: Each bit's carry goes to separate output, not to next bit - **Constant depth**: All bits computed in parallel (1 level of logic) - **Requires final addition**: Must add sum and carry vectors to get actual result

4.4.3 Final Addition Step

To get the actual sum, add sum to carry shifted left by one — the carry from bit i has weight 2^(i+1) (`Key Property` above: Sum + 2xCarry = A + B + C):

```
// CSA produces two vectors
math_adder_carry_save_nbit #(.N(N)) u_csa (
    .i_a(a), .i_b(b), .i_c(c),
    .ow_sum(sum_vec),
    .ow_carry(carry_vec)
);

// Add sum to (carry << 1) -- carry has weight 2^(i+1).
// Three N-bit operands sum to at most 3*(2^N - 1): that needs N+2
bits.
logic [N+1:0] final_result;
assign final_result = {2'b0, sum_vec} + {carry_vec, 1'b0};
```

Common mistake: adding the vectors without shifting the carry. `Sum + 2xCarry = A + B + C` (the `Key Property` above) — the factor of two IS a left shift by one. `sum_vec + carry_vec` is wrong; `sum_vec + (carry_vec << 1)` is correct. I've debugged this exact bug in someone else's multiplier at least twice.

4.5 Timing Characteristics

4.5.1 Propagation Delays

Module	Logic Levels	Typical Delay (ns)
Single-bit CSA	2	~0.4
N-bit CSA	2	~0.4 (independent of N!)

Key Advantage: Delay is constant regardless of bit width. That's the whole point of the architecture.

Critical Path:

```
i_a/i_b/i_c → ow_sum    (2 XOR gates)
i_a/i_b/i_c → ow_carry  (2 gates: AND + OR)
```

Comparison: | Adder Type | N-bit Delay | |———–|———–| | **CSA** | **O(1)** - constant | | Ripple Carry | O(N) - linear | | Carry Lookahead | O(N) - linear (simplified) | | Parallel Prefix | O(log N) |

4.6 Usage Examples

4.6.1 Adding 3 Numbers (Single CSA Stage)

```
logic [7:0] a, b, c;
logic [7:0] sum_vec, carry_vec;
logic [9:0] final_result;

// Stage 1: CSA reduces 3 numbers to 2
math_adder_carry_save_nbit #(.N(8)) u_csa (
    .i_a(a),
    .i_b(b),
    .i_c(c),
    .ow_sum(sum_vec),
    .ow_carry(carry_vec)
);

// Stage 2: conventional adder, carry shifted left 1
assign final_result = {2'b0, sum_vec} + {1'b0, carry_vec, 1'b0};

// Example: 10 + 20 + 30 = 60
// CSA gives sum_vec = 0, carry_vec = 30; 0 + (30 << 1) = 60
initial begin
    a = 8'd10;
    b = 8'd20;
    c = 8'd30;
    #1;
    assert(final_result == 10'd60);
end
```

4.6.2 Adding 4 Numbers (2-Level CSA Tree)

```
logic [7:0] a, b, c, d;
logic [7:0] sum1, carry1;
logic [9:0] final_result;  // sum2/carry2 declared 9-bit below

// Level 1: CSA reduces 4 numbers to 2 (actually 3)
// CSA1: A + B + C → Sum1, Carry1
math_adder_carry_save_nbit #(.N(8)) u_csa1 (
    .i_a(a), .i_b(b), .i_c(c),
    .ow_sum(sum1),
    .ow_carry(carry1)
);
```

```
// Level 2: CSA reduces 3 numbers to 2
// CSA2: Sum1 + (Carry1<<1) + D → Sum2, Carry2
// carry1 has weight 2^(i+1): it enters the next level SHIFTED, so
level 2
// runs one bit wider.
logic [8:0] sum2, carry2;
math_adder_carry_save_nbit #(.N(9)) u_csa2 (
    .i_a({1'b0, sum1}),
    .i_b({carry1, 1'b0}),
    .i_c({1'b0, d}),
    .ow_sum(sum2),
    .ow_carry(carry2)
);

// Final: Conventional adder. 4 x 8-bit operands: max 4*255 = 1020 (10
bits).
assign final_result = {1'b0, sum2} + {carry2, 1'b0};
```

4.6.3 Adding 7 Numbers (Wallace Tree Structure)

```
// Reduce 7 numbers to 2 using CSA tree
logic [7:0] n1, n2, n3, n4, n5, n6, n7;

// Level 1: 7 → 5 (two CSAs)
logic [7:0] l1_s1, l1_c1, l1_s2, l1_c2;
math_adder_carry_save_nbit #(.N(8)) u_csa_l1_1 (
    .i_a(n1), .i_b(n2), .i_c(n3),
    .ow_sum(l1_s1), .ow_carry(l1_c1)
);
math_adder_carry_save_nbit #(.N(8)) u_csa_l1_2 (
    .i_a(n4), .i_b(n5), .i_c(n6),
    .ow_sum(l1_s2), .ow_carry(l1_c2)
);
// n7 passes through

// Carries have weight 2^(i+1): every carry enters the next level
SHIFTED
// left one, so each level runs one bit wider than the last.

// Level 2: 5 → 4 (one CSA, 9 bits)
logic [8:0] l2_s1, l2_c1;
math_adder_carry_save_nbit #(.N(9)) u_csa_l2_1 (
    .i_a({1'b0, l1_s1}), .i_b({l1_c1, 1'b0}), .i_c({1'b0, l1_s2}),
    .ow_sum(l2_s1), .ow_carry(l2_c1)
);
// l1_c2, n7 pass through
```

```
// Level 3: 4 → 3 (one CSA, 10 bits)
logic [9:0] l3_s1, l3_c1;
math_adder_carry_save_nbit #(.N(10)) u_csa_l3_1 (
    .i_a({1'b0, l2_s1}), .i_b({l2_c1, 1'b0}), .i_c({1'b0, l1_c2,
1'b0}),  // weight-2 value: shift left 1 like every carry
    .ow_sum(l3_s1), .ow_carry(l3_c1)
);
// n7 passes through

// Level 4: 3 → 2 (one CSA, 11 bits)
logic [10:0] l4_s1, l4_c1;
math_adder_carry_save_nbit #(.N(11)) u_csa_l4_1 (
    .i_a({1'b0, l3_s1}), .i_b({l3_c1, 1'b0}), .i_c({3'b0, n7}),
    .ow_sum(l4_s1), .ow_carry(l4_c1)
);

// Final addition. 7 x 8-bit operands: max 7*255 = 1785 (11 bits).
logic [10:0] final_result;
assign final_result = l4_s1 + {l4_c1, 1'b0};
```

4.6.4 8-bit×8-bit Multiplier Partial Product Reduction

```
// Generate partial products (8×8 = 8 partial products)
logic [7:0] pp [0:7];  // Partial products

// CSA tree to reduce 8 partial products to 2
// (Simplified - real implementation more complex)

// Level 1: 8 → 6 (two CSAs)
logic [15:0] l1_s1, l1_c1, l1_s2, l1_c2;
math_adder_carry_save_nbit #(.N(16)) u_csa_pp_l1_1 (
    .i_a({8'b0, pp[0]}),
    .i_b({7'b0, pp[1], 1'b0}),
    .i_c({6'b0, pp[2], 2'b0}),
    .ow_sum(l1_s1), .ow_carry(l1_c1)
);
// ... continue tree reduction
// Final: Fast adder for last two vectors
```

4.7 Design Notes

4.7.1 When to Use Carry-Save Adders

Ideal Use Cases: - **Multipliers**: Wallace tree, Dadda tree - **Multi-operand addition**: Add 3+ numbers - **DSP applications**: MAC units, filters - **Dot products**: Sum of products - **Compression trees**: Reduce many numbers to few

4.7.2 When NOT to Use CSA

Inappropriate Use Cases: - **Two-operand addition**: Use conventional adder (simpler) - **Final result needed immediately**: CSA requires additional adder stage - **Single addition**: No benefit over regular adder

4.7.3 CSA Tree Design Guidelines

Number of Stages: each CSA stage reduces the operand count to ceil(2N/3) (groups of three become pairs, remainders pass through); stages to reach 2 is roughly ceil(log_1.5(N/2)) — and remainder effects add a stage at some sizes, so count the sequence, not just the bound.

Example: - 3 operands: 1 CSA + 1 adder (3 → 2) - 7 operands: 4 CSA stages / 5 CSAs + 1 adder (7 → 5 → 4 → 3 → 2; each CSA is 3→2, removing exactly one operand, so N operands take N-2 CSAs) - 15 operands: 6 CSA stages / 13 CSAs + 1 adder (15 → 10 → 7 → 5 → 4 → 3 → 2; ceil(log_1.5(15/2)) = 5 underestimates — the passthroughs cost a stage)

Optimization: Minimize final adder width by aligning partial products carefully.

4.7.4 Final Adder Selection

After CSA tree reduction, choose the final adder wisely — it sits on the critical path of every multiplier you'll build:

Width	Best Final Adder
≤ 8 bits	Ripple carry (small, adequate)
8-16 bits	Carry lookahead
16-32 bits	Brent-Kung
≥ 32 bits	Brent-Kung (to 64-bit) or Han-Carlson (to 72-bit); pipeline beyond

4.7.5 Performance

Module	LUTs	Description
Single-bit CSA	2	1 XOR + 1 Majority
N-bit CSA	~2N	N single-bit CSAs

Comparison to Full Adder: - **Logic**: Identical (both use XOR + Majority) - **Usage**: Different (parallel vs chained)

CSA vs Ripple Carry Adder (8-bit example):

Operation	Depth	Delay (ns)
CSA (parallel)	2 levels	~0.4

Operation	Depth	Delay (ns)
Ripple (sequential)	16 levels	~3.0
Speedup	**8×**	**7.5×**

Why CSA is Faster: - No carry propagation between bits - All bits computed in parallel - Constant depth regardless of width

4.7.6 Common Pitfalls

Anti-Pattern 1: Reducing without shifting the carry

```
// WRONG: carry has weight 2^(i+1), so this drops a factor of two
assign final_result = sum_vec + carry_vec;  // INCORRECT!

// RIGHT: shift carry left by one
assign final_result = sum_vec + {carry_vec, 1'b0};
```

Anti-Pattern 2: Using CSA for two-operand addition

```
// WRONG: Overkill for simple addition
math_adder_carry_save_nbit u_csa (
    .i_a(a), .i_b(b), .i_c(8'b0),  // Third input wasted!
    ...
);
assign result = sum_vec + carry_vec;  // Extra adder stage!

// RIGHT: Use conventional adder
assign result = a + b;  // Simpler, same result
```

Anti-Pattern 3: Ignoring final addition

```
// WRONG: CSA outputs are NOT the final result!
math_adder_carry_save_nbit u_csa (
    .i_a(a), .i_b(b), .i_c(c),
    .ow_sum(result),        // INCOMPLETE!
    .ow_carry(ignored)      // Missing carry!
);

// RIGHT: add sum and carry, carry shifted left one
assign final_result = ow_sum + {ow_carry, 1'b0};
```

Anti-Pattern 4: Incorrect width for final addition

```
// WRONG: Overflow possible
logic [7:0] a, b, c;
logic [7:0] sum_vec, carry_vec;
logic [7:0] result;  // TOO NARROW!
assign result = sum_vec + carry_vec;  // May overflow!
```

```
// RIGHT: carry shifts left one (weight 2^(i+1)), and the sum of three
// 8-bit operands needs 10 bits, not 9 (3*255 = 765).
logic [9:0] result;
assign result = {2'b0, sum_vec} + {carry_vec, 1'b0};
```

4.8 Related Modules

- **math_adder_full.sv** - Identical logic, used in ripple chains
- **math_multiplier_wallace_tree_*.sv** - Uses CSA for partial product reduction
- **math_multiplier_dadda_tree_*.sv** - Uses CSA with optimized schedule
- **math_multiplier_carry_save.sv** - Multiplier variant using CSA stages

4.9 Testing

Covered by 2 test suites:

- `val/math/test_math_adder_carry_save.py`
- `val/math/test_math_adder_carry_save_nbit.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

4.10 References

- Wallace, C.S. “A Suggestion for a Fast Multiplier.” IEEE Transactions on Electronic Computers, 1964.
- Dadda, L. “Some Schemes for Parallel Multipliers.” Alta Frequenza, 1965.
- Deschamps, J.P., Bioul, G., Sutter, G. “Synthesis of Arithmetic Circuits.” Wiley, 2006.
- Hennessy, J.L., Patterson, D.A. “Computer Architecture: A Quantitative Approach.” Morgan Kaufmann, 2011.

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5 Full Adder

`math_adder_full` is the fundamental building block of binary arithmetic: a single-bit full adder that adds two input bits plus a carry-in and produces a sum and carry-out.

5.1 Overview

The `math_adder_full` module adds three single-bit inputs (two operands and a carry-in) and produces a sum bit and carry-out bit. This is the basis for multi-bit adders and more complex arithmetic units — if you understand this module cold, everything else in the adder family is structure, not new logic.

5.1.1 Module Declaration

```
module math_adder_full #(parameter int N=1) (
    input  logic i_a,
    input  logic i_b,
    input  logic i_c,
    output logic ow_sum,
    output logic ow_carry
);
```

5.2 Parameters

Parameter	Type	Default	Description
N	int	1	Parameter for potential future extensions (currently unused)

5.3 Ports

Port	Direction	Width	Description
i_a	Input	1	First input operand bit
i_b	Input	1	Second input operand bit
i_c	Input	1	Carry input bit
ow_sum	Output	1	Sum output bit (i_a ⊕ i_b ⊕ i_c)
ow_carry	Output	1	Carry output bit

5.4 Functional Description

5.4.1 Full Adder Logic

The full adder implements the following Boolean functions:

- **Sum Output**: `ow_sum = i_a ⊕ i_b ⊕ i_c`
- **Carry Output**: `ow_carry = (i_a & i_b) | (i_c & (i_a ⊕ i_b))`

5.4.2 Truth Table

i_a	i_b	i_c	ow_sum	ow_carry	Decimal
0	0	0	0	0	0 + 0 + 0 = 0
0	0	1	1	0	0 + 0 + 1 = 1
0	1	0	1	0	0 + 1 + 0 = 1
0	1	1	0	1	0 + 1 + 1 = 2
1	0	0	1	0	1 + 0 + 0 = 1
1	0	1	0	1	1 + 0 + 1 = 2
1	1	0	0	1	1 + 1 + 0 = 2
1	1	1	1	1	1 + 1 + 1 = 3

5.4.3 Implementation Details

Sum Generation

The sum output uses a three-input XOR gate:

```
assign ow_sum = i_a ^ i_b ^ i_c;
```

This produces 1 when an odd number of inputs are 1, and 0 when an even number are 1.

Carry Generation

The carry output uses optimized logic:

```
assign ow_carry = (i_a & i_b) | (i_c & (i_a ^ i_b));
```

This can be broken down as: - `(i_a & i_b)`: Carry generated when both primary inputs are 1 - `(i_c & (i_a ^ i_b))`: Carry propagated when exactly one primary input is 1 and carry-in is 1

5.4.4 Logic Gate Implementation

Traditional Gate-Level View

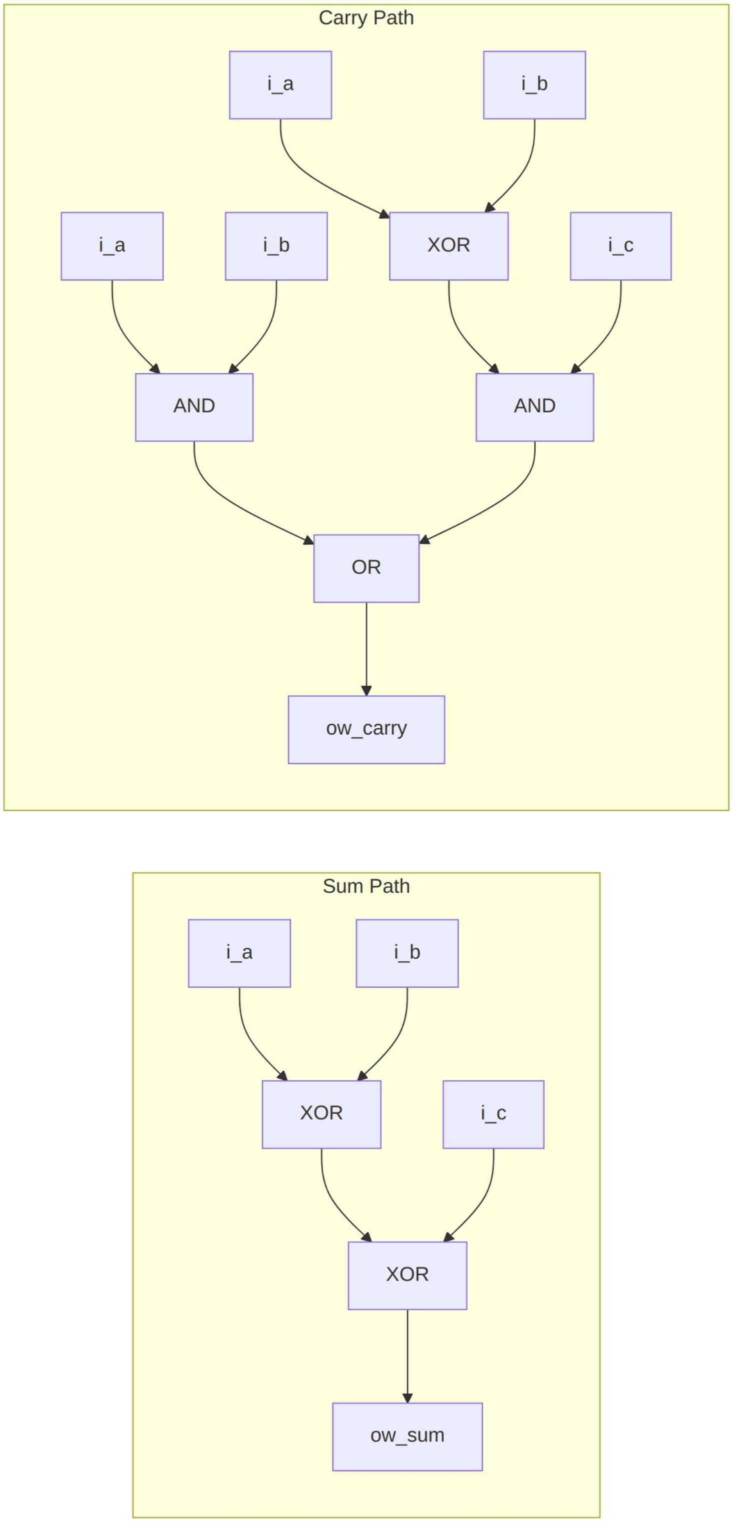
Carry Path
i_a
i_b
i_a
i_b
XOR
i_c
AND
AND
OR
ow_carry
Sum Path
i_a
i_b
XOR
i_c
XOR
ow_sum

Diagram 1

Optimized Implementation

The actual implementation uses shared XOR logic for efficiency:

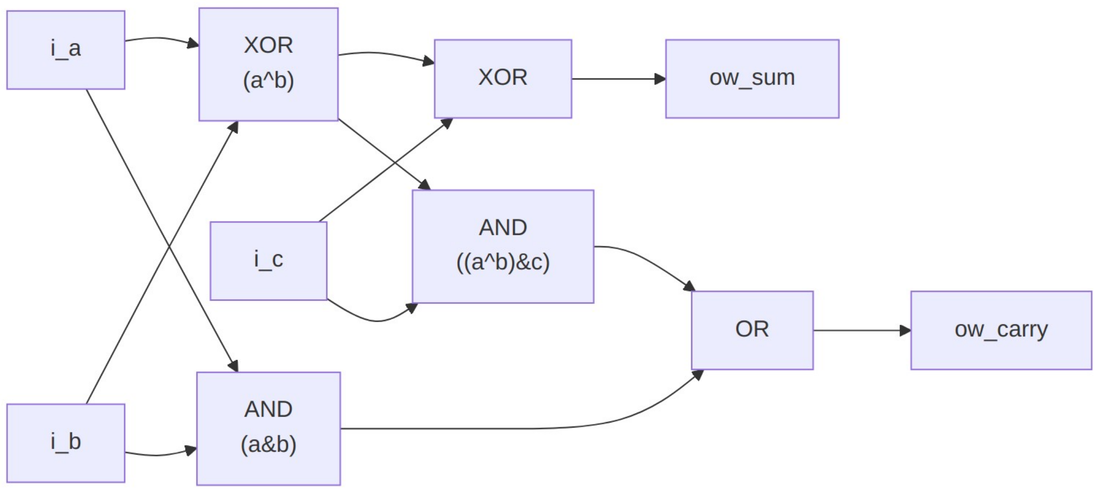

Diagram 2

Key optimization: The `a^b` XOR result is shared between sum calculation and carry propagation. One gate, two jobs.

5.5 Timing Characteristics

Characteristic	Typical Value	Description
Propagation Delay (Sum)	2 × t_XOR	Through 2 XOR gates
Propagation Delay (Carry)	t_AND + t_OR	Through AND-OR path
Setup Time	0	Purely combinational
Hold Time	0	Purely combinational

5.6 Usage Examples

5.6.1 Basic Full Adder

```
logic a_bit, b_bit, cin;
logic sum_bit, cout;

math_adder_full u_full_adder (
```

```
    .i_a      (a_bit),
    .i_b      (b_bit),
    .i_c      (cin),
    .ow_sum   (sum_bit),
    .ow_carry (cout)
);
```

5.6.2 Building a 4-Bit Ripple Carry Adder

```
logic [3:0] a, b, sum;
logic cin, cout;
logic [3:0] carry_chain;

// Bit 0
math_adder_full u_add0 (
    .i_a      (a[0]),
    .i_b      (b[0]),
    .i_c      (cin),
    .ow_sum   (sum[0]),
    .ow_carry (carry_chain[0])
);

// Bit 1
math_adder_full u_add1 (
    .i_a      (a[1]),
    .i_b      (b[1]),
    .i_c      (carry_chain[0]),
    .ow_sum   (sum[1]),
    .ow_carry (carry_chain[1])
);

// Bit 2
math_adder_full u_add2 (
    .i_a      (a[2]),
    .i_b      (b[2]),
    .i_c      (carry_chain[1]),
    .ow_sum   (sum[2]),
    .ow_carry (carry_chain[2])
);

// Bit 3
math_adder_full u_add3 (
    .i_a      (a[3]),
    .i_b      (b[3]),
    .i_c      (carry_chain[2]),
    .ow_sum   (sum[3]),
    .ow_carry (cout)
);
```

5.6.3 Part of a Carry-Save Adder

```
// In a 3:2 carry-save adder stage
math_adder_full u_csa_stage (
    .i_a      (partial_sum1[i]),
    .i_b      (partial_sum2[i]),
    .i_c      (partial_sum3[i]),
    .ow_sum   (sum_vector[i]),
    .ow_carry (carry_vector[i+1])
);
```

5.6.4 Testbench

```
module tb_math_adder_full;
    logic i_a, i_b, i_c;
    logic ow_sum, ow_carry;
    logic [1:0] expected_result;

    math_adder_full dut (.*);

    initial begin
        // Test all combinations
        for (int i = 0; i < 8; i++) begin
            {i_a, i_b, i_c} = i;
            #1;
            expected_result = i_a + i_b + i_c;

            assert ({ow_carry, ow_sum} == expected_result)
            else $error("Mismatch: %b + %b + %b = %b, expected %b",
                           i_a, i_b, i_c, {ow_carry, ow_sum},
expected_result);
        end
        $display("All tests passed!");
    end
endmodule
```

5.7 Design Notes

5.7.1 Advantages

- **Simplicity**: Minimal gate count and complexity
- **Modularity**: Perfect building block for larger arithmetic units
- **Predictable**: Well-defined timing and behavior
- **Efficient**: Optimized carry generation logic

5.7.2 Synthesis Considerations

Technology Mapping

Most synthesis tools will: - Map XOR gates to efficient library cells - Optimize the carry logic for the target technology - May use dedicated adder primitives in some technologies

Optimization Notes

```
// Alternative carry implementation (equivalent but different
structure)
assign ow_carry = (i_a & i_b) | (i_a & i_c) | (i_b & i_c);
```

This alternative has higher gate count but may have different timing characteristics.

5.7.3 Performance

- **Area**: 5 logic gates (2 XOR, 2 AND, 1 OR)
- **Power**: Low static power, dynamic power proportional to switching activity
- **Speed**: Limited by XOR gate delays (typically slower than AND/OR)

5.7.4 Applications

- **Multi-bit Adders**: Building block for ripple carry adders
- **Carry-Save Adders**: Used in parallel multiplication
- **ALU Design**: Fundamental component in arithmetic logic units
- **Accumulator Circuits**: Used in digital signal processing

Bottom line: `math_adder_full` provides the essential functionality for binary addition, and it earns its place as the critical building block of digital arithmetic circuits.

5.8 Related Modules

- `math_adder_half`: Half adder (2 inputs, no carry-in)
- `math_adder_full_nbit`: N-bit full adder using ripple carry
- `math_adder_ripple_carry`: Multi-bit ripple carry adder
- `math_adder_carry_save`: Carry-save adder for multiple operand addition

5.9 Testing

Covered by 2 test suites:

- `val/math/test_math_adder_full.py`
- `val/math/test_math_adder_full_nbit.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

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6 Han-Carlson Parallel Prefix Adder

A hybrid parallel prefix adder that combines the low wiring complexity of Brent-Kung with near-Kogge-Stone speed — a well-chosen balance of area, wire routing, and logic depth for advanced process nodes.

6.1 Overview

The `math_adder_han_carlson` module family implements Han-Carlson parallel prefix adders. Han-Carlson is a “sparsity-2” architecture: it computes prefix operations only on even bit positions in the intermediate stages, then fills in the odd positions in a final stage using area-efficient gray cells.

Available widths: 16, 22, 32, 44, 48 and 72 bits. All six are auto-generated by `bin/rtl_generators/ieee754/generate_all.py`; each width exists because a specific floating-point datapath needs it.

Width	Generated for	Measured usage (rtl/ instantiation grep, 2026-07-30)
16	final CPA of `math_multiplier_dadda_4to2_008`	exactly that
22	FP16 product CPA (11 x 11 = 22); final CPA of `math_multiplier_dadda_4to2_011`	exactly that
32	FP32 exponent paths	UNUSED: `math_ieee754_2008_fp32_adder` adds exponents behaviorally (assign, no prefix instance)
44	FP16 FMA accumulator	UNUSED: `math_ieee754_2008_fp16_fma` accumulates behaviorally

Width	Generated for	Measured usage (rtl/ instantiation grep, 2026-07-30)
48	BF16 FMA wide addition; final CPA of `math_multiplier_dadda_4to2_024`	those two (no FP32-product-CPA user exists)
72	FP32 FMA accumulator (`math_ieee754_2008_fp32_fma`)	exactly that

The exponent paths of the ieee754 adders are behavioral (`assign`), so the earlier "16: FP16 exponent paths / 32: FP32 exponent paths" entries were aspirational, not real — see math_library.md's (correct) Core-operators table. Widths on paper are not widths in silicon; the grep doesn't lie.

: Han-Carlson widths present in rtl/math and what uses each

Key Features: - **O(log N + 1) depth** - Only one level more than Kogge-Stone - **~35% fewer cells** than Kogge-Stone at 16 bits (32 vs 49) - **Constant fanout of 2** - Better for advanced process nodes - **Reduced wiring** - 4 tracks vs 8 for Kogge-Stone (16-bit) - **Optimal for** multiplier final CPA and FMA wide addition

6.1.1 Module Hierarchy

Top-Level Modules: - `math_adder_han_carlson_016.sv` - 16-bit adder - `math_adder_han_carlson_022.sv` - 22-bit adder - `math_adder_han_carlson_032.sv` - 32-bit adder - `math_adder_han_carlson_044.sv` - 44-bit adder - `math_adder_han_carlson_048.sv` - 48-bit adder - `math_adder_han_carlson_072.sv` - 72-bit adder

Building Blocks: - `math_prefix_cell.sv` - Black cell (outputs G and P) - `math_prefix_cell_gray.sv` - Gray cell (outputs G only)

6.1.2 Module Declaration

16-bit Han-Carlson Adder

```
module math_adder_han_carlson_016 #(
    parameter int N = 16
) (
    input  logic [N-1:0] i_a,
    input  logic [N-1:0] i_b,
    input  logic         i_cin,
    output logic [N-1:0] ow_sum,
    output logic         ow_cout
);
```

48-bit Han-Carlson Adder

```
module math_adder_han_carlson_048 #(
    parameter int N = 48
) (
    input  logic [N-1:0] i_a,
    input  logic [N-1:0] i_b,
    input  logic         i_cin,
    output logic [N-1:0] ow_sum,
    output logic         ow_cout
);
```

6.2 Parameters

Parameter	Type	Default	Description
N	int	matches the variant	Adder width (fixed per variant)

Note: `N` defaults to the width in the module name (`_016` defaults to 16, `_072` to 72) and must not be overridden — the prefix network is generated for that exact width. Pick the variant that matches your width instead.

6.3 Ports

Port	Direction	Width	Description
i_a	Input	N	First operand
i_b	Input	N	Second operand
i_cin	Input	1	Carry input
ow_sum	Output	N	Sum result (A + B + Cin)
ow_cout	Output	1	Carry output

6.4 Functional Description

6.4.1 Han-Carlson Structure

Han-Carlson uses a “sparsity-2” approach:

1. **Stage 0:** Compute initial P and G for all positions
2. **Stage 1:** Combine adjacent pairs on even positions only
3. **Stages 2-N:** Kogge-Stone style prefix on even positions (doubling span)

4. **Final Stage:** Fill odd positions from even neighbors using gray cells

6.4.2 Stage-by-Stage Breakdown (16-bit)

```
Stage 0: Initial P/G computation
  P[i] = A[i] XOR B[i]
  G[i] = A[i] AND B[i]

Stage 1: Span-2 prefix on even positions
  Position 0: G[0:-1] (incorporate cin)
  Even positions: G[i:i-1] = combine(G[i], P[i], G[i-1], P[i-1])
  Odd positions: pass through

Stage 2: Span-4 prefix on even positions (step 2)
  Even positions >= 2: G[i:i-3] = combine(G[i:i-1], G[i-2:i-3])
  Others: pass through

Stage 3: Span-8 prefix on even positions (step 4)
  Even positions >= 4: G[i:i-7] = combine(G[i:i-3], G[i-4:i-7])
  Others: pass through

Stage 4: Span-16 prefix on even positions (step 8)
  Even positions >= 8: G[i:i-15] = combine(...)
  Others: pass through

Stage 5: Fill odd positions (gray cells)
  Odd positions: G[i:-1] = combine(G[i], P[i], G[i-1:-1])
  Even positions: pass through

Sum computation:
  Sum[0] = P[0] XOR Cin
  Sum[i] = P[i] XOR G[i-1:-1]
```

6.4.3 Prefix Network Visualization (16-bit simplified)

```
Position:   0   1   2   3   4   5   6   7   8   9  10  11  12  13  14
15

Stage 0:    o   o   o   o   o   o   o   o   o   o   o   o   o   o   o
o
            Initial P/G generation

Stage 1:    *   |   *   |   *   |   *   |   *   |   *   |   *   |   *
|
            Even positions combine with odd neighbor (*)
            Odd positions pass through (|)
```

```
Stage 2:    *   |   |   |   *   |   |   |   *   |   |   |   *   |   |
|
            Even positions combine with step=2 (positions
2,4,6,8,10,12,14)

Stage 3:    *   |   |   |   |   |   |   |   *   |   |   |   |   |   |
|
            Even positions >= 4 combine (cells at 4,6,8,10,12,14)

Stage 4:    *   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|
            Even positions >= 8 combine (cells at 8,10,12,14)

Stage 5:    o   *   o   *   o   *   o   *   o   *   o   *   o   *   o
*
            Fill odd positions with gray cells (*)

Legend: o=pass through, |=pass through, *=prefix cell
```

6.4.4 Why Han-Carlson is Optimal

The Brent-Kung and Han-Carlson columns are counted from this library's RTL; the Kogge-Stone column is a textbook reference, as no Kogge-Stone adder is implemented here.

16-bit prefix adder comparison

Metric	Kogge-Stone	Brent-Kung	Han-Carlson
Prefix Depth (16-bit)	4	6	5
Prefix Cells (16-bit)	49	27	32
Wiring Tracks	8	2	4
Fanout	Varies	2	2
Best For	Max speed	Min area	Balanced

Han-Carlson achieves: - 1 level slower than Kogge-Stone (5 vs 4) - ~35% fewer cells than Kogge-Stone (32 vs 49) - 2x fewer wiring tracks than Kogge-Stone (4 vs 8) - Constant fanout of 2 (important for advanced nodes)

One honest caveat: Brent-Kung is both smaller *and*, in this library, only one level deeper than Han-Carlson at N=16. Han-Carlson's advantage is in wiring tracks and fanout regularity, and it widens as N grows — not in cell count at this width.

6.5 Timing Characteristics

6.5.1 16-bit Adder

Metric	Value
Logic Levels	5 prefix stages + sum XOR
Prefix Cells	32 (24 black + 8 gray)
Critical Path	cin -> Stage 1 -> Stage 5 -> sum[15]

6.5.2 48-bit Adder

Metric	Value
Logic Levels	7 prefix stages + sum XOR
Prefix Cells	136 (112 black + 24 gray)
Critical Path	cin -> Stage 1 -> Stage 7 -> sum[47]

6.5.3 Depth Formula

For N-bit Han-Carlson: - **Prefix-tree depth** = ceil(log2(N)) + 1 stages - **Total depth** = prefix tree + 1 sum stage = ceil(log2(N)) + 2 stages - **16-bit:** prefix = 4 + 1 = 5, total = 6 stages (5 prefix + 1 sum) - **48-bit:** prefix = 6 + 1 = 7, total = 8 stages (7 prefix + 1 sum)

The +2 is **total** depth, not prefix depth — the sparsity-2 Han-Carlson prefix network is `log2(N) + 1` deep, and the final sum stage adds one more. Quoting +2 without saying which is being counted invites an off-by-one when comparing against the Harris taxonomy, which tabulates prefix depth. Say which number you're quoting.

6.6 Usage Examples

6.6.1 Basic 16-bit Addition

```
logic [15:0] a, b, sum;
logic cout;

math_adder_han_carlson_016 u_add (
    .i_a(a),
    .i_b(b),
    .i_cin(1'b0),
    .ow_sum(sum),
    .ow_cout(cout)
);
```

6.6.2 Subtraction (A - B)

```
logic [15:0] a, b, diff;
logic borrow;

// A - B = A + (~B) + 1
math_adder_han_carlson_016 u_sub (
    .i_a(a),
    .i_b(~b),
    .i_cin(1'b1),  // +1 for two's complement
    .ow_sum(diff),
    .ow_cout(borrow)
);
```

6.6.3 In BF16 Multiplier (Final CPA)

```
// Final carry-propagate addition after Dadda reduction
wire [15:0] w_cpa_a, w_cpa_b;  // From Dadda tree
wire [15:0] w_product;
wire        w_cout;

math_adder_han_carlson_016 u_final_cpa (
    .i_a(w_cpa_a),
    .i_b(w_cpa_b),
    .i_cin(1'b0),
    .ow_sum(w_product),
    .ow_cout(w_cout)  // Unused for unsigned multiply
);
```

6.6.4 In BF16 FMA (Wide Addition)

```
// 48-bit mantissa addition in FMA
wire [47:0] w_mant_larger, w_mant_smaller_aligned;
wire [47:0] w_sum;
wire        w_cout;

// For subtraction, invert smaller operand
wire [47:0] w_adder_b = w_effective_sub ? ~w_mant_smaller_aligned
                                        : w_mant_smaller_aligned;

math_adder_han_carlson_048 u_wide_add (
    .i_a(w_mant_larger),
    .i_b(w_adder_b),
    .i_cin(w_effective_sub),  // +1 for two's complement
    .ow_sum(w_sum),
    .ow_cout(w_cout)
);
```

6.7 Design Notes

6.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Fewer cells than Kogge-Stone via sparsity-2 structure 2. **Wire complexity** - Reduced wiring tracks and constant fanout 3. **Logic depth** - Near-optimal O(log N + 1) depth

6.7.2 When to Use Han-Carlson

Appropriate Use Cases: - Multiplier final CPA (balanced speed/area) - FMA wide addition (48+ bits) - Designs targeting advanced process nodes where wire delay matters - Area-constrained high-speed arithmetic

Consider Alternatives When: - Absolute minimum depth needed -> Kogge-Stone - Absolute minimum area needed -> Brent-Kung or Ripple Carry - Width not power-of-2 -> Consider behavioral or Ripple Carry

6.7.3 Auto-Generated Code

These modules are auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/ieee754/han_carlson_adder.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/ieee754/generate_all.py rtl/math`

This matches the `// Regenerate:` banner of the generated files, and it is the one that emits all six widths. A second, older generator still exists at `bin/rtl_generators/bf16/`, but it emits only the 16- and 48-bit variants; running it will not reproduce the current file set.

Do not edit the generated .sv files manually. Modify the generator script instead.

6.7.4 Performance

Cell counts are exact, obtained by elaborating the generate stages of each file. Gray cells always number N/2 (the final fill-in stage). LUT figures are estimates derived from the cell counts, not synthesis results.

Han-Carlson prefix-cell counts, elaborated from the generated RTL

Width	Prefix Stages	Black Cells	Gray Cells	Total Cells	LUTs (Est.)
16-bit	5	24	8	32	~90
22-bit	6	39	11	50	~140
32-bit	6	64	16	80	~225
44-bit	7	100	22	122	~345
48-bit	7	112	24	136	~385
72-bit	8	188	36	224	~630

Comparison with Other Architectures:

16-bit architecture comparison, prefix depth and cell count

Architecture	16-bit Depth	16-bit Cells	Wiring Complexity
Ripple Carry	16	16	Minimal
Brent-Kung	6	27	Low
Han-Carlson	**5**	**32**	**Medium**
Kogge-Stone (not implemented here)	4	49	High

6.7.5 Common Pitfalls

Anti-Pattern 1: Expecting parameterizable width

```
// WRONG: N is fixed per variant
math_adder_han_carlson_016 #(.N(24)) u_add (...);  // Won't work!

// RIGHT: Use appropriate variant or extend inputs
math_adder_han_carlson_048 u_add (
    .i_a({24'b0, a[23:0]}),  // Zero-extend
    .i_b({24'b0, b[23:0]}),
    ...
);
```

Anti-Pattern 2: Ignoring that outputs are combinational

```
// Remember: outputs are purely combinational
// Add pipeline registers if needed for timing closure
always_ff @(posedge clk) begin
    sum_reg <= ow_sum;  // Register the output
end
```

6.8 Related Modules

- **math_prefix_cell** - Black cell building block
- **math_prefix_cell_gray** - Gray cell building block
- **math_multiplier_dadda_4to2_008/011/024** - Use the 16/22/48-bit Han-Carlson as final CPA
- **math_bf16_fma** - Uses 48-bit Han-Carlson for wide addition
- **math_ieee754_2008_fp32_fma** - Uses 72-bit Han-Carlson for wide addition

- **math_adder_brent_kung_008/016/032/064** - Alternative area-optimized prefix adders

6.9 Testing

From the test suite (`val/math/test_math_adder_han_carlson.py`):

- **Key test scenarios**:
  - Structural coverage follows the MEASURED usage table above (the authoritative list): 16-bit lives only inside `math_multiplier_dadda_4to2_008`'s final CPA; 48-bit serves the BF16 FMA wide addition and `dadda_4to2_024`'s CPA; 72-bit the FP32 FMA accumulator

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

6.10 References

- Han, T., Carlson, D.A. "Fast Area-Efficient VLSI Adders." IEEE Symposium on Computer Arithmetic, 1987.
- Harris, D. "A Taxonomy of Parallel Prefix Networks." Asilomar Conference, 2003.
- Knowles, S. "A Family of Adders." IEEE Symposium on Computer Arithmetic, 1999.

6.11 Navigation

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7 Carry Lookahead Adder

> **Naming caveat – read this before selecting this module.** This is **not** a true Weinberger & Smith carry-lookahead adder. A real CLA uses recursive group generate/propagate to reach **O(log N)** depth; this implementation is a simplified P/G carry chain optimised for area and simplicity, with **O(N)** depth. For genuine logarithmic-depth adders use `math_adder_brent_kung` or `math_adder_han_carlson`.

A parameterized carry lookahead adder supporting arbitrary bit widths with O(N) depth, using generate and propagate logic to cut carry propagation delay compared to ripple carry adders.

7.1 Overview

The `math_adder_pg_chain` module implements a carry lookahead adder (CLA) with parameterizable width. It's not a true parallel-prefix lookahead (which would have O(log N) depth), but it uses propagate (P) and generate (G) signals to compute carries more efficiently than a simple ripple-carry adder. Think of it as a good middle shelf: a sensible balance between speed and area for small to medium width additions (4-16 bits).

7.1.1 Module Declaration

```
module math_adder_pg_chain #(
    parameter int N = 4      // Adder width in bits (any width ≥ 1)
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic         i_c,       // Carry input
    output logic [N-1:0] ow_sum,    // Sum output
    output logic         ow_carry   // Carry output
);
```

7.2 Parameters

Parameter	Type	Default	Description
N	int	4	Adder width in bits (range: 1-64, typical: 4-16)

Width Guidelines: - **Typical**: 4, 8, 16 bits (good speed/area balance) - **Minimum**: 1 bit (degenerates to a single full adder — it has a carry input, which a half adder lacks) - **Maximum**: 64 bits (practical synthesis limit, but consider parallel prefix for >16 bits)

7.3 Ports

Port	Direction	Width	Description
i_a	Input	N	Operand A (addend)
i_b	Input	N	Operand B (addend)
i_c	Input	1	Carry input (0 for addition, 1

Port	Direction	Width	Description
			for +1 increment)
ow_sum	Output	N	Sum output (A + B + Cin)
ow_carry	Output	1	Carry output (overflow)

7.4 Functional Description

7.4.1 Algorithm Overview

The carry lookahead adder uses **propagate (P)** and **generate (G)** signals to compute carries more efficiently than ripple carry:

7.4.1.1 Bit-Level Definitions

For each bit position i:

```
Generate:  G[i] = A[i] & B[i]   (carry is generated at this position)
Propagate: P[i] = A[i] ^ B[i]   (carry is propagated through this
position)
```

7.4.1.2 Carry Calculation

Initial carry:

```
C[0] = Cin  (input carry)
```

Carry recurrence (for i = 1 to N):

```
C[i] = G[i-1] | (P[i-1] & C[i-1])
```

Interpretation: - Carry is generated (G[i-1] = 1) at position i-1, OR - Carry is propagated (P[i-1] = 1) from position i-1 if there was an incoming carry (C[i-1] = 1)

7.4.1.3 Sum Calculation

```
Sum[i] = P[i] ^ C[i]
```

Sum is XOR of propagate signal and incoming carry.

7.4.2 Implementation

```
// Stage 1: Generate P and G signals (parallel, 1 level of logic)
for (i = 0; i < N; i++) begin
    w_g[i] = i_a[i] & i_b[i];  // Generate
    w_p[i] = i_a[i] ^ i_b[i];  // Propagate
```

```
end

// Stage 2: Calculate carries (sequential dependency, N levels)
w_c[0] = i_c;
for (i = 1; i <= N; i++) begin
    w_c[i] = w_g[i-1] | (w_p[i-1] & w_c[i-1]);
end

// Stage 3: Calculate sum (parallel, 1 level of logic)
for (i = 0; i < N; i++) begin
    ow_sum[i] = w_p[i] ^ w_c[i];
end

// Final carry out
ow_carry = w_c[N];
```

7.5 Timing Characteristics

7.5.1 Timing Analysis

Logic Depth: - **P/G generation**: 1 level (AND/XOR) - **Carry chain**: N levels (each carry depends on previous) - **Sum calculation**: 1 level (XOR) - **Total**: N + 2 levels

Critical Path:

```
i_a/i_b → P[0]/G[0] → C[1] → C[2] → ... → C[N-1] → C[N] → Sum[N-1]
```

7.5.2 Combinational Delay Analysis

Width	Logic Levels	Typical Delay (ns) @ 1.0V	Max Frequency
4-bit	6	~1.2	~800 MHz
8-bit	10	~2.0	~500 MHz
16-bit	18	~3.5	~285 MHz
32-bit	34	~6.5	~155 MHz

Logic Level Breakdown (8-bit example): 1. P/G generation: 1 level (AND/XOR) 2. Carry chain: 8 levels (C[1] → C[2] → … → C[8]) 3. Sum calculation: 1 level (XOR) 4. **Total**: 10 levels

7.5.3 Critical Paths

Main Critical Path (Longest):

```
i_a[0]/i_b[0] → P[0]/G[0] → C[1] → C[2] → ... → C[N] → ow_carry
```

Sum Output Path:

```
i_a[i]/i_b[i] → P[i] → ow_sum[i]  (depends on C[i] from carry chain)
```

7.5.4 Performance vs Width

Width	Delay (relative)	Frequency (relative)
4-bit	1.0× (baseline)	1.00× (baseline)
8-bit	1.7×	0.60×
16-bit	3.0×	0.35×
32-bit	5.7×	0.19×

Observation: Delay scales linearly with width. For N > 16, that's your cue to reach for a parallel prefix adder (Brent-Kung, Kogge-Stone).

7.6 Usage Examples

7.6.1 Basic 8-bit Addition

```
logic [7:0] a, b, sum;
logic carry_out;

math_adder_pg_chain #(
    .N(8)
) u_adder (
    .i_a      (a),
    .i_b      (b),
    .i_c      (1'b0),        // No carry input
    .ow_sum   (sum),
    .ow_carry (carry_out)
);

// Example: 200 + 100
initial begin
    a = 8'd200;
    b = 8'd100;
    #1;  // Wait for combinational propagation
    assert(sum == 8'd44);       // 300 % 256 = 44
    assert(carry_out == 1'b1);  // Overflow
end
```

7.6.2 4-bit Incrementer

```
logic [3:0] count, count_plus_1;

math_adder_pg_chain #(
    .N(4)
) u_incrementer (
```

```
    .i_a      (count),
    .i_b      (4'b0),        // B = 0
    .i_c      (1'b1),        // Carry in = 1 → +1
    .ow_sum   (count_plus_1),
    .ow_carry ()             // Unused
);
```

7.6.3 16-bit ALU Component

```
typedef enum logic [2:0] {
    ALU_ADD  = 3'b000,
    ALU_SUB  = 3'b001,
    ALU_INC  = 3'b010,
    ALU_DEC  = 3'b011
} alu_op_t;

module simple_alu (
    input  logic [15:0]  a,
    input  logic [15:0]  b,
    input  alu_op_t      op,
    output logic [15:0]  result,
    output logic         carry,
    output logic         zero
);

    logic [15:0] b_mux;
    logic        cin_mux;

    // Operation selection
    always_comb begin
        case (op)
            ALU_ADD: begin
                b_mux = b;
                cin_mux = 1'b0;
            end
            ALU_SUB: begin
                b_mux = ~b;          // Two's complement: A + (~B) + 1
                cin_mux = 1'b1;
            end
            ALU_INC: begin
                b_mux = 16'b0;
                cin_mux = 1'b1;      // A + 0 + 1 = A + 1
            end
            ALU_DEC: begin
                b_mux = 16'hFFFF;    // -1 in two's complement
                cin_mux = 1'b0;      // A + (-1) + 0 = A - 1
            end
            default: begin
```

```
                b_mux = 16'b0;
                cin_mux = 1'b0;
            end
        endcase
    end

    // Adder
    math_adder_pg_chain #(
        .N(16)
    ) u_adder (
        .i_a      (a),
        .i_b      (b_mux),
        .i_c      (cin_mux),
        .ow_sum   (result),
        .ow_carry (carry)
    );

    // Zero flag
    assign zero = ~|result;

endmodule
```

7.6.4 Multi-Precision Addition (32-bit via 2×16-bit)

```
logic [15:0] a_low, a_high, b_low, b_high;
logic [15:0] sum_low, sum_high;
logic carry_low, carry_high;

// Low 16 bits
math_adder_pg_chain #(.N(16)) u_add_low (
    .i_a      (a_low),
    .i_b      (b_low),
    .i_c      (1'b0),
    .ow_sum   (sum_low),
    .ow_carry (carry_low)
);

// High 16 bits (chain carry from low)
math_adder_pg_chain #(.N(16)) u_add_high (
    .i_a      (a_high),
    .i_b      (b_high),
    .i_c      (carry_low),     // Carry from low adder
    .ow_sum   (sum_high),
    .ow_carry (carry_high)
);
```

```
// Concatenate results
logic [31:0] sum_32 = {sum_high, sum_low};
```

7.6.5 Conditional Increment (For Loop Counters)

```
logic [7:0] loop_counter;
logic loop_active, loop_done;

math_adder_pg_chain #(.N(8)) u_loop_inc (
    .i_a      (loop_counter),
    .i_b      (8'b0),
    .i_c      (loop_active),   // Increment only when active
    .ow_sum   (loop_counter_next),
    .ow_carry ()
);

always_ff @(posedge clk) begin
    if (rst_n == 0) begin
        loop_counter <= 8'b0;
    end else if (loop_active) begin
        loop_counter <= loop_counter_next;
    end
end

assign loop_done = (loop_counter == 8'd99);
```

7.7 Design Notes

7.7.1 When to Use This Adder

Appropriate Use Cases: - Small to medium width additions (4-16 bits) - Area-constrained designs with moderate speed requirements - ALU components in simple processors - Address arithmetic (incrementers, offsets) - Simple calculators or counters - FPGA designs with limited LUT budget

Consider Alternatives When: - **N > 16**: Use Brent-Kung or Kogge-Stone for better speed - **N ≤ 4**: Use ripple carry for minimal area - **Critical datapath**: Use parallel prefix adders (Brent-Kung/Kogge-Stone) - **Very high frequency**: Pipeline or use faster adder architecture

7.7.2 Width Selection Guidelines

Recommended Widths: - **4-bit**: Good for nibble operations, BCD arithmetic - **8-bit**: Standard byte operations, small ALU - **16-bit**: Address arithmetic, medium-width ALU - **32-bit**: Only if area is critical (prefer Brent-Kung)

7.7.3 Synthesis Considerations

Optimization Tips:

1. **Flatten hierarchy** for better carry chain optimization:

   ```
   set_flatten true -effort high
   ```

2. **Mark as combinational** to prevent retiming issues:

   ```
   set_dont_touch u_adder/w_c* false
   ```

3. **Critical path optimization**:

   ```
   set_max_delay -from [get_ports i_a] -to [get_ports ow_carry] 2.0
   ```

4. **Consider pipelining** for N > 16:

   ```
   // Pipeline example: Break carry chain in middle
   logic [N/2:0] r_c_mid;
   logic [N-1:0] r_sum_high;

   always_ff @(posedge clk) begin
       r_c_mid <= w_c[N/2];  // Register mid-point carry
   end
   ```

7.7.4 Verilator Lint Note

The module includes a Verilator pragma:

```
/* verilator lint_off UNOPTFLAT */
// ... carry chain logic ...
/* verilator lint_on UNOPTFLAT */
```

Reason: The carry chain creates combinational loops that Verilator warns about but are intentional for CLA operation.

7.7.5 Performance

Width	LUTs (Est.)	FFs (Pipeline)	Description
4-bit	~12	0 (combinational)	Minimal
8-bit	~24	0 (combinational)	Small
16-bit	~48	0 (combinational)	Medium
32-bit	~96	0	Large

Width	LUTs (Est.)	FFs (Pipeline)	Description
		(combinational)	(consider alternatives)

Comparison to Other Adders:

Adder Type	Logic Depth	Area	Best Use Case
Ripple Carry	O(N)	Minimal	N ≤ 4, area-critical
CLA (this)	**O(N)**	**Small**	**4 ≤ N ≤ 16, balanced**
Brent-Kung	O(log N)	Medium	N ≥ 16, speed-critical
Kogge-Stone	O(log N)	Large	N ≥ 32, max speed

Note: True carry lookahead adders (with group generate/propagate) can achieve O(log N) depth but require more complex logic. This implementation is a hybrid approach optimized for simplicity and area.

Speed vs Area Trade-offs:

Adder Architecture	Relative Speed	Relative Area	Best Width Range
Ripple Carry	1.0× (slowest)	1.0× (smallest)	N ≤ 4
CLA (this)	**1.5×**	**1.2×**	**4 ≤ N ≤ 16**
Brent-Kung	4.0×	4.0×	16 ≤ N ≤ 32
Kogge-Stone	6.0× (fastest)	6.0× (largest)	N ≥ 32

7.7.6 Common Pitfalls

Anti-Pattern 1: Using for very wide additions without pipelining

```
// WRONG: 64-bit addition with CLA (poor performance)
math_adder_pg_chain #(.N(64)) u_add (...);
// Result: 66 logic levels, very slow!

// RIGHT: Use parallel prefix adder
math_adder_brent_kung_032 u_add_low (...)  // Lower 32 bits
math_adder_brent_kung_032 u_add_high (...)  // Upper 32 bits
```

Anti-Pattern 2: Expecting registered outputs

```
// WRONG: Assuming outputs are registered
always_ff @(posedge clk) begin
    data <= ow_sum;  // ow_sum is combinational!
end

// RIGHT: Add explicit output register
logic [N-1:0] r_sum;
always_ff @(posedge clk) begin
    r_sum <= ow_sum;
end
```

Anti-Pattern 3: Chaining adders without pipelining

```
// WRONG: Long chain without pipeline
logic [15:0] a, b, c, d, result;
logic [15:0] ab_sum, abc_sum;

math_adder_pg_chain #(.N(16)) u1
(.i_a(a), .i_b(b), .ow_sum(ab_sum), ...);
math_adder_pg_chain #(.N(16)) u2
(.i_a(ab_sum), .i_b(c), .ow_sum(abc_sum), ...);
math_adder_pg_chain #(.N(16)) u3
(.i_a(abc_sum), .i_b(d), .ow_sum(result), ...);

// 3 × 18 levels = 54 logic levels!

// RIGHT: Pipeline the chain
always_ff @(posedge clk) begin
    r_ab_sum <= ab_sum;
    r_abc_sum <= abc_sum;
end
```

Anti-Pattern 4: Ignoring synthesis warnings

```
// Synthesis warning: "Combinational loop detected"
// This is expected for CLA carry chain, but verify:
// - Loop is intentional
// - No actual circular dependency
// - All signals eventually resolve
```

7.8 Related Modules

- **math_adder_brent_kung_*.sv** - Faster parallel prefix adder (O(log N) depth)
- **math_adder_ripple_carry.sv** - Minimal area adder (slower)
- **math_adder_carry_save_nbit.sv** - For multi-operand addition (multipliers)

- **math_subtractor_carry_lookahead.sv** - CLA-based subtractor variant

7.9 Testing

From the test suite (`val/math/test_math_adder_pg_chain.py`):

Key Test Scenarios: - Random stimulus (all widths) - Corner cases: 0+0, 0+MAX, MAX+MAX, MAX+1 - Carry propagation: All 1's + 1 (tests full carry chain) - Incrementer mode: A + 0 + 1 - Subtraction mode: A + (~B) + 1 - Multi-precision chaining

Test Command:

```
pytest val/math/test_math_adder_pg_chain.py -v
```

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

7.10 References

- Weinberger, A., Smith, J.L. "A Logic for High-Speed Addition." National Bureau of Standards, 1958.
- Sklanski, J. "Conditional-Sum Addition Logic." IRE Transactions on Electronic Computers, 1960.
- Deschamps, J.P., Bioul, G., Sutter, G. "Synthesis of Arithmetic Circuits: FPGA, ASIC and Embedded Systems." Wiley, 2006.

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8 Ripple Carry Adder

A parameterized N-bit ripple carry adder built from a chain of full adders — minimal area at the cost of O(N) propagation delay.

8.1 Overview

The `math_adder_ripple_carry` module implements the classic ripple carry adder, where the carry propagates sequentially from LSB to MSB through a chain of full adders. That gives you O(N) delay — the slowest of the adder architectures — but the smallest area and the simplest implementation. For low-speed, area-critical work, that's often the right trade.

8.1.1 Module Declaration

```
module math_adder_ripple_carry #(
    parameter int N = 4      // Adder width in bits (any width ≥ 1)
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic         i_c,       // Carry input
    output logic [N-1:0] ow_sum,    // Sum output
    output logic         ow_carry   // Carry output
);
```

Related Module: `math_adder_full_nbit` provides functionally identical behavior with slightly different internal structure (uniform generate loop vs explicit first full adder).

8.2 Parameters

Parameter	Type	Default	Description
N	int	4	Adder width in bits (range: 1-64, typical: 2-8)

Width Guidelines: - **Optimal**: 2-8 bits (best area/speed trade-off for this architecture) - **Minimum**: 1 bit (degenerates to single full adder) - **Maximum**: 64 bits (practical limit, but very slow) - **Above 16 bits**: Consider carry lookahead or parallel prefix adders

8.3 Ports

Port	Direction	Width	Description
i_a	Input	N	Operand A (addend)
i_b	Input	N	Operand B (addend)
i_c	Input	1	Carry input (0 for addition, 1 for +1 increment)
ow_sum	Output	N	Sum output (A + B + Cin)
ow_carry	Output	1	Carry output (final carry from MSB)

8.4 Functional Description

8.4.1 Algorithm: Sequential Carry Propagation

The ripple carry adder computes addition bit-by-bit from LSB to MSB, with each bit's carry output feeding the next bit's carry input:

```
Bit 0 (LSB):  Sum[0], C[0] = FullAdder(A[0], B[0], Cin)
Bit 1:        Sum[1], C[1] = FullAdder(A[1], B[1], C[0])
Bit 2:        Sum[2], C[2] = FullAdder(A[2], B[2], C[1])
...
Bit N-1 (MSB): Sum[N-1], Cout = FullAdder(A[N-1], B[N-1], C[N-2])
```

8.4.2 Full Adder Building Block

Each bit position uses a full adder with three inputs:

```
// Full adder logic (implemented in math_adder_full.sv)
assign sum   = a ^ b ^ c_in;                    // XOR for sum
assign c_out = (a & b) | (c_in & (a ^ b));     // Majority function for carry
```

Truth Table: | A | B | Cin | Sum | Cout | |—|—|—–|—–|——| | 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 1 | 1 | 0 | | 0 | 1 | 0 | 1 | 0 | | 0 | 1 | 1 | 0 | 1 | | 1 | 0 | 0 | 1 | 0 | | 1 | 0 | 1 | 0 | 1 | | 1 | 1 | 0 | 0 | 1 | | 1 | 1 | 1 | 1 | 1 |

8.4.3 Implementation

```
// Intermediate carries between full adders
logic [N-1:0] w_c;

// First full adder (LSB) - uses input carry i_c
math_adder_full fa0 (
    .i_a(i_a[0]),
    .i_b(i_b[0]),
    .i_c(i_c),
    .ow_sum(ow_sum[0]),
    .ow_carry(w_c[0])
);

// Remaining full adders - chain carry from previous stage
genvar i;
generate
    for (i = 1; i < N; i++) begin : gen_full_adders
        math_adder_full fa (
            .i_a(i_a[i]),
            .i_b(i_b[i]),
            .i_c(w_c[i-1]),        // Carry from previous bit
            .ow_sum(ow_sum[i]),
```

```
            .ow_carry(w_c[i])
        );
    end
endgenerate

// Final carry out
assign ow_carry = w_c[N-1];
```

8.5 Timing Characteristics

8.5.1 Timing Analysis

Logic Depth: - **Per bit**: 2 levels (XOR for sum + Carry logic) - **Carry chain**: N × 2 levels (sequential dependency) - **Total**: 2N levels

Critical Path (Carry Propagation):

```
i_c → FA[0].Cout → FA[1].Cout → ... → FA[N-1].Cout → ow_carry
```

Why It's Slow: The critical path traverses all N full adders sequentially. Each full adder adds 2 gate delays, so total delay grows linearly with N. No way around that in this topology.

8.5.2 Combinational Delay Analysis

Width	Logic Levels	Typical Delay (ns) @ 1.0V	Max Frequency
2-bit	4	~0.8	~1250 MHz
4-bit	8	~1.5	~650 MHz
8-bit	16	~3.0	~330 MHz
16-bit	32	~6.0	~165 MHz
32-bit	64	~12.0	~83 MHz

Logic Level Breakdown (8-bit example): - Carry chain: 8 full adders × 2 levels each = 16 levels - Sum outputs: Available after carry arrives at each position

Observation: Delay doubles when width doubles (linear scaling).

8.5.3 Critical Paths

Longest Path (Carry Out):

```
i_c → FA[0] → FA[1] → FA[2] → ... → FA[N-1] → ow_carry
2N gate delays
```

Sum Output Paths (variable length):

```
LSB (Sum[0]): i_a[0]/i_b[0] → 2 levels (fastest)
Mid (Sum[4]): Depends on C[3] → 10 levels (8-bit example)
MSB (Sum[7]): Depends on C[6] → 16 levels (slowest)
```

8.5.4 Performance vs Width

Width	Relative Delay	Relative Frequency
4-bit	1.0×	1.00×
8-bit	2.0×	0.50×
16-bit	4.0×	0.25×
32-bit	8.0×	0.125×

Why Linear Scaling is Bad: For wide adders, delay becomes prohibitive. Use a faster architecture for N > 8.

8.6 Usage Examples

8.6.1 Basic 8-bit Addition

```
logic [7:0] a, b, sum;
logic carry_out;

math_adder_ripple_carry #(
    .N(8)
) u_adder (
    .i_a      (a),
    .i_b      (b),
    .i_c      (1'b0),        // No carry input
    .ow_sum   (sum),
    .ow_carry (carry_out)
);

// Example: 100 + 50
initial begin
    a = 8'd100;
    b = 8'd50;
    #1;  // Wait for combinational propagation
    assert(sum == 8'd150);
end
```

8.6.2 4-bit Incrementer

```
logic [3:0] count, count_plus_1;

math_adder_ripple_carry #(
    .N(4)
```

```
) u_incrementer (
    .i_a      (count),
    .i_b      (4'b0),        // B = 0
    .i_c      (1'b1),        // Carry in = 1 → +1
    .ow_sum   (count_plus_1),
    .ow_carry ()             // Unused
);
```

8.6.3 BCD Digit Adder (4-bit)

```
// Add two BCD digits (0-9)
logic [3:0] bcd_a, bcd_b, bcd_sum_raw;
logic [3:0] bcd_sum_corrected;
logic carry_raw, carry_corrected;

// Raw binary addition
math_adder_ripple_carry #(.N(4)) u_bcd_add (
    .i_a      (bcd_a),
    .i_b      (bcd_b),
    .i_c      (1'b0),
    .ow_sum   (bcd_sum_raw),
    .ow_carry (carry_raw)
);

// BCD correction: if sum > 9, add 6
logic bcd_overflow;
assign bcd_overflow = carry_raw | (bcd_sum_raw > 4'd9);

math_adder_ripple_carry #(.N(4)) u_bcd_correct (
    .i_a      (bcd_sum_raw),
    .i_b      (bcd_overflow ? 4'd6 : 4'd0),
    .i_c      (1'b0),
    .ow_sum   (bcd_sum_corrected),
    .ow_carry ()
);

assign bcd_carry_out = bcd_overflow;
```

8.6.4 Multi-Precision Addition (16-bit via 2×8-bit)

```
logic [7:0] a_low, a_high, b_low, b_high;
logic [7:0] sum_low, sum_high;
logic carry_low, carry_high;

// Low byte
math_adder_ripple_carry #(.N(8)) u_add_low (
    .i_a      (a_low),
    .i_b      (b_low),
```

```
    .i_c      (1'b0),
    .ow_sum   (sum_low),
    .ow_carry (carry_low)
);

// High byte (chain carry from low)
math_adder_ripple_carry #(.N(8)) u_add_high (
    .i_a      (a_high),
    .i_b      (b_high),
    .i_c      (carry_low),
    .ow_sum   (sum_high),
    .ow_carry (carry_high)
);

logic [15:0] sum_16 = {sum_high, sum_low};
```

8.6.5 Simple ALU (Add/Subtract)

```
logic [7:0] a, b, result;
logic sub;  // 1 = subtract, 0 = add
logic [7:0] b_mux;
logic carry_out;

// Two's complement subtraction: A - B = A + (~B) + 1
assign b_mux = sub ? ~b : b;

math_adder_ripple_carry #(.N(8)) u_alu (
    .i_a      (a),
    .i_b      (b_mux),
    .i_c      (sub),         // Carry in = 1 for subtraction
    .ow_sum   (result),
    .ow_carry (carry_out)
);
```

8.7 Design Notes

8.7.1 When to Use Ripple Carry Adder

Ideal Use Cases: - Width ≤ 8 bits - Area is critical constraint - Speed is not critical (< 100 MHz) - Simple control logic - Low-power designs (minimal gates = minimal switching) - BCD arithmetic (4-bit digits) - Small incrementers/decrementers

8.7.2 When to Use Alternatives

Use Carry Lookahead if: - 4 ≤ N ≤ 16 - Need moderate speed improvement - Can afford 20-50% more area

Use Brent-Kung if: - N = 8, 16, or 32 - Speed is critical - Can afford 4× area

Use Han-Carlson if: - N ≥ 32 - Maximum speed required — the library's Kogge-Stone-class option (no Kogge-Stone module exists in the library)

8.7.3 Width Selection Trade-offs

Application	Recommended Width	Rationale
Counter increment	4-8 bits	Minimal area, adequate speed
BCD digit	4 bits	Natural fit for BCD
Byte operations	8 bits	Good balance
Word operations	16 bits	Marginal - consider CLA
Double word	32 bits	Too slow - use faster adder

8.7.4 Synthesis Considerations

Optimization Tips:

1. **Allow inference of fast carry chains**:

```
# Let tool map to dedicated carry logic
set_dont_touch false
```

2. **Consider FPGA carry chains**:

    - Modern FPGAs have dedicated fast carry chains
    - Ripple carry maps well to these primitives
    - May achieve better performance than expected

3. **For ASIC, use standard cell library**:

```
# Use full adder cells from library
set_dont_use [get_lib_cells */ADDF*] false
```

4. **Break long chains for high frequency**:

```
// Pipeline every 8 bits
logic [7:0] r_sum_low, r_carry_mid;
always_ff @(posedge clk) begin
    r_sum_low <= ow_sum[7:0];
    r_carry_mid <= w_c[7];
end
```

8.7.5 Performance

Width	LUTs (Est.)	FFs (Pipeline)	Description
4-bit	~8	0 (combinational)	Minimal
8-bit	~16	0 (combinational)	Small
16-bit	~32	0 (combinational)	Medium (consider alternatives)
32-bit	~64	0 (combinational)	Large (use faster adder)

Why It's Smallest: No additional logic beyond full adders — just a direct carry chain.

Comparison to Other Adders:

Adder Type	Logic Depth	Area (relative)	Speed (relative)	Best Use Case
Ripple Carry	**O(N)**	**1.0× (min)**	**1.0× (slowest)**	**N ≤ 8, area-critical**
Carry Lookahead	O(N)	1.2×	1.5×	4 ≤ N ≤ 16
Brent-Kung	O(log N)	4.0×	4.0×	16 ≤ N ≤ 32
Kogge-Stone	O(log N)	6.0×	6.0×	N ≥ 32, critical path

Key Advantage: Smallest possible adder area — only N full adders, no additional logic.

Comparison: Area vs Speed

Width	Ripple	CLA	Brent-Kung
8-bit Area	16	24	60
8-bit Speed	1×	1.7×	4×
Area Advantage	Baseline	+50%	+275%

8.7.6 Common Pitfalls

Anti-Pattern 1: Using for wide additions

```
// WRONG: 64-bit ripple carry (extremely slow!)
math_adder_ripple_carry #(.N(64)) u_add (...);
// Result: 128 logic levels, ~25ns delay @ 1.0V!

// RIGHT: Use faster adder for wide operations
math_adder_brent_kung_032 u_add_low (...);   // Lower 32 bits
math_adder_brent_kung_032 u_add_high (...);  // Upper 32 bits
```

Anti-Pattern 2: In critical timing paths

```
// WRONG: Ripple adder in critical datapath
always_ff @(posedge clk) begin
    data_reg <= ripple_adder_output;  // May not meet timing!
end

// RIGHT: Use faster adder or pipeline
```

Anti-Pattern 3: Ignoring FPGA carry chains

```
// WRONG: Custom carry logic that doesn't infer carry chain
// (Defeats FPGA optimization)

// RIGHT: Use standard ripple carry structure
// Synthesis tools will map to fast carry primitives
```

Anti-Pattern 4: Chaining multiple adders

```
// WRONG: Cascading multiple ripple adders in one cycle
logic [7:0] sum1, sum2, sum3;
math_adder_ripple_carry #(.N(8)) u1
(.i_a(a), .i_b(b), .ow_sum(sum1), ...);
math_adder_ripple_carry #(.N(8)) u2
(.i_a(sum1), .i_b(c), .ow_sum(sum2), ...);
math_adder_ripple_carry #(.N(8)) u3
(.i_a(sum2), .i_b(d), .ow_sum(sum3), ...);
// 3 × 16 levels = 48 gate delays!

// RIGHT: Pipeline or use carry-save adder tree
```

8.8 Related Modules

- **math_adder_full.sv** - Single-bit full adder building block
- **math_adder_half.sv** - Single-bit half adder (no carry input)
- **math_adder_full_nbit.sv** - Functionally equivalent to this module

- **math_adder_pg_chain.sv** - Faster alternative (1.5× speed, +20% area)
- **math_adder_brent_kung_*.sv** - Much faster (4× speed for 32-bit)

8.9 Testing

From the test suite (`val/math/test_math_adder_ripple_carry.py`):

Key Test Scenarios: - Random stimulus (all widths) - Corner cases: 0+0, 0+MAX, MAX+MAX - Full carry propagation: 0x00 + 0xFF (all carries ripple) - Incrementer mode: A + 0 + 1 - Subtraction mode: A + (~B) + 1 - BCD operations (4-bit)

Test Command:

```
pytest val/math/test_math_adder_ripple_carry.py -v
```

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

8.10 References

- Mano, M.M. "Digital Design." Prentice Hall, 2002. (Chapter 4: Combinational Logic)
- Weste, N., Harris, D. "CMOS VLSI Design." Addison Wesley, 2010.
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9 Add/Subtract Unit

A unified N-bit add/subtract unit built on two's complement arithmetic — both operations, one control signal, shared hardware.

9.1 Overview

The `math_addsub_full_nbit` module is a combined adder/subtractor that performs either A + B or A - B based on a control signal. Subtraction uses two's complement (A - B = A + (~B) + 1), so the adder hardware serves both operations.

Key Feature: One arithmetic unit handles both operations using XOR gates and a control signal — noticeably less hardware than separate adder and subtractor modules.

9.1.1 Module Declaration

```
module math_addsub_full_nbit #(
    parameter int N = 4       // Bit width
) (
    input  logic [N-1:0] i_a,       // Operand A
    input  logic [N-1:0] i_b,       // Operand B
    input  logic         i_c,       // Control: 0=add, 1=subtract
    output logic [N-1:0] ow_sum,    // Result output
    output logic         ow_carry   // Carry/borrow output
);
```

9.2 Parameters

Parameter	Type	Default	Description
N	int	4	Bit width (range: 1-64, typical: 8-32)

9.3 Ports

Port	Direction	Width	Description
i_a	Input	N	Operand A
i_b	Input	N	Operand B
i_c	Input	1	Control signal (0=add, 1=subtract)
ow_sum	Output	N	Result (A+B or A-B)
ow_carry	Output	1	Carry (add) or inverted borrow (subtract)

Control Signal (i_c): - **i_c = 0**: Addition mode → Result = A + B - **i_c = 1**: Subtraction mode → Result = A - B

Carry Output Interpretation: - **Add mode (i_c=0)**: ow_carry = carry output (overflow for unsigned) - **Subtract mode (i_c=1)**: ow_carry = ~borrow (1 if A≥B, 0 if A<B)

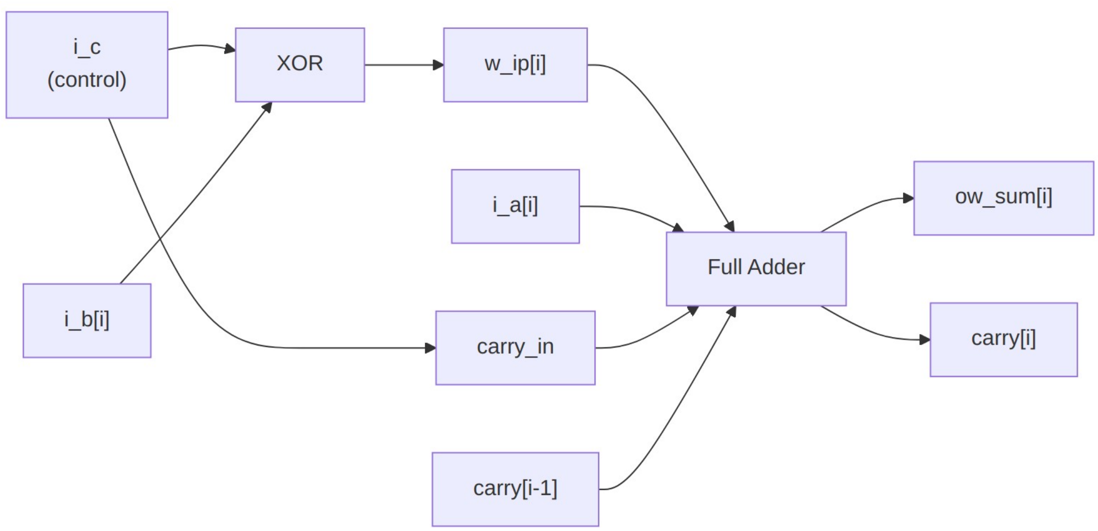
9.4 Functional Description

9.4.1 Two's Complement Subtraction

The module implements subtraction using two's complement arithmetic:

```
A - B = A + (~B) + 1
```

Implementation Strategy: 1. **XOR B with control signal**: `w_ip[i] = i_b[i] ^ i_c` - If i_c = 0 (add): w_ip = B (unchanged) - If i_c = 1 (sub): w_ip = ~B (one's complement) 2. **Feed i_c as carry input**: Completes two's complement (+1) 3. **Use standard adder**: A + w_ip + i_c



Logic Diagram

9.4.2 Implementation

```
// XOR B with control signal (inverts B for subtraction)
genvar i;
generate
    for (i = 0; i < N; i++) begin : gen_xor
        assign w_ip[i] = i_b[i] ^ i_c;
    end
endgenerate

// Carry chain initialization
assign w_c[0] = i_c;  // i_c = 0 for add, 1 for subtract

// Full adder chain
generate
    for (i = 0; i < N; i++) begin : gen_full_adders
        math_adder_full fa (
            .i_a(i_a[i]),
```

```
            .i_b(w_ip[i]),      // XORed B
            .i_c(w_c[i]),
            .ow_sum(ow_sum[i]),
            .ow_carry(w_c[i+1])
        );
    end
endgenerate

assign ow_carry = w_c[N];
```

9.4.3 Operation Modes

9.4.3.1 Addition Mode (i_c = 0)

```
w_ip[i] = i_b[i] ^ 0 = i_b[i]  (B unchanged)
w_c[0] = 0                      (no initial carry)
Result = A + B + 0 = A + B
```

9.4.3.2 Subtraction Mode (i_c = 1)

```
w_ip[i] = i_b[i] ^ 1 = ~i_b[i]  (B inverted = one's complement)
w_c[0] = 1                       (carry in = 1)
Result = A + (~B) + 1 = A - B   (two's complement)
```

9.5 Timing Characteristics

Metric	Value
Logic Depth	2N + 1 levels (1 XOR + N full adders x 2)
Typical Delay (8-bit)	~3.2 ns @ 1.0V
Max Frequency (8-bit)	~300 MHz

Logic Level Breakdown (8-bit): 1. XOR stage: 1 level (B ^ control) 2. Carry chain: 16 levels (8 full adders × 2) 3. **Total**: 17 levels

Critical Path:

```
i_b[0] → XOR → FA[0] → FA[1] → ... → FA[7] → ow_carry
```

9.6 Usage Examples

9.6.1 Simple Add/Subtract ALU

```
logic [7:0] a, b, result;
logic sub;  // 0=add, 1=subtract
logic carry_borrow;

math_addsub_full_nbit #(.N(8)) u_alu (
```

```
    .i_a(a),
    .i_b(b),
    .i_c(sub),
    .ow_sum(result),
    .ow_carry(carry_borrow)
);

// Example: Addition
initial begin
    a = 8'd100;
    b = 8'd50;
    sub = 1'b0;  // Add mode
    #1;
    assert(result == 8'd150);
    assert(carry_borrow == 1'b0);  // No overflow
end

// Example: Subtraction
initial begin
    a = 8'd100;
    b = 8'd50;
    sub = 1'b1;  // Subtract mode
    #1;
    assert(result == 8'd50);
    assert(carry_borrow == 1'b1);  // A >= B (no borrow)
end
```

9.6.2 Detecting Underflow in Subtraction

```
logic [7:0] a, b, result;
logic underflow;

math_addsub_full_nbit #(.N(8)) u_alu (
    .i_a(a),
    .i_b(b),
    .i_c(1'b1),  // Subtract mode
    .ow_sum(result),
    .ow_carry(carry_out)
);

// For subtraction, borrow = ~carry_out
// Underflow (A < B) when carry_out = 0
assign underflow = ~carry_out;

// Example: Underflow case
initial begin
    a = 8'd50;
    b = 8'd100;
```

```
    #1;
    assert(underflow == 1'b1);  // A < B
    assert(result == 8'd206);   // Wraps: 50 - 100 + 256 = 206
end
```

9.6.3 Simple Calculator

```
module calculator (
    input  logic [7:0] operand_a,
    input  logic [7:0] operand_b,
    input  logic       operation,  // 0=add, 1=subtract
    output logic [7:0] result,
    output logic       overflow,   // Overflow (add) or underflow
(sub)
    output logic       zero        // Result is zero
);

    logic carry_borrow;

    math_addsub_full_nbit #(.N(8)) u_alu (
        .i_a(operand_a),
        .i_b(operand_b),
        .i_c(operation),
        .ow_sum(result),
        .ow_carry(carry_borrow)
    );

    // Overflow/underflow detection
    assign overflow = operation ? ~carry_borrow : carry_borrow;

    // Zero flag
    assign zero = ~|result;

endmodule
```

9.6.4 Multi-Function ALU

```
typedef enum logic [1:0] {
    ALU_ADD = 2'b00,
    ALU_SUB = 2'b01,
    ALU_INC = 2'b10,
    ALU_DEC = 2'b11
} alu_op_t;

module multi_alu (
    input  logic [7:0] a, b,
    input  alu_op_t    op,
    output logic [7:0] result,
```

```
    output logic       carry_flag
);

    logic [7:0] b_mux;
    logic       ctrl;

    // Operand and control selection
    always_comb begin
        case (op)
            ALU_ADD: begin
                b_mux = b;
                ctrl = 1'b0;      // Add mode
            end
            ALU_SUB: begin
                b_mux = b;
                ctrl = 1'b1;      // Subtract mode
            end
            ALU_INC: begin
                b_mux = 8'd1;
                ctrl = 1'b0;      // A + 1 + 0 = A + 1 -- with ctrl=1,
i_c would
                                  // invert B AND carry in, giving A +
~0 + 1 = A
            end
            ALU_DEC: begin
                b_mux = 8'b1;
                ctrl = 1'b1;      // A - 1 = A + (~1) + 1
            end
        endcase
    end

    math_addsub_full_nbit #(.N(8)) u_alu (
        .i_a(a),
        .i_b(b_mux),
        .i_c(ctrl),
        .ow_sum(result),
        .ow_carry(carry_flag)
    );

endmodule
```

9.7 Design Notes

9.7.1 Advantages

Hardware efficiency: Single unit for both operations **Simple control**: One bit selects operation **Standard practice**: Common in ALU designs **No separate subtractor**: Reuses adder logic

9.7.2 When to Use This Module

Appropriate Use Cases: - ALU designs requiring add/subtract - Arithmetic units with mode control - Simple calculators - Control logic needing both operations

Alternative Approaches: - **Manual control**: Use standard adder with external XOR and control (saves one XOR level if control is early) - **Behavioral**: `assign result = sub ? (a - b) : (a + b);` (synthesis tool optimizes)

9.7.3 Synthesis Considerations

Optimization Tips:

1. **Let synthesis infer fast carry chains** (FPGAs):

   ```
   set_dont_touch false
   ```

2. **XOR placement**: XOR gates before adder may allow optimization

3. **Consider pipelining** for high frequency:

   ```
   logic [N-1:0] r_b_conditioned;
   always_ff @(posedge clk) begin
       r_b_conditioned <= i_b ^ {N{i_c}};
   end
   ```

9.7.4 Performance

Width	LUTs (Est.)	Description
8-bit	~18	N XORs + N full adders
16-bit	~34	Linear scaling
32-bit	~66	Linear scaling

Comparison: | Architecture | Area (relative) | Operations | |————–|—————–|————| | Separate Add + Sub | 2.0× | Add or Sub | | **Add/Sub Unit** | **1.1×** | **Add and Sub** | | Adder only (manual control) | 1.0× | Add and Sub (external invert) |

Advantage: ~10% area overhead vs adder-only, but integrated control simplifies system design. That's a trade I'll take every time in control logic.

9.7.5 Common Pitfalls

Anti-Pattern 1: Misinterpreting carry output

```
// WRONG: Carry means different things for add vs subtract!
if (carry) begin
    // overflow? underflow? Depends on operation!
```

```
end

// RIGHT: Interpret based on operation
logic overflow_underflow;
assign overflow_underflow = (i_c == 0) ? carry :  // overflow (add)
                                         ~carry;  // underflow (sub)
```

Anti-Pattern 2: Using for signed arithmetic without overflow detection

```
// WRONG: Missing signed overflow detection
logic [7:0] result_signed;
math_addsub_full_nbit u_alu (.i_c(1'b0), ...);  // Add mode
// Result may overflow for signed!

// RIGHT: Check signed overflow
logic signed_overflow;
assign signed_overflow = (i_a[N-1] == i_b[N-1]) &&
                         (ow_sum[N-1] != i_a[N-1]);
```

Anti-Pattern 3: Forgetting XOR overhead

```
// Be aware: XOR stage adds one logic level
// If timing is critical, consider pipelining the XOR stage
```

9.8 Related Modules

- **math_adder_ripple_carry.sv** - Add-only (no XOR overhead)
- **math_subtractor_ripple_carry.sv** - Subtract-only (rarely used)
- **math_adder_pg_chain.sv** - Faster alternative for adder stage

9.9 Testing

From the test suite (`val/math/test_math_addsub_full_nbit.py`):

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

9.10 References

- Mano, M.M. "Digital Design." Prentice Hall, 2002.
- Hennessy, J.L., Patterson, D.A. "Computer Architecture: A Quantitative Approach." Morgan Kaufmann, 2011.

9.11 Navigation

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10 BF16 Adder

A pipelined BF16 (Brain Float 16) floating-point adder with IEEE 754-compliant special case handling, Round-to-Nearest-Even rounding, and flush-to-zero support for subnormal numbers.

10.1 Overview

The `math_bf16_adder` module implements full BF16 addition and subtraction with configurable pipeline stages for frequency/latency tradeoffs. It follows industry-standard practices used in AI/ML accelerators for efficient 16-bit floating-point operations.

Key Features: - **BF16 format** - Same exponent range as FP32, reduced mantissa precision - **Configurable pipeline** - 4 optional pipeline stages for frequency optimization - **IEEE 754 special cases** - Zero, infinity, NaN handling - **RNE rounding** - Round-to-Nearest-Even for unbiased results - **FTZ mode** - Flush-to-Zero for subnormal inputs and outputs - **Status flags** - Overflow, underflow, and invalid operation indicators - **Valid handshaking** - Input/output valid signals for pipeline control

10.1.1 Module Declaration

```
module math_bf16_adder #(
    parameter int PIPE_STAGE_1 = 0,  // After exponent diff + swap
    parameter int PIPE_STAGE_2 = 0,  // After alignment shifter
    parameter int PIPE_STAGE_3 = 0,  // After mantissa add/sub
    parameter int PIPE_STAGE_4 = 0   // After normalize
) (
    input  logic        i_clk,
    input  logic        i_rst_n,
    input  logic [15:0] i_a,            // BF16 operand A
    input  logic [15:0] i_b,            // BF16 operand B
    input  logic        i_valid,        // Input valid
    output logic [15:0] ow_result,      // BF16 sum/difference
    output logic        ow_overflow,    // Overflow to infinity
    output logic        ow_underflow,   // Underflow to zero
    output logic        ow_invalid,     // Invalid operation (NaN)
    output logic        ow_valid        // Output valid
);
```

10.2 Parameters

Parameter	Type	Default	Description
PIPE_STAGE_1	int	0	Register after exponent compare and operand swap
PIPE_STAGE_2	int	0	Register after mantissa alignment
PIPE_STAGE_3	int	0	Register after mantissa add/subtract
PIPE_STAGE_4	int	0	Register after normalization

Latency Formula: `PIPE_STAGE_1 + PIPE_STAGE_2 + PIPE_STAGE_3 + PIPE_STAGE_4` cycles (each enabled stage is one register bank; all-disabled is combinational, same-cycle)

10.3 Ports

Port	Direction	Width	Description
i_clk	Input	1	Clock signal
i_rst_n	Input	1	Active-low asynchronous reset
i_a	Input	16	BF16 addend A
i_b	Input	16	BF16 addend B
i_valid	Input	1	Input operands are valid
ow_result	Output	16	BF16 sum (A + B)
ow_overflow	Output	1	1 if result overflowed to infinity
ow_underflow	Output	1	1 if result underflowed to zero
ow_invalid	Output	1	1 if operation was invalid

Port	Direction	Width	Description
			(inf - inf)
ow_valid	Output	1	Output result is valid

10.4 Functional Description

10.4.1 BF16 Format

```
BF16: [15]=sign, [14:7]=exponent (8 bits, bias=127), [6:0]=mantissa (7
bits)

Special values:
  +0:    0x0000 (sign=0, exp=0, mant=0)
  -0:    0x8000 (sign=1, exp=0, mant=0)
  +inf:  0x7F80 (sign=0, exp=255, mant=0)
  -inf:  0xFF80 (sign=1, exp=255, mant=0)
  NaN:   0x7FC0 (exp=255, mant!=0, canonical quiet NaN)
```

10.4.2 Architecture

```
+-------------------------------------------------------+
                    |              math_bf16_adder
|
                    |
|
    i_a[15:0] ------+--+-- sign_a, exp_a, mant_a
|
                    |  |
|
    i_b[15:0] ------+--+-- sign_b, exp_b, mant_b
|
                    |  |
|
    i_valid --------+--+
|
                    |  |
|
                    |  v
|
                    |
+-------------------------------------------------+|
                    | | Stage A: Unpack + Special Cases + Compare +
Swap ||
                    | |   - Exponent difference calculation
||
```

```
                    | |   - Magnitude comparison and operand swap
||
                    | |   - Effective operation determination
||
                    |
+------------------------------------------------+|
                    |                     | [PIPE_STAGE_1]
|
                    |                     v
|
                    |
+------------------------------------------------+|
                    | | Stage B: Mantissa Alignment
||
                    | |   - shifter_barrel (right shift smaller
operand) ||
                    | |   - Sticky bit computation
||
                    |
+------------------------------------------------+|
                    |                     | [PIPE_STAGE_2]
|
                    |                     v
|
                    |
+------------------------------------------------+|
                    | | Stage C: Mantissa Add/Subtract
||
                    | |   - 12-bit addition or subtraction
||
                    | |   - Overflow detection
||
                    |
+------------------------------------------------+|
                    |                     | [PIPE_STAGE_3]
|
                    |                     v
|
                    |
+------------------------------------------------+|
                    | | Stage D: Normalization
||
                    | |   - count_leading_zeros for shift amount
||
                    | |   - shifter_barrel for normalization
||
                    | |   - Exponent adjustment
||
```

```
                    |
+--------------------------------------------------+|
                    |                     | [PIPE_STAGE_4]
|
                    |                     v
|
                    |
+--------------------------------------------------+|
                    | | Stage E: Rounding + Result Assembly
||
                    | |   - RNE rounding with guard/round/sticky
||
                    | |   - Special case priority selection
||
                    |
+--------------------------------------------------+|
                    |
|
    ow_result[15:0]
<-----------------------------------------------------+
    ow_overflow
<---------------------------------------------------------+
    ow_underflow
<--------------------------------------------------------+
    ow_invalid
<----------------------------------------------------------+
    ow_valid
<------------------------------------------------------------+

+-----------------------------------------------------+
```

10.4.3 Processing Pipeline

1. **Stage A: Unpack + Compare + Swap**
   - Parse sign, exponent, mantissa from both inputs
   - Detect special cases (zero, subnormal, infinity, NaN)
   - Compare magnitudes and swap so larger operand is always “L”
   - Determine effective operation (add or subtract based on signs)
2. **Stage B: Mantissa Alignment**
   - Extend mantissas with implied bit and guard bits (11 bits)
   - Right-shift smaller mantissa by exponent difference
   - Compute sticky bit from shifted-out bits
3. **Stage C: Mantissa Add/Subtract**
   - Add or subtract aligned mantissas (12-bit operation)

   - Detect addition overflow (sum >= 2.0)
4. **Stage D: Normalization**
   - Count leading zeros for subtraction results
   - Shift to normalize (implied 1 at correct position)
   - Adjust exponent accordingly
5. **Stage E: Rounding + Result Assembly**
   - Apply RNE rounding using guard/round/sticky bits
   - Handle rounding overflow
   - Select result based on special case priority

10.4.4 Exponent Comparison and Swap

```
// Compare magnitudes: larger operand becomes "L", smaller becomes "S"
wire [8:0]  w_exp_diff_raw = {1'b0, w_exp_a} - {1'b0, w_exp_b};
wire        w_a_larger     = w_exp_a_larger && (~w_exp_equal ||
w_mant_a_larger);

// Absolute exponent difference for alignment shift
wire [7:0]  w_exp_diff     = w_a_larger ? (w_exp_a - w_exp_b) :
(w_exp_b - w_exp_a);
```

10.4.5 Mantissa Alignment

```
// Extend mantissas: {implied_1, mant[6:0], guard, round, sticky} = 11
bits
wire [10:0] w_mant_l_ext = {r1_l_is_normal, r1_mant_l, 3'b000};
wire [10:0] w_mant_s_ext = {r1_s_is_normal, r1_mant_s, 3'b000};

// Right-shift smaller mantissa using barrel shifter
shifter_barrel #(.WIDTH(11)) u_align_shifter (
    .data        (w_mant_s_ext),
    .ctrl        (SHIFT_RIGHT_LOGIC),
    .shift_amount(w_shift_amt),
    .data_out    (w_mant_s_aligned)
);
```

10.4.6 Round-to-Nearest-Even

```
// Extract rounding bits from normalized mantissa
wire [6:0] w_mant_final_raw = r4_mant_normalized[9:3];
wire       w_guard_bit      = r4_mant_normalized[2];
wire       w_round_bit      = r4_mant_normalized[1];
wire       w_sticky_bit     = r4_mant_normalized[0] | r4_norm_sticky;

// RNE: Round up if guard && (round || sticky || LSB)
```

```
wire w_round_up = w_guard_bit && (w_round_bit || w_sticky_bit ||
w_mant_final_raw[0]);
```

10.4.7 Special Case Priority

```
always_comb begin
    // Default: normal result
    ow_result = {w_final_sign, w_exp_out, w_mant_out};

    // Priority order (highest to lowest):
    if (r4_any_nan || r4_inf_minus_inf) begin
        // 1. NaN: any NaN input, or inf - inf
        ow_result  = {1'b0, 8'hFF, 7'h40};  // Canonical qNaN
        ow_invalid = r4_inf_minus_inf;
    end else if (r4_any_inf) begin
        // 2. Infinity: inf input (not inf-inf case)
        ow_result = {inf_sign, 8'hFF, 7'h00};
    end else if (r4_a_eff_zero && r4_b_eff_zero) begin
        // 3. Both inputs zero: -0 + -0 = -0, otherwise +0
        ow_result = {r4_sign_a & r4_sign_b, 8'h00, 7'h00};
    end else if (r4_a_eff_zero) begin
        // 4. A zero: result is B
        ow_result = {r4_sign_b, r4_exp_adjusted, w_mant_out};
    end else if (r4_b_eff_zero) begin
        // 5. B zero: result is A
        ow_result = {r4_sign_a, r4_exp_adjusted, w_mant_out};
    end else if (r4_sum_is_zero) begin
        // 6. Exact zero from subtraction
        ow_result = {1'b0, 8'h00, 7'h00};  // +0 per IEEE 754 RNE
    end else if (w_final_overflow) begin
        // 4. Overflow to infinity
        ow_result   = {w_final_sign, 8'hFF, 7'h00};
        ow_overflow = 1'b1;
    end else if (r4_exp_underflow) begin
        // 5. Underflow to zero (FTZ)
        ow_result    = {w_final_sign, 8'h00, 7'h00};
        ow_underflow = 1'b1;
    end
end
```

10.5 Timing Characteristics

10.5.1 Pipeline Latency Configurations

Configuration	Latency	Use Case
[0,0,0,0]	0 cycles (combinational)	Area-constrained, low frequency

Configuration	Latency	Use Case
[1,0,0,0]	1 cycle	Balance input timing
[0,0,0,1]	1 cycle	Balance output timing
[1,1,1,1]	4 cycles	Maximum frequency

10.5.2 Logic Depth by Stage

Stage	Logic Depth
Exponent compare	~3 gates
Alignment shift	~4 gates (log2)
Mantissa add	~5 gates
CLZ + normalize	~6 gates
Rounding	~3 gates
Total (comb)	~20-25 gates

10.6 Usage Examples

10.6.1 Basic Addition (Combinational)

```
logic [15:0] a, b, sum;
logic overflow, underflow, invalid, valid;

math_bf16_adder u_adder (
    .i_clk(clk),
    .i_rst_n(rst_n),
    .i_a(a),
    .i_b(b),
    .i_valid(1'b1),
    .ow_result(sum),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid),
    .ow_valid(valid)
);

// Example: 1.0 + 1.0 = 2.0
// 1.0 in BF16: 0x3F80 (sign=0, exp=127, mant=0)
// 2.0 in BF16: 0x4000 (sign=0, exp=128, mant=0)
initial begin
    a = 16'h3F80;  // 1.0
    b = 16'h3F80;  // 1.0
    #1;
```

```
    // sum should be 0x4000 (2.0)
end
```

10.6.2 Pipelined Configuration

```
// 4-stage pipeline for high frequency
math_bf16_adder #(
    .PIPE_STAGE_1(1),
    .PIPE_STAGE_2(1),
    .PIPE_STAGE_3(1),
    .PIPE_STAGE_4(1)
) u_adder_pipelined (
    .i_clk(clk),
    .i_rst_n(rst_n),
    .i_a(a),
    .i_b(b),
    .i_valid(in_valid),
    .ow_result(sum),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid),
    .ow_valid(out_valid)  // Valid 4 cycles after in_valid, for ONE
cycle
);
```

10.6.3 Subtraction via Sign Manipulation

```
// To compute A - B, negate B's sign and add
logic [15:0] a_minus_b;
logic [15:0] b_negated;

assign b_negated = {~b[15], b[14:0]};  // Flip sign bit

math_bf16_adder u_subtractor (
    .i_clk(clk),
    .i_rst_n(rst_n),
    .i_a(a),
    .i_b(b_negated),
    .i_valid(1'b1),
    .ow_result(a_minus_b),
    // ...
);
```

10.6.4 With Status Checking

```
always_ff @(posedge clk) begin
    if (out_valid) begin
        if (invalid)
            $display("Invalid operation: inf - inf");
        else if (overflow)
```

```
            $display("Overflow: result is infinity");
        else if (underflow)
            $display("Underflow: result is zero");
    end
end
```

10.7 Design Notes

10.7.1 Flush-to-Zero (FTZ)

Subnormal numbers are flushed to zero: - **Input subnormals** - Treated as zero (effective zero) - **Output subnormals** - Not generated (result goes to zero)

This simplifies hardware and matches AI accelerator conventions.

10.7.2 Invalid Operation (inf - inf)

When adding +inf and -inf (or -inf and +inf): - Result is canonical quiet NaN (0x7FC0) - `ow_invalid` flag is asserted

10.7.3 Sign of Zero

The sign of zero follows IEEE 754: - **x - x** = +**0** (in RNE rounding mode) - **-0** + **-0** = **-0** - +**0** + **-0** = +**0**

10.7.4 Rounding Mode

Only Round-to-Nearest-Even (RNE) is supported. This is standard for BF16 in AI applications.

10.7.5 Performance

Component	LUTs (est.)
Unpack + compare	~40
Alignment shifter	~50
Mantissa adder	~30
CLZ + normalize	~60
Rounding logic	~20
Result MUX	~30
Pipeline regs (per stage)	~50

10.7.6 Common Pitfalls

Anti-Pattern 1: Ignoring valid signals

```
// WRONG: Capturing result without checking valid
result <= ow_result;  // May capture invalid data during pipeline fill
```

```verilog
// RIGHT: Only capture when valid
if (ow_valid)
    result <= ow_result;
```

Anti-Pattern 2: Forgetting pipeline latency

```verilog
// WRONG: Expecting immediate result with pipeline
i_a <= operand_a;
i_b <= operand_b;
i_valid <= 1'b1;
@(posedge clk);
sum <= ow_result;  // Result not ready yet!

// RIGHT: Account for latency
// With [1,1,1,1] config, wait 4 cycles for result (ow_valid pulses
once; waiting 5 misses it)
```

Anti-Pattern 3: Not handling special flags

```verilog
// WRONG: Using result without checking flags
accumulator <= accumulator + ow_result;  // May be NaN or infinity!

// RIGHT: Check flags first
if (ow_valid && !ow_invalid && !ow_overflow)
    accumulator <= accumulator + ow_result;
```

10.8 Related Modules

- **math_bf16_multiplier** - BF16 multiplication
- **math_bf16_fma** - Fused Multiply-Add with FP32 accumulator
- **shifter_barrel** - Used internally for mantissa alignment
- **count_leading_zeros** - Used internally for normalization

This module instantiates **shifter_barrel** (alignment shifting and normalization) and **count_leading_zeros** (determining the normalization shift amount).

10.9 Testing

From the test suite (`val/math/test_math_bf16_adder.py`):

- **Key test scenarios**:
    - Tests the complete BF16 adder including:
    - Normal addition and subtraction with RNE rounding
    - Special value handling (zero, infinity, NaN, subnormal)

- Overflow and underflow detection
- Invalid operation detection (inf - inf)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

10.10 References

- Google Brain Float (BF16) specification
- IEEE 754-2019 Standard for Floating-Point Arithmetic
- Intel BFloat16 documentation
- NVIDIA TensorFloat documentation
- Muller, J.M. et al. "Handbook of Floating-Point Arithmetic." Birkhauser, 2018.

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11 BF16 Exponent Adder

The exponent path for BF16 multiplication — bias subtraction, normalization adjustment, overflow/underflow detection, and special-value identification, all in one small combinational block.

11.1 Overview

The `math_bf16_exponent_adder` module computes the result exponent for BF16 multiplication using the formula: `exp_result = exp_a + exp_b - bias + norm_adjust`. It does the arithmetic in extended precision so it can catch overflow and underflow before they happen — which is exactly where you want to detect them.

Key Features: - **Bias-compensated addition** - Handles 127 bias subtraction - **Normalization adjustment** - +1 when mantissa product >= 2.0 - **Overflow detection** - Flags when result > 254 - **Underflow detection** - Flags when result <= 0 - **Special case flags** - Identifies zero, infinity, and NaN inputs

11.2 Parameters

None — the module is hardwired to the 8-bit BF16 exponent.

11.3 Ports

```
module math_bf16_exponent_adder(
    input  logic [7:0] i_exp_a,        // 8-bit exponent A
    input  logic [7:0] i_exp_b,        // 8-bit exponent B
    input  logic       i_norm_adjust,  // +1 if mantissa needs
normalization
    output logic [7:0] ow_exp_out,     // 8-bit result exponent
    output logic       ow_overflow,    // Exponent overflow (result =
inf)
    output logic       ow_underflow,   // Exponent underflow (result =
zero)
    output logic       ow_a_is_zero,   // Input A has zero exponent
    output logic       ow_b_is_zero,   // Input B has zero exponent
    output logic       ow_a_is_inf,    // Input A has max exponent
    output logic       ow_b_is_inf,    // Input B has max exponent
    output logic       ow_a_is_nan,    // Input A has max exponent
(NaN check)
    output logic       ow_b_is_nan     // Input B has max exponent
(NaN check)
);
```

Port	Direction	Width	Description
i_exp_a	Input	8	Biased exponent of operand A
i_exp_b	Input	8	Biased exponent of operand B
i_norm_adjust	Input	1	+1 adjustment when mantissa product >= 2.0
ow_exp_out	Output	8	Result exponent (saturated on overflow/underflow)
ow_overflow	Output	1	1 if result exponent > 254 (infinity)
ow_underflow	Output	1	1 if result exponent <= 0 (zero)
ow_a_is_zero	Output	1	1 if exp_a == 0 (caller checks mantissa)
ow_b_is_zero	Output	1	1 if exp_b == 0 (caller checks mantissa)
ow_a_is_in	Output	1	1 if exp_a == 255 (caller

Port	Direction	Width	Description
f			checks mantissa for NaN)
ow_b_is_inf	Output	1	1 if exp_b == 255 (caller checks mantissa for NaN)
ow_a_is_nan	Output	1	Same as ow_a_is_inf (actual NaN needs mantissa != 0)
ow_b_is_nan	Output	1	Same as ow_b_is_inf (actual NaN needs mantissa != 0)

11.4 Functional Description

11.4.1 BF16 Exponent Background

The encoding recap, because everything else in this module follows from it:

```
BF16: [15]=sign, [14:7]=exponent (8 bits, bias=127), [6:0]=mantissa (7 bits)

Exponent encoding:
  exp = 0:   Zero (if mantissa=0) or Subnormal (if mantissa!=0)
  exp = 255: Infinity (if mantissa=0) or NaN (if mantissa!=0)
  exp = 1-254: Normal number, actual exponent = exp - 127
```

11.4.2 Multiplication Exponent Formula

For floating-point multiplication:

```
result = A * B
       = (2^(exp_a - 127) * mant_a) * (2^(exp_b - 127) * mant_b)
       = 2^(exp_a + exp_b - 254) * (mant_a * mant_b)

If mantissa product >= 2.0, normalize by right-shifting mantissa and adding 1 to exponent:
       = 2^(exp_a + exp_b - 254 + 1) * (mant_a * mant_b / 2)

Biased result exponent = exp_a + exp_b - 127 + norm_adjust
```

11.4.3 Special Case Detection

```
// Zero exponent: input is zero or subnormal
assign ow_a_is_zero = (i_exp_a == 8'h00);
assign ow_b_is_zero = (i_exp_b == 8'h00);
```

```
// Max exponent: input is infinity or NaN
assign ow_a_is_inf = (i_exp_a == 8'hFF);
assign ow_b_is_inf = (i_exp_b == 8'hFF);

// NaN detection requires mantissa check (done by caller)
assign ow_a_is_nan = (i_exp_a == 8'hFF);
assign ow_b_is_nan = (i_exp_b == 8'hFF);
```

Note: The module reports exp==255 as both “inf” and “nan” — the caller must check the mantissa to distinguish.

11.4.4 Extended Precision Arithmetic

```
// Use 10-bit arithmetic to detect overflow/underflow before they
happen
wire [9:0] w_exp_sum_raw;

// exp_a + exp_b + norm_adjust - 127
assign w_exp_sum_raw = {2'b0, i_exp_a} + {2'b0, i_exp_b} +
                       {9'b0, i_norm_adjust} - 10'd127;
```

Why 10 bits? - Maximum sum: 255 + 255 + 1 = 511 (needs 9 bits) - After subtracting 127: 511 - 127 = 384 (still fits in 9 bits) - Need 10th bit to detect negative results (underflow)

11.4.5 Overflow/Underflow Detection

```
// Underflow: result <= 0 (MSB set means negative, or result is
exactly 0)
wire w_underflow_raw = w_exp_sum_raw[9] | (w_exp_sum_raw == 10'd0);

// Overflow: result > 254 (255 is reserved for inf/nan)
// Only valid if not underflow (negative values wrap to large
positive)
wire w_overflow_raw = ~w_underflow_raw & (w_exp_sum_raw > 10'd254);

// Special cases override normal overflow/underflow
wire w_either_special = ow_a_is_inf | ow_b_is_inf | ow_a_is_zero |
ow_b_is_zero;

assign ow_overflow  = w_overflow_raw & ~w_either_special;
assign ow_underflow = w_underflow_raw & ~w_either_special;
```

Key insight: Check underflow BEFORE overflow. Negative values (underflow) appear as large positive values in unsigned arithmetic, so an overflow check alone will misfire on them. This is the ordering bug I see most often in hand-rolled exponent logic.

11.4.6 Result Saturation

```
always_comb begin
    if (ow_overflow) begin
        ow_exp_out = 8'hFF;  // Saturate to infinity
    end else if (ow_underflow) begin
        ow_exp_out = 8'h00;  // Saturate to zero
    end else begin
        ow_exp_out = w_exp_sum_raw[7:0];  // Normal result
    end
end
```

11.5 Timing Characteristics

Path	Logic Depth
Special case detection	1 gate (comparator)
Exponent addition	2-3 gates (10-bit add)
Overflow/underflow detection	2 gates (compare + AND)
Result MUX	1 gate
Total	~5-7 gate levels

11.5.1 Resource Utilization

Component	LUTs (est.)
10-bit adder	~15
Comparators	~10
Output MUX	~8
Special case flags	~4

11.6 Usage Examples

11.6.1 Basic Usage

```
logic [7:0] exp_a, exp_b;
logic       norm_adjust;
logic [7:0] exp_result;
logic       overflow, underflow;
logic       a_zero, b_zero, a_inf, b_inf;

math_bf16_exponent_adder u_exp_add (
    .i_exp_a(exp_a),
    .i_exp_b(exp_b),
    .i_norm_adjust(norm_adjust),
```

```
    .ow_exp_out(exp_result),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_a_is_zero(a_zero),
    .ow_b_is_zero(b_zero),
    .ow_a_is_inf(a_inf),
    .ow_b_is_inf(b_inf),
    .ow_a_is_nan(),   // Use with mantissa check
    .ow_b_is_nan()
);
```

11.6.2 In BF16 Multiplier

```
// Get normalization flag from mantissa multiplier
wire w_needs_norm;
math_bf16_mantissa_mult u_mant_mult
(..., .ow_needs_norm(w_needs_norm), ...);

// Compute exponent
math_bf16_exponent_adder u_exp_add (
    .i_exp_a(i_a[14:7]),
    .i_exp_b(i_b[14:7]),
    .i_norm_adjust(w_needs_norm),
    .ow_exp_out(w_exp_result),
    .ow_overflow(w_exp_overflow),
    .ow_underflow(w_exp_underflow),
    ...
);

// Handle rounding-induced exponent overflow
wire w_mant_round_overflow = ...;  // From mantissa rounding
wire [7:0] w_exp_final = w_mant_round_overflow ? (w_exp_result + 8'd1)
                                                : w_exp_result;
wire w_final_overflow = w_exp_overflow | (w_exp_final == 8'hFF);
```

11.6.3 Example Calculations

exp_a	exp_b	norm_adj	Calculation	Result	Status
127	127	0	127 + 127 - 127 + 0 = 127	127	Normal
127	127	1	127 + 127 - 127 + 1 = 128	128	Normal
200	200	0	200 + 200	255	Overflow

exp_a	exp_b	norm_adj	Calculation	Result	Status
			- 127 + 0 = 273		
50	50	0	50 + 50 - 127 + 0 = -27	0	Underflow
1	1	0	1 + 1 - 127 + 0 = -125	0	Underflow

11.7 Design Notes

11.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Simple 10-bit arithmetic instead of full-width operations 2. **Wire complexity** - Minimal signal routing 3. **Logic depth** - Parallel overflow/underflow detection

11.7.2 Why Override Flags for Special Cases?

When either input is zero or infinity, the overflow/underflow flags shouldn't trigger the normal saturation path: - **Zero * anything** -> Zero (handled by caller, not underflow) - **Infinity * anything** -> Infinity (handled by caller, not overflow) - **0 * Infinity** -> NaN (invalid operation, handled by caller)

The flags are suppressed to let the caller handle these cases explicitly.

11.7.3 Exponent Ranges

Biased exp	Meaning	Result
0	Zero/Subnormal	Treat as zero (FTZ)
1-254	Normal	Valid computation
255	Inf/NaN	Special handling

11.7.4 Order of Checks

The module checks underflow BEFORE overflow because: 1. Negative results (e.g., -25) appear as large positive values (e.g., 999) in unsigned math 2. Without underflow check first, -25 would incorrectly trigger overflow

11.7.5 Auto-Generated Code

This module is auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/bf16/bf16_exponent_adder.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/bf16/generate_all.py rtl/math`

Do not edit the generated .sv file manually.

11.8 Related Modules

- **math_bf16_mantissa_mult** - Companion mantissa handling
- **math_bf16_multiplier** - Complete BF16 multiplier using this module
- **math_bf16_fma** - BF16 FMA with similar exponent logic

11.9 Testing

From the test suite (`val/math/test_math_bf16_exponent_adder.py`):

- **Key test scenarios**:
  - This module handles exponent addition with bias handling, overflow/underflow
  - detection, and special case (zero/inf/nan) detection.
  - BF16 exponent: 8 bits, bias = 127
  - Result = exp_a + exp_b - 127 + norm_adjust

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

11.10 References

- IEEE 754-2019 Standard for Floating-Point Arithmetic
- Google Brain Float (BF16) specification
- Muller, J.M. et al. “Handbook of Floating-Point Arithmetic.” Birkhauser, 2018.

11.11 Navigation

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12 BF16 Extended Modules

The extended BF16 (Brain Float 16) module set for AI/ML accelerators: activation functions, mathematical operations, comparison/selection, and format conversions.

12.1 Overview

The core BF16 arithmetic (adder, multiplier, FMA) only gets you multiply-accumulate. Everything else a neural network needs lives here — a comprehensive set of modules for inference and training. All modules follow IEEE 754-like conventions with Flush-to-Zero (FTZ) for subnormals.

Key Features: - **Piecewise linear approximations** for transcendental functions - **LUT-based implementations** for high accuracy where needed - **Consistent interface** across all activation functions - **NaN propagation** throughout all operations - **Auto-generated** from Python generators for consistency

12.2 Functional Description

12.2.1 Activation Functions

The standard neural network activation functions, BF16 in and BF16 out.

Module	Function	Description
`math_bf16_relu`	max(0, x)	Rectified Linear Unit - passes positive values, zeros negative
`math_bf16_leaky_relu`	max(0.01x, x)	Leaky ReLU - small gradient for negative values
`math_bf16_gelu`	x * Phi(x)	Gaussian Error Linear Unit - smooth approximation
`math_bf16_sigmoid`	1/(1+exp(-x))	Logistic function - output in (0, 1)
`math_bf16_tanh`	(exp(x)-exp(-x))/(exp(x)+exp(-x))	Hyperbolic tangent - output in (-1, 1)
`math_bf16_silu`	x * sigmoid(x)	Sigmoid Linear Unit (Swish)
`math_bf16_softmax_8`	exp(xi)/sum(exp(xj))	8-input softmax normalization

Every SCALAR activation function in this group uses the same two-port pattern:

```
module math_bf16_{activation} (
    input  logic [15:0] i_a,        // BF16 input
```

```
    output logic [15:0] ow_result   // BF16 output
);
```

12.2.2 Mathematical Operations

The extended math operations for neural network computations.

Module	Function	Description
math_bf16_exp2	2^x	Base-2 exponential (LUT + interpolation)
math_bf16_log2	log2(x)	Base-2 logarithm
math_bf16_log2_scale	log2(x) + scale	Scaled logarithm for normalization
math_bf16_reciprocal	1/x	Full-precision reciprocal
math_bf16_fast_reciprocal	~1/x	LUT-based fast approximation
math_bf16_divider	a/b	Full BF16 division
math_bf16_goldschmidt_div	a/b	Iterative Goldschmidt division
math_bf16_newton_raphson_recip	1/x	Newton-Raphson iterative reciprocal

The reciprocal and division modules give you an accuracy/latency trade-off — fast LUT-based single-cycle on one end, iterative Goldschmidt on the other:

```
// Fast reciprocal (single-cycle, LUT-based)
module math_bf16_fast_reciprocal #(
    parameter int LUT_DEPTH = 128    // 32, 64, or 128 entries
) (
    input  logic [15:0] i_a,
    output logic [15:0] ow_result,
    output logic        ow_overflow,
    output logic        ow_underflow,
    output logic        ow_invalid,
    output logic        ow_div_by_zero
);

// Goldschmidt division (iterative, higher accuracy)
module math_bf16_goldschmidt_div #(
    parameter int ITERATIONS = 2,     // 1 or 2 iterations
    parameter int PIPELINED = 1,      // 0=combinational, 1=pipelined
    parameter int LUT_DEPTH = 128     // Initial estimate LUT size
```

```
) (
    input  logic        i_clk,
    input  logic        i_rst_n,
    input  logic [15:0] i_a,          // Dividend
    input  logic [15:0] i_b,          // Divisor
    input  logic        i_valid,
    output logic [15:0] ow_result,
    output logic        ow_valid,
    output logic        ow_overflow,
    output logic        ow_underflow,
    output logic        ow_invalid,
    output logic        ow_div_by_zero
);
```

12.2.3 Comparison and Selection

Modules for comparing and selecting BF16 values.

Module	Function	Description
`math_bf16_comparator`	a <=> b	Full comparison with flags
`math_bf16_max`	max(a, b)	Two-input maximum
`math_bf16_min`	min(a, b)	Two-input minimum
`math_bf16_max_tree`	max(a[])	N-input maximum tree
`math_bf16_max_tree_8`	max(a[7:0])	8-input maximum
`math_bf16_min_tree_8`	min(a[7:0])	8-input minimum
`math_bf16_clamp`	clamp(x, lo, hi)	Value clamping to range

The comparator interface:

```
module math_bf16_comparator (
    input  logic [15:0] i_a,
    input  logic [15:0] i_b,
    output logic        ow_eq,        // a == b
    output logic        ow_lt,        // a < b
    output logic        ow_gt,        // a > b
    output logic        ow_unordered  // NaN comparison
);
```

12.2.4 Format Conversions

Convert between BF16 and other floating-point formats.

Module	Conversion	Description
`math_bf16_to_fp16`	BF16 -> FP16	Requires mantissa rounding
`math_bf16_to_fp32`	BF16 -> FP32	Lossless upconversion
`math_bf16_to_fp8_e4m3`	BF16 -> FP8 E4M3	ML inference format
`math_bf16_to_fp8_e5m2`	BF16 -> FP8 E5M2	ML training format
`math_bf16_to_int`	BF16 -> INT	Fixed-point conversion
`math_int_to_bf16`	INT -> BF16	Integer to float conversion
`math_bf16_scale_to_int8`	BF16 -> INT8	Quantization with scaling

The converters all follow the same shape — value in, converted value out, plus the three status flags:

```
module math_bf16_to_{format} (
    input  logic [15:0] i_a,          // BF16 input
    output logic [N:0]  ow_result,    // Target format output
    output logic        ow_overflow,  // Value too large
    output logic        ow_underflow, // Value too small (rounds to 0)
    output logic        ow_invalid    // NaN input
);
```

12.2.5 Special Value Handling

Special values behave the same way across every module:

Input	ReLU	Sigmoid	Max/Min	Converters
+0	+0	0.5	Compares as 0	Format-specific 0
-0	+0	0.5	Compares as 0	Format-specific 0
+Inf	+Inf	1.0	Largest	Passes through where the target has Inf; saturates + overflow flag where

Input	ReLU	Sigmoid	Max/Min	Converters
				it does not (e4m3)
-Inf	0	0.0	Smallest	Same, sign-preserved – NO underflow flag (underflow means too-small magnitude)
NaN	NaN	NaN	Propagates	Invalid flag
Subnormal	FTZ	FTZ	FTZ	FTZ

12.3 Usage Examples

12.3.1 Applying an Activation

```
// Apply ReLU activation to layer output
math_bf16_relu u_relu (
    .i_a(layer_output),
    .ow_result(activated_output)
);

// Apply sigmoid for binary classification
math_bf16_sigmoid u_sigmoid (
    .i_a(logit),
    .ow_result(probability)
);
```

12.3.2 Max/Min Tree Pooling

```
// Find maximum across 8 values (e.g., for pooling)
logic [15:0] pool_inputs [8];
logic [15:0] pool_max;

math_bf16_max_tree_8 u_max_pool (
    .i_data(pool_inputs),
    .ow_max(pool_max)
);
```

12.3.3 Quantization

```
// Convert BF16 weights to INT8 for inference
logic [15:0] bf16_weight;
logic [7:0]  int8_weight;
logic [15:0] scale_factor;  // Pre-computed scale
```

```
math_bf16_scale_to_int8 u_quantize (
    .i_value(bf16_weight),
    .i_scale(scale_factor),
    .ow_int8(int8_weight),
    .ow_overflow(quant_overflow),
    .ow_underflow()
);
```

12.4 Design Notes

12.4.1 Piecewise Linear Approximations

Transcendental functions (sigmoid, tanh, GELU) use piecewise linear approximation:

```
Sigmoid approximation regions:
  x <= -4:     sigmoid(x) = 0
  -4 < x < 0:  linear interpolation
  0 <= x < 4:  linear interpolation
  x >= 4:      sigmoid(x) = 1
```

That buys you ~1-2% accuracy while maintaining single-cycle latency.

12.4.2 LUT-Based Operations

When you need higher accuracy (exp2, log2, reciprocal), the implementation switches to LUT + linear interpolation:

```
// LUT stores function values at regular intervals
// Linear interpolation between adjacent entries
// Configurable LUT depth: 32, 64, or 128 entries
// Accuracy: ~0.1% for 128-entry LUT
```

12.4.3 Auto-Generation

All modules are auto-generated for consistency:

```
# Regenerate all BF16 modules
PYTHONPATH=bin:$PYTHONPATH python3
bin/rtl_generators/ieee754/generate_all.py rtl/math
```

Generator files: - `bin/rtl_generators/ieee754/fp_activations.py` - Activation functions - `bin/rtl_generators/ieee754/fp_conversions.py` - Format converters - `bin/rtl_generators/ieee754/fp_comparisons.py` - Comparison modules - `bin/rtl_generators/ieee754/bf16_reciprocal.py` - Reciprocal/division

12.5 Related Modules

- **math_bf16_multiplier** - Core BF16 multiplication

- **math_bf16_adder** - Core BF16 addition
- **math_bf16_fma** - Fused Multiply-Add
- **math_fp16_modules** - FP16 equivalents
- **math_fp8_modules** - FP8 E4M3/E5M2 modules

12.6 Testing

Covered by 31 test suites:

- `val/math/test_math_bf16_adder.py`
- `val/math/test_math_bf16_clamp.py`
- `val/math/test_math_bf16_comparator.py`
- `val/math/test_math_bf16_divider.py`
- `val/math/test_math_bf16_exp2.py`
- `val/math/test_math_bf16_exponent_adder.py`
- `val/math/test_math_bf16_fast_reciprocal.py`
- `val/math/test_math_bf16_fma.py`
- … and 23 more (`ls val/math/test_math_*.py`)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

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13 BF16 Fused Multiply-Add (FMA)

A BF16 Fused Multiply-Add unit with an FP32 accumulator, built for AI training workloads where holding precision through accumulation is critical.

13.1 Overview

The `math_bf16_fma` module implements `result = (A * B) + C` as a single fused operation where A and B are BF16 inputs and C is an FP32 accumulator. This matches the computation pattern used in neural network training where weights and activations are BF16 but accumulated gradients need FP32 precision.

Key Features: - **Mixed precision** - BF16 inputs (A, B), FP32 accumulator (C), FP32 output - **Single fused operation** - No intermediate rounding between multiply and add - **48-bit internal precision** - Captures full product + alignment bits - **IEEE 754 special cases** - Zero, infinity, NaN handling - **RNE rounding** - Round-to-Nearest-Even for unbiased results - **FTZ mode** - Flush-to-Zero for subnormal inputs

13.2 Parameters

None — the module is fixed at BF16 inputs and an FP32 accumulator.

13.3 Ports

```
module math_bf16_fma(
    input  logic [15:0] i_a,          // BF16 multiplicand
    input  logic [15:0] i_b,          // BF16 multiplier
    input  logic [31:0] i_c,          // FP32 addend (accumulator)
    output logic [31:0] ow_result,    // FP32 result
    output logic        ow_overflow,  // Overflow to infinity
    output logic        ow_underflow, // Underflow to zero
    output logic        ow_invalid    // Invalid operation
);
```

Port	Direction	Width	Description
i_a	Input	16	BF16 multiplicand A
i_b	Input	16	BF16 multiplier B
i_c	Input	32	FP32 addend C (accumulator)
ow_result	Output	32	FP32 result = A * B + C
ow_overflow	Output	1	1 if result overflowed to infinity
ow_underflow	Output	1	1 if result underflowed to zero
ow_invalid	Output	1	1 if operation was invalid

13.4 Functional Description

13.4.1 Format Specifications

13.4.1.1 BF16 Format

```
BF16: [15]=sign, [14:7]=exponent (8 bits, bias=127), [6:0]=mantissa (7
bits)

Special values:
  +0:    0x0000 (sign=0, exp=0, mant=0)
  -0:    0x8000 (sign=1, exp=0, mant=0)
  +inf:  0x7F80 (sign=0, exp=255, mant=0)
  -inf:  0xFF80 (sign=1, exp=255, mant=0)
  NaN:   0x7FC0 (exp=255, mant!=0)
```

13.4.1.2 FP32 Format

```
FP32: [31]=sign, [30:23]=exponent (8 bits, bias=127), [22:0]=mantissa
(23 bits)

Special values:
  +0:    0x00000000
  -0:    0x80000000
  +inf:  0x7F800000
  -inf:  0xFF800000
  qNaN:  0x7FC00000 (canonical quiet NaN)
```

13.4.2 Architecture

```
+------------------------------------------------------------+
                    |                   math_bf16_fma
|
                    |
|
    i_a[15:0] ------+--+-- sign_a ----+
|
    (BF16)          |  +-- exp_a -----+---------+
|
                    |  +-- mant_a ---+|         |
|
                    |                ||         |
|
    i_b[15:0] ------+--+-- sign_b ---++-XOR-----+---- prod_sign
|
    (BF16)          |  +-- exp_b ----+|         |
|
                    |  +-- mant_b --+||         |
```

```
|                   |               |||         |
|                   |   +-----------+||         |
|                   |   | 8x8 Dadda  ||         |
|                   |   | Multiplier ||         |    +-----------+
|                   |   +-----------+||         +--->| Exponent  |
|                   |        |      ||              | Compare & |
|                   |        v      ||              | Alignment |
|                   |   [15:0] prod ||              +-----------+
|                   |        |      ||                    |
|                   |        v      ||                    v
|                   |   Extend to   ++----> exp_diff --> Shift smaller
|                   |   48-bit      |                     operand
|                   |        |      |                       |
|   i_c[31:0] ------+--------+------+                       |
|   (FP32)          |        |                              |
|                   |        v                              v
|                   |   +-----------------------------------------------+
|                   |   |          48-bit Han-Carlson Adder              |
|                   |   |  (Structural, handles add/subtract)           |
|                   |   +-----------------------------------------------+
|                   |                       |
|                   |                       v
|                   |   +-----------------------------------------------+
|                   |   |     Absolute Value (conditional negate)       |
```

```
|
                    |   |         (48-bit Han-Carlson)            |
|
                    |   +--------------------------------------------+
|
                    |                       |
|
                    |                       v
|
                    |   +--------------------------------------------+
|
                    |   |    Count Leading Zeros (Normalization)    |
|
                    |   |          (MSB-down CLZ scan)              |
|
                    |   +--------------------------------------------+
|
                    |                       |
|
                    |                       v
|
                    |   +--------------------------------------------+
|
                    |   |    Round-to-Nearest-Even (RNE)            |
|
                    |   |    Guard, Round, Sticky extraction        |
|
                    |   +--------------------------------------------+
|
                    |                       |
|
                    |                       v
|
                    |   +--------------------------------------------+
|
                    |   |        Special Case Priority MUX          |
|
                    |   |   NaN > Inf > Zero > Overflow > Normal    |
|
                    |   +--------------------------------------------+
|
                    |                       |
|
    ow_result[31:0] <-----------------------+
|
    ow_overflow
<---------------------------------------------------------------|
    ow_underflow
```

```
<-------------------------------------------------------------|
    ow_invalid
<---------------------------------------------------------------|

+-----------------------------------------------------------+
```

13.4.3 Processing Pipeline

1. **Field Extraction** - Parse sign, exponent, mantissa from BF16 and FP32 inputs
2. **Special Case Detection** - Identify zero, subnormal, infinity, NaN for all inputs
3. **Product Computation** - 8x8 Dadda multiply for mantissas, exponent addition with bias correction
4. **Alignment** - Compare exponents, shift smaller operand to align
5. **Wide Addition** - 48-bit Han-Carlson add/subtract based on effective sign
6. **Absolute Value** - Conditional negate if subtraction result is negative
7. **Normalization** - Count leading zeros, shift left, adjust exponent
8. **RNE Rounding** - Extract guard/round/sticky, apply rounding
9. **Result Assembly** - Select output based on special case priority

13.4.4 Mixed Precision Computation

```
// BF16 product extended to 48-bit internal precision
wire [7:0] w_mant_a_ext = {w_a_is_normal, w_mant_a};  // 8-bit with
implied 1
wire [7:0] w_mant_b_ext = {w_b_is_normal, w_mant_b};  // 8-bit with
implied 1

// 8x8 -> 16-bit product, then extended to 48 bits
wire [15:0] w_prod_mant_raw;
wire [47:0] w_prod_mant_48 = {w_prod_mant_ext, 24'b0};

// FP32 addend extended to 48 bits
wire [23:0] w_c_mant_ext = w_c_is_normal ? {1'b1, w_mant_c} : 24'h0;
wire [47:0] w_c_mant_48  = {w_c_mant_ext, 24'b0};
```

13.4.5 Exponent Alignment

```
// Signed exponent difference for alignment
wire signed [10:0] w_exp_diff = $signed({1'b0, w_prod_exp}) -
                                $signed({1'b0, w_exp_c_ext});

// Shift smaller operand to align mantissas
wire [5:0] w_shift_clamped = (w_shift_amt > 10'd48) ? 6'd48 :
```

```
w_shift_amt[5:0];
wire [47:0] w_mant_smaller_shifted = ... >> w_shift_clamped;
```

13.4.6 Effective Addition/Subtraction

```
// Effective operation based on signs
wire w_effective_sub = w_sign_larger ^ w_sign_smaller;

// Two's complement for subtraction
wire [47:0] w_adder_b = w_effective_sub ? ~w_mant_smaller_shifted
                                        : w_mant_smaller_shifted;

// 48-bit structural addition
math_adder_han_carlson_048 u_wide_adder (
    .i_a(w_mant_larger),
    .i_b(w_adder_b),
    .i_cin(w_effective_sub),  // +1 for two's complement
    .ow_sum(w_adder_sum),
    .ow_cout(w_adder_cout)
);
```

13.4.7 Normalization with CLZ

```
// count_leading_zeros scans from the MSB down, which is exactly what
normalization
// needs, so w_sum_abs is connected directly - no bit reversal.
count_leading_zeros #(.WIDTH(48)) u_clz (
    .data(w_sum_abs),
    .clz(w_lz_count_raw)
);

// Normalize: shift left, adjust exponent
wire [47:0] w_mant_normalized = w_sum_abs << w_lz_count;
wire signed [10:0] w_exp_adjusted = w_exp_result_pre - w_lz_count_raw
+ w_add_overflow;
```

13.4.8 Special Case Priority

```
always_comb begin
    // Priority order (highest to lowest):
    if (w_any_nan | w_invalid) begin
        // 1. NaN: any NaN input, or 0*inf, or inf-inf
        ow_result = {1'b0, 8'hFF, 23'h400000};  // Canonical qNaN
        ow_invalid = w_invalid;
    end else if (w_prod_is_inf & ~w_c_is_inf) begin
        // 2. Product infinity
        ow_result = {w_prod_sign, 8'hFF, 23'h0};
    end else if (w_c_is_inf) begin
        // 3. Addend infinity
```

```
        ow_result = {w_sign_c, 8'hFF, 23'h0};
    end else if (w_prod_is_zero & w_c_eff_zero) begin
        // 4. 0 * 0 + 0 = 0 (sign from IEEE rules: & of the two signs)
        ow_result = {w_prod_sign & w_sign_c, 8'h00, 23'h0};
    end else if (w_prod_is_zero) begin
        // 5. Zero product: pass-through addend
        ow_result = i_c;
    end else if (w_c_eff_zero) begin
        // 6. Zero addend: product only. Product over/underflow MUST
be checked
        //    here -- the general checks below act on the addition
result, not
        //    the raw product exponent.
        if (w_prod_exp > 10'd254) begin
            ow_result = {w_prod_sign, 8'hFF, 23'h0};  ow_overflow =
1'b1;
        end else if (w_prod_underflow) begin
            ow_result = {w_prod_sign, 8'h00, 23'h0};  ow_underflow =
1'b1;
        end else begin
            ow_result = {w_prod_sign, w_prod_exp[7:0],
w_prod_mant_ext[22:0]};
        end
    end else if (w_sum_abs == 48'h0) begin
        // 7. Exact zero result -> +0 (RNE)
        ow_result = 32'h0;
    end else if (w_overflow_cond) begin
        // 8. Overflow to infinity
        ow_result = {w_result_sign, 8'hFF, 23'h0};
        ow_overflow = 1'b1;
    end else if (w_underflow_cond) begin
        // 9. Underflow to zero
        ow_result = {w_result_sign, 8'h00, 23'h0};
        ow_underflow = 1'b1;
    end
    // Default: normal result
end
```

13.5 Timing Characteristics

Stage	Logic Depth
Field extraction	0 (wiring)
Special case detection	2 gates
8x8 Dadda multiply	~15 gates
Exponent compare/align	~5 gates

Stage	Logic Depth
48-bit Han-Carlson add	~8 stages
Absolute value (48-bit add)	~8 stages
CLZ + normalization shift	~6 gates
RNE rounding	~3 gates
Result MUX	~3 gates
Total	~40-50 gates

13.5.1 Resource Utilization

Component	LUTs (est.)
8x8 Dadda multiplier	~140
48-bit Han-Carlson adder (main)	~250
48-bit Han-Carlson adder (negate)	~250
CLZ (48-bit)	~50
Alignment shifter	~100
Rounding logic	~30
Special case MUX	~50

13.6 Usage Examples

13.6.1 Basic FMA Operation

```
logic [15:0] a, b;       // BF16 inputs
logic [31:0] c;          // FP32 accumulator
logic [31:0] result;
logic overflow, underflow, invalid;

math_bf16_fma u_fma (
    .i_a(a),
    .i_b(b),
    .i_c(c),
    .ow_result(result),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid)
);

// Example: 2.0 * 3.0 + 1.0 = 7.0
```

```
// 2.0 in BF16: 0x4000
// 3.0 in BF16: 0x4040
// 1.0 in FP32: 0x3F800000
initial begin
    a = 16'h4000;           // 2.0
    b = 16'h4040;           // 3.0
    c = 32'h3F800000;       // 1.0
    #1;
    // result should be ~7.0 (0x40E00000)
end
```

13.6.2 Neural Network Dot Product

```
// Typical AI training pattern: accumulate weight * activation
products
logic [15:0] weights[0:N-1];      // BF16 weights
logic [15:0] activations[0:N-1]; // BF16 activations
logic [31:0] accumulator;         // FP32 accumulator
logic [31:0] fma_result;

math_bf16_fma u_fma (
    .i_a(weights[idx]),
    .i_b(activations[idx]),
    .i_c(accumulator),
    .ow_result(fma_result),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid)
);

// Accumulation loop
always_ff @(posedge clk) begin
    if (start)
        accumulator <= 32'h0;  // Initialize to zero
    else if (valid)
        accumulator <= fma_result;  // Accumulate
end
```

13.6.3 With Pipeline Register

The FMA is purely combinational, so pipeline it as needed for timing:

```
// FMA is purely combinational - pipeline as needed for timing
logic [15:0] a_reg, b_reg;
logic [31:0] c_reg, result_reg;

always_ff @(posedge clk) begin
    // Input register stage
    a_reg <= a;
```

```
    b_reg <= b;
    c_reg <= c;
end

math_bf16_fma u_fma (
    .i_a(a_reg),
    .i_b(b_reg),
    .i_c(c_reg),
    .ow_result(result_comb),
    ...
);

always_ff @(posedge clk) begin
    // Output register stage
    result_reg <= result_comb;
end
```

13.7 Design Notes

13.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Efficient submodule reuse (Dadda multiplier, Han-Carlson adders) 2. **Wire complexity** - Clean datapath with aligned internal widths 3. **Logic depth** - Parallel operations where possible

13.7.2 Why FP32 Accumulator?

AI training requires maintaining precision during gradient accumulation: - BF16 has only 7 mantissa bits (8-bit precision) - Accumulating thousands of small values loses precision - FP32 accumulator provides 23 mantissa bits - Result can be truncated back to BF16 for storage if needed

13.7.3 Fusion Benefits

A fused operation is more accurate than separate multiply-then-add: - **Unfused:** `temp = A * B` (rounded), `result = temp + C` (rounded) - 2 rounding errors - **Fused:** `result = A * B + C` (single rounding) - 1 rounding error - Extra precision is maintained throughout the pipeline

13.7.4 Product Underflow Handling

When the BF16 product exponent is too small:

```
wire w_prod_underflow = w_prod_exp_adj[10] | (w_prod_exp_adj <
11'sd1);
wire w_prod_is_zero = w_a_eff_zero | w_b_eff_zero | w_prod_underflow;
```

Product underflow causes pass-through of the addend C.

13.7.5 Invalid Operation Cases

Invalid operations produce NaN: 1. **0 * Infinity** - Indeterminate form 2. **Infinity - Infinity** - Indeterminate form (same magnitude, opposite signs)

```
wire w_invalid_mul = (w_a_eff_zero & w_b_is_inf) | (w_b_eff_zero &
w_a_is_inf);
wire w_invalid_add = w_prod_is_inf & w_c_is_inf & (w_prod_sign !=
w_sign_c);
```

13.7.6 Common Pitfalls

Anti-Pattern 1: Using BF16 accumulator

```
// WRONG: Precision loss in accumulation
logic [15:0] bf16_accum;  // Only 7 mantissa bits!

// RIGHT: Use FP32 accumulator
logic [31:0] fp32_accum;  // 23 mantissa bits
```

Anti-Pattern 2: Ignoring special cases

```
// WRONG: Assuming result is always valid
output_reg <= result;

// RIGHT: Check status flags -- noting that 'invalid' means 0*inf or
inf-inf
// only; NaN INPUTS produce a qNaN result with invalid = 0, so test
the
// result for NaN separately if that matters to you
if (invalid)
    handle_invalid_op();
else if (overflow)
    handle_infinity();
else
    output_reg <= result;
```

Anti-Pattern 3: Expecting subnormal outputs

```
// NOTE: FTZ mode flushes tiny results to zero
// If gradients become very small, they flush to zero
// This is intentional for AI training efficiency
```

13.7.7 Auto-Generated Code

This module is auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/bf16/bf16_fma.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/bf16/generate_all.py rtl/math`

Do not edit the generated .sv file manually.

13.8 Related Modules

- **math_multiplier_dadda_4to2_008** - 8x8 Dadda multiplier used for mantissa
- **math_adder_han_carlson_048** - 48-bit adder for wide addition
- **count_leading_zeros** - Normalization support
- **math_bf16_multiplier** - Standalone BF16 multiplier (simpler, no accumulate)
- **math_bf16_mantissa_mult** - Mantissa multiplication submodule
- **math_bf16_exponent_adder** - Exponent computation submodule

13.9 Testing

Covered by 2 test suites:

- `val/math/test_math_bf16_fma.py`
- `val/math/test_math_bf16_fma_systematic.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

13.10 References

- Google Brain Float (BF16) specification
- IEEE 754-2019 Standard for Floating-Point Arithmetic
- Intel BFloat16 documentation
- NVIDIA TensorFloat documentation
- Muller, J.M. et al. "Handbook of Floating-Point Arithmetic." Birkhauser, 2018.

13.11 Navigation

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14 BF16 Mantissa Multiplier

A mantissa multiplier specialized for BF16 (Brain Float 16): it handles the implied leading 1, detects when normalization is needed, and extracts the rounding bits for IEEE 754-compliant floating-point multiplication.

14.1 Overview

The `math_bf16_mantissa_mult` module wraps an 8x8 Dadda multiplier with the BF16-specific logic — implicit leading-bit handling, normalization detection, and guard/round/sticky extraction for Round-to-Nearest-Even (RNE) rounding.

Key Features: - **8x8 unsigned multiply** - 7-bit mantissa + 1 implied bit - **Normalization detection** - Flags when product >= 2.0 - **Mantissa extraction** - Selects correct 7 bits based on normalization - **Rounding support** - Provides round and sticky bits for RNE - **FTZ mode** - Handles subnormal inputs as zero

14.2 Parameters

None — the module is fixed at the BF16 mantissa width.

14.3 Ports

```
module math_bf16_mantissa_mult(
    input  logic [6:0]  i_mant_a,      // 7-bit explicit mantissa A
    input  logic [6:0]  i_mant_b,      // 7-bit explicit mantissa B
    input  logic        i_a_is_normal, // A has implied leading 1
    input  logic        i_b_is_normal, // B has implied leading 1
    output logic [15:0] ow_product,    // 16-bit raw product
    output logic        ow_needs_norm, // 1 if product >= 2.0
    output logic [6:0]  ow_mant_out,   // 7-bit result mantissa
    output logic        ow_guard_bit,  // Guard bit (first bit below
mantissa)
    output logic        ow_round_bit,  // Round bit for RNE
    output logic        ow_sticky_bit  // Sticky bit for RNE
);
```

Port	Direction	Width	Description
i_mant_a	Input	7	Explicit mantissa of operand A (bits [6:0] of BF16)
i_mant_b	Input	7	Explicit mantissa of operand B (bits [6:0] of BF16)
i_a_is_normal	Input	1	1 if A is normalized (has implied leading 1)
i_b_is_normal	Input	1	1 if B is normalized (has implied leading 1)
ow_produc	Output	16	Full 16-bit raw product

Port	Direction	Width	Description
t			from 8x8 multiply
ow_needs_ norm	Output	1	1 if product >= 2.0 (MSB of product is 1)
ow_mant_ out	Output	7	Pre-rounded 7-bit mantissa result
ow_guard_ bit	Output	1	Guard bit (first bit below the kept mantissa)
ow_round_ bit	Output	1	Round bit for RNE rounding
ow_sticky_ bit	Output	1	Sticky bit for RNE rounding

14.4 Functional Description

14.4.1 BF16 Structure

```
BF16: [15]=sign, [14:7]=exponent (8 bits), [6:0]=mantissa (7 bits)

For normalized numbers (exp != 0 and exp != 255):
  Value = (-1)^sign * 2^(exp-127) * 1.mantissa

The "1." is implicit (not stored), so actual mantissa is 8 bits: {1,
mantissa[6:0]}
```

14.4.2 Mantissa Multiplication

When multiplying two normalized BF16 numbers: - Operand A mantissa: `1.aaaaaaa` (8 bits with implicit 1) - Operand B mantissa: `1.bbbbbbb` (8 bits with implicit 1) - Product: 16-bit result in range [1.0, 4.0)

Product format: - If product >= 2.0: `1x.xxxxxx_xxxxxxxx` (needs right shift + exp++) - If product < 2.0: `01.xxxxxx_xxxxxxxx` (no shift needed)

14.4.3 Mantissa Extension

```
// Extend 7-bit explicit mantissa with implied leading 1
wire [7:0] w_mant_a_ext = {i_a_is_normal, i_mant_a};
wire [7:0] w_mant_b_ext = {i_b_is_normal, i_mant_b};
```

For normal numbers, this creates `1.mantissa`. For zeros/subnormals (FTZ mode), it creates `0.mantissa`, which effectively treats them as zero.

14.4.4 8x8 Multiplication

```
math_multiplier_dadda_4to2_008 u_mult (
    .i_multiplier(w_mant_a_ext),
    .i_multiplicand(w_mant_b_ext),
    .ow_product(ow_product)
);
```

14.4.5 Normalization Detection

```
// MSB set means product >= 2.0, needs right shift
assign ow_needs_norm = ow_product[15];
```

14.4.6 Mantissa Extraction

```
// Extract 7-bit mantissa based on normalization state
assign ow_mant_out = ow_needs_norm ? ow_product[14:8]  // Shifted
right
                                   : ow_product[13:7]; // No shift
```

Product bit mapping:

Product format	Implied 1	Mantissa bits	Guard	Round	Sticky
Needs norm (MSB=1)	[15]	[14:8]	[7]	[6]	[5:0]
No norm (MSB=0)	[14]	[13:7]	[6]	[5]	[4:0]

14.4.7 Rounding Bits for RNE

```
// Guard, round, sticky for RNE rounding
wire w_guard_norm    = ow_product[7];
wire w_guard_nonorm  = ow_product[6];

wire w_round_norm    = ow_product[6];
wire w_round_nonorm  = ow_product[5];

wire w_sticky_norm   = |ow_product[5:0];
wire w_sticky_nonorm = |ow_product[4:0];

assign ow_guard_bit  = ow_needs_norm ? w_guard_norm  : w_guard_nonorm;
assign ow_round_bit  = ow_needs_norm ? w_round_norm  : w_round_nonorm;
assign ow_sticky_bit = ow_needs_norm ? w_sticky_norm :
w_sticky_nonorm;
```

Note: Guard, round, and sticky are exported separately, and the sticky is the TRUE sticky (no guard folded in). The pre-MATH-001 version folded the guard into the sticky, which is harmless under the old `R & (G|S|LSB)` decision but breaks the correct `G & (R|S|LSB)`: with the fold, G=1 forces the parenthesized term to 1, so ties-at-even wrongly round up. See `math_bf16_multiplier` for the rounding decision that consumes these bits.

14.5 Timing Characteristics

Stage	Logic Depth
Mantissa extension	0 (just wiring)
8x8 Dadda multiply	~13-15 gates
Normalization detection	1 gate (wire to MSB)
Mantissa MUX	1 gate
Round/sticky OR	1 gate
Total	~15-17 gates

14.5.1 Resource Utilization

Component	Resource

14.6 Usage Examples

14.6.1 Basic Usage

```
logic [6:0] mant_a, mant_b;
logic       a_normal, b_normal;
logic [15:0] product;
logic        needs_norm;
logic [6:0]  mant_result;
logic        round_bit, sticky_bit;

math_bf16_mantissa_mult u_mant_mult (
    .i_mant_a(mant_a),
    .i_mant_b(mant_b),
    .i_a_is_normal(a_normal),
    .i_b_is_normal(b_normal),
    .ow_product(product),
    .ow_needs_norm(needs_norm),
    .ow_mant_out(mant_result),
    .ow_round_bit(round_bit),
    .ow_sticky_bit(sticky_bit)
);
```

14.6.2 In BF16 Multiplier

```
// Extract fields from BF16 inputs
wire [6:0] w_mant_a = i_a[6:0];
wire [6:0] w_mant_b = i_b[6:0];
wire       w_a_is_normal = (i_a[14:7] != 8'h00) & (i_a[14:7] !=
8'hFF);
wire       w_b_is_normal = (i_b[14:7] != 8'h00) & (i_b[14:7] !=
8'hFF);

// Mantissa multiplication
math_bf16_mantissa_mult u_mant_mult (
    .i_mant_a(w_mant_a),
    .i_mant_b(w_mant_b),
    .i_a_is_normal(w_a_is_normal),
    .i_b_is_normal(w_b_is_normal),
    .ow_product(w_product),
    .ow_needs_norm(w_needs_norm),
    .ow_mant_out(w_mant_mult_out),
    .ow_guard_bit(w_guard_bit),
    .ow_round_bit(w_round_bit),
    .ow_sticky_bit(w_sticky_bit)
);

// Apply RNE rounding: round up iff guard=1 AND (round | sticky | LSB)
wire w_lsb = w_mant_mult_out[0];
wire w_round_up = w_guard_bit & (w_round_bit | w_sticky_bit | w_lsb);
wire [7:0] w_mant_rounded = {1'b0, w_mant_mult_out} + {7'b0,
w_round_up};

// Check for mantissa overflow from rounding
wire w_mant_overflow = w_mant_rounded[7];
wire [6:0] w_mant_final = w_mant_overflow ? 7'h00 :
w_mant_rounded[6:0];
```

14.7 Design Notes

14.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Minimal overhead over raw multiplier 2. **Wire complexity** - Simple MUXing for normalization 3. **Logic depth** - Normalization adds only 1 gate level

14.7.2 Flush-to-Zero (FTZ) Mode

Subnormal inputs (exp=0, mantissa!=0) are treated as zero: - `i_a_is_normal` or `i_b_is_normal` will be 0 - Extended mantissa becomes `{0, mantissa}` instead of `{1, mantissa}` - Product will be effectively zero

This matches standard BF16 behavior in AI accelerators.

14.7.3 Why Separate from Multiplier?

This wrapper is separate from the raw multiplier because: 1. **Reusability** - Raw multiplier can be used elsewhere 2. **Testability** - Each component can be tested independently 3. **Clarity** - BF16-specific logic is isolated

14.7.4 Round-to-Nearest-Even

This module emits `ow_guard_bit`, `ow_round_bit` and `ow_sticky_bit` (the true sticky, unfolded since MATH-001). The consumer (`math_bf16_multiplier`) rounds up on `guard_bit & (round_bit | sticky_bit | lsb)` = `G & (R | S | LSB)` – textbook RNE; see the rounding note in math_bf16_multiplier.md. (History: this module once folded the guard into sticky and the consumer rounded on `R & (G | S | LSB)`; the fold makes ties-at-even round up, which the MATH-001 sweep caught at ~2.4% of random pairs.)

14.7.5 Auto-Generated Code

This module is auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/bf16/bf16_mantissa_mult.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/bf16/generate_all.py rtl/math`

Do not edit the generated .sv file manually.

14.8 Related Modules

- **math_multiplier_dadda_4to2_008** - Underlying 8x8 multiplier
- **math_bf16_exponent_adder** - Companion exponent handling
- **math_bf16_multiplier** - Complete BF16 multiplier using this module
- **math_bf16_fma** - BF16 FMA also uses Dadda multiplier directly

14.9 Testing

From the test suite (`val/math/test_math_bf16_mantissa_mult.py`):

- **Key test scenarios**:
  - This multiplier handles 7-bit mantissa inputs with implied 1 handling,

- producing a 16-bit product with normalization detection and rounding bits.

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

14.10 References

- Google Brain Float (BF16) specification
- IEEE 754-2019 Standard for Floating-Point Arithmetic
- Muller, J.M. et al. "Handbook of Floating-Point Arithmetic." Birkhauser, 2018.

14.11 Navigation

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15 BF16 Multiplier

A complete BF16 (Brain Float 16) multiplier with IEEE 754-compliant special case handling, Round-to-Nearest-Even rounding, and flush-to-zero support for subnormal numbers.

15.1 Overview

The `math_bf16_multiplier` module implements full BF16 multiplication by integrating the mantissa multiplier, exponent adder, and special case handling logic. It follows industry-standard practices used in AI/ML accelerators for efficient 16-bit floating-point operations.

Key Features: - **BF16 format** - Same exponent range as FP32, reduced mantissa precision - **IEEE 754 special cases** - Zero, infinity, NaN handling - **Rounding** - textbook round-to-nearest-even, `G & (R | S | LSB)` (fixed in MATH-001) - **FTZ mode** - Flush-to-Zero for subnormal inputs and outputs - **Status flags** - Overflow, underflow, and invalid operation indicators

15.2 Parameters

None — the module is fixed at the BF16 format.

15.3 Ports

```
module math_bf16_multiplier(
    input  logic [15:0] i_a,           // BF16 operand A
    input  logic [15:0] i_b,           // BF16 operand B
```

```
    output logic [15:0] ow_result,      // BF16 product
    output logic        ow_overflow,    // Overflow to infinity
    output logic        ow_underflow,   // Underflow to zero (post-round detection)
    output logic        ow_invalid      // Invalid operation (0 * inf = NaN)
);
```

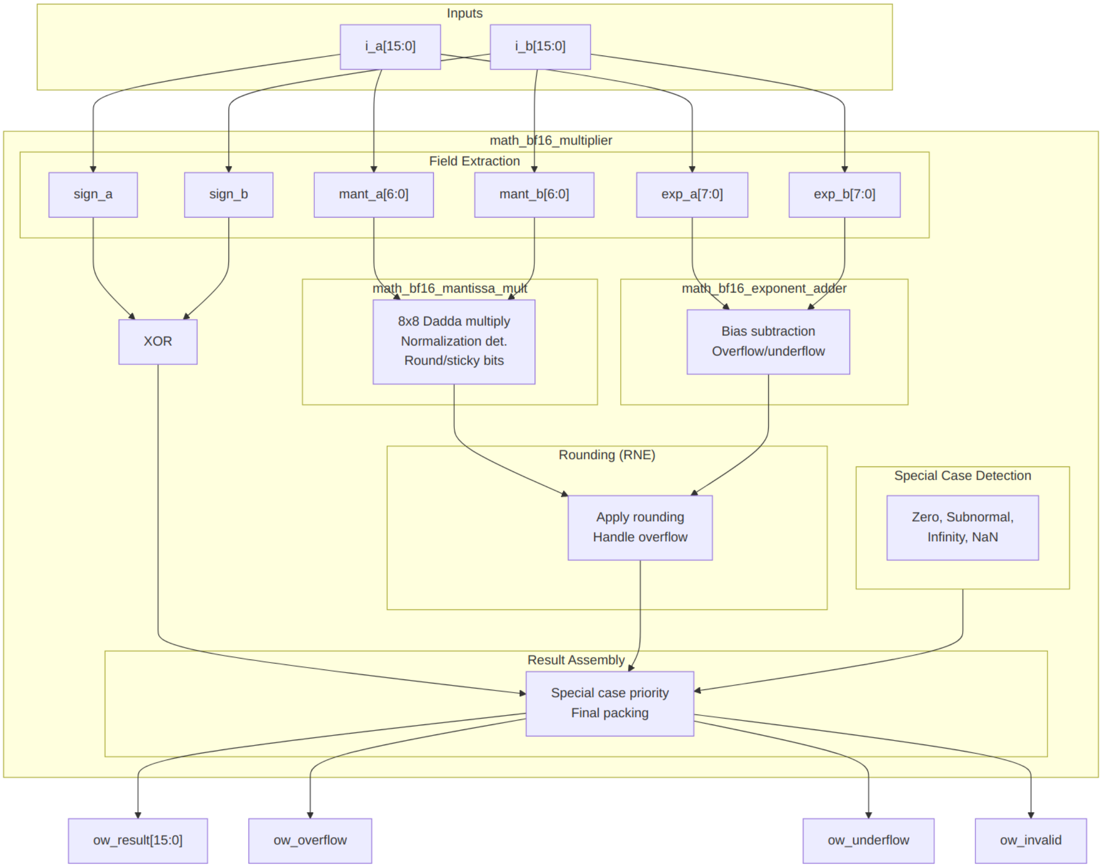
Port	Direction	Width	Description
i_a	Input	16	BF16 multiplicand A
i_b	Input	16	BF16 multiplier B
ow_result	Output	16	BF16 product
ow_overflow	Output	1	1 if result overflowed to infinity
ow_underflow	Output	1	1 if result underflowed to zero
ow_invalid	Output	1	1 if operation was invalid (0 * inf)

15.4 Functional Description

15.4.1 BF16 Format

```
BF16: [15]=sign, [14:7]=exponent (8 bits, bias=127), [6:0]=mantissa (7 bits)

Special values:
  +0:    0x0000 (sign=0, exp=0, mant=0)
  -0:    0x8000 (sign=1, exp=0, mant=0)
  +inf:  0x7F80 (sign=0, exp=255, mant=0)
  -inf:  0xFF80 (sign=1, exp=255, mant=0)
  NaN:   0x7FC0 (exp=255, mant!=0, canonical quiet NaN)
```



Architecture

15.4.2 Processing Pipeline

1. **Field Extraction** - Parse sign, exponent, mantissa from inputs
2. **Special Case Detection** - Identify zero, subnormal, infinity, NaN
3. **Sign Computation** - XOR of input signs
4. **Mantissa Multiplication** - 8x8 Dadda tree with normalization
5. **Exponent Addition** - Add exponents, subtract bias, adjust for normalization
6. **Rounding** - Apply round-to-nearest-even (`G & (R | S | LSB)`) to the mantissa
7. **Result Assembly** - Select output based on special case priority

15.4.3 Special Case Detection

```
// Zero: exp=0, mant=0
wire w_a_is_zero = (w_exp_a == 8'h00) & (w_mant_a == 7'h00);
```

```
// Subnormal: exp=0, mant!=0 (treated as zero in FTZ mode)
wire w_a_is_subnormal = (w_exp_a == 8'h00) & (w_mant_a != 7'h00);

// Infinity: exp=FF, mant=0
wire w_a_is_inf = (w_exp_a == 8'hFF) & (w_mant_a == 7'h00);

// NaN: exp=FF, mant!=0
wire w_a_is_nan = (w_exp_a == 8'hFF) & (w_mant_a != 7'h00);

// Effective zero (includes subnormals)
wire w_a_eff_zero = w_a_is_zero | w_a_is_subnormal;

// Normal number (has implied leading 1)
wire w_a_is_normal = ~w_a_eff_zero & ~w_a_is_inf & ~w_a_is_nan;
```

15.4.4 Round-to-Nearest-Even

> **Rounding boolean, as implemented (fixed 2026-08-09, MATH-001):** `w_round_up = w_guard_bit & (w_round_bit | w_sticky_bit | w_lsb)` – textbook RNE, verified by sweep: 0 mismatches in 5000 random pairs against an exact behavioral reference. This page previously documented the pre-fix form (`R & (G | S | LSB)`, with the guard folded into sticky), which diverged from RNE in 6 of 16 guard patterns – five ordinary inexact cases plus the exact-half tie. The fold is gone: `math_bf16_mantissa_mult` now exports the guard bit and the true sticky separately.

```
// Textbook RNE: round up iff guard=1 AND (round OR sticky OR lsb)
wire w_lsb = w_mant_mult_out[0];
wire w_round_up = w_guard_bit & (w_round_bit | w_sticky_bit | w_lsb);

// Apply rounding
wire [7:0] w_mant_rounded = {1'b0, w_mant_mult_out} + {7'b0,
w_round_up};

// Handle mantissa overflow from rounding (1.1111111 + 1 = 10.0000000)
wire w_mant_round_overflow = w_mant_rounded[7];
wire [6:0] w_mant_final = w_mant_round_overflow ? 7'h00 :
w_mant_rounded[6:0];

// Adjust exponent for rounding overflow
wire [7:0] w_exp_final = w_mant_round_overflow ? (w_exp_sum + 8'd1) :
w_exp_sum;
```

15.4.5 Special Case Priority

```
always_comb begin
    // Default: normal result
    ow_result = {w_sign_result, w_exp_final, w_mant_final};
    ow_overflow = 1'b0;
    ow_underflow = 1'b0;
    ow_invalid = 1'b0;

    // Priority order (highest to lowest):
    if (w_any_nan | w_invalid_op) begin
        // 1. NaN: any NaN input, or 0 * inf
        ow_result = {w_sign_result, 8'hFF, 7'h40};  // Canonical qNaN
        ow_invalid = w_invalid_op;
    end else if (w_result_zero) begin
        // 2. Zero: checked BEFORE overflow. A zero input makes the exponent
        //    adder emit a garbage value (e.g. 0xFF) that would otherwise
        //    falsely trigger the overflow path.
        ow_result = {w_sign_result, 8'h00, 7'h00};
    end else if (w_result_inf | w_final_overflow) begin
        // 3. Infinity: inf input or overflow
        ow_result = {w_sign_result, 8'hFF, 7'h00};
        ow_overflow = w_final_overflow & ~w_result_inf;
    end else if (w_exp_underflow & ~w_uf_rescued) begin
        // 4. Underflow to zero -- detected AFTER rounding, per IEEE 754.
        //    w_uf_rescued covers the one edge this distinction changes: a
        //    pre-round exponent sum of exactly 0 whose mantissa rounds with
        //    carry is exactly the minimum normal (0x0080) and falls through
        //    to the default assignment instead of flushing (MATH-008).
        ow_result = {w_sign_result, 8'h00, 7'h00};
        ow_underflow = 1'b1;
    end
end
```

15.5 Timing Characteristics

Stage	Logic Depth
Field extraction	0 (wiring)
Special case detection	2 gates
Mantissa multiply	~15 gates

Stage	Logic Depth
Exponent add	~5 gates
Rounding	~3 gates
Result MUX	~2 gates
Total	~25-30 gates

15.5.1 Resource Utilization

Component	LUTs (est.)
Special case detection	~30
Mantissa multiplier	~160
Exponent adder	~40
Rounding logic	~15
Result MUX	~25

15.6 Usage Examples

15.6.1 Basic Multiplication

```
logic [15:0] a, b, product;
logic overflow, underflow, invalid;

math_bf16_multiplier u_mult (
    .i_a(a),
    .i_b(b),
    .ow_result(product),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid)
);

// Example: 2.0 * 3.0 = 6.0
// 2.0 in BF16: 0x4000 (sign=0, exp=128, mant=0)
// 3.0 in BF16: 0x4040 (sign=0, exp=128, mant=0x40)
initial begin
    a = 16'h4000;  // 2.0
    b = 16'h4040;  // 3.0
    #1;
    // product should be ~6.0 (0x40C0)
end
```

15.6.2 With Status Checking

```
always_ff @(posedge clk) begin
    if (overflow)
        $display("Overflow: result is infinity");
    else if (underflow)
        $display("Underflow: result is zero");
    else if (invalid)
        $display("Invalid: 0 * inf (NaN inputs give qNaN but do NOT assert invalid)");
end
```

15.6.3 In Neural Network Layer

```
// Multiply weight by activation
logic [15:0] weight, activation, product;
logic [15:0] accumulator;  // Would typically be FP32 for accuracy

math_bf16_multiplier u_mult (
    .i_a(weight),
    .i_b(activation),
    .ow_result(product),
    .ow_overflow(),
    .ow_underflow(),
    .ow_invalid()
);

// Accumulate (simplified - real implementations use FP32 accumulator)
always_ff @(posedge clk) begin
    if (start)
        accumulator <= product;
    else
        accumulator <= accumulator + product;  // Would need FP adder
end
```

15.7 Design Notes

15.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Efficient submodule integration 2. **Wire complexity** - Clean datapath structure 3. **Logic depth** - Parallel special case and normal path computation

15.7.2 Flush-to-Zero (FTZ)

Subnormal numbers are flushed to zero: - **Input subnormals** - Treated as zero (effective zero) - **Output subnormals** - Not generated (result goes to zero)

This simplifies hardware and matches AI accelerator conventions.

15.7.3 NaN Handling

- **Input NaN** - Propagated to output as canonical quiet NaN
- **0 * Infinity** - Produces NaN with invalid flag set
- **Canonical qNaN** - exp=FF, mant=0x40. The sign is the product sign (`w_sign_result`, the XOR of the input signs), NOT forced to 0 – a negative-product NaN is 0xFFC0, not 0x7FC0.

15.7.4 Sign of Zero

The sign of zero follows IEEE 754: - **+0 * +0** = +0 - **+0 * -0** = -0 - **-0 * -0** = +0

15.7.5 Rounding Mode

Only Round-to-Nearest-Even (RNE) is supported. This is standard for BF16 in AI applications.

15.7.6 Common Pitfalls

Anti-Pattern 1: Ignoring status flags

```
// WRONG: Assuming result is always valid
result <= product;  // May be NaN or infinity!

// RIGHT: Check flags
if (!invalid && !overflow)
    result <= product;
else
    handle_exception();
```

Anti-Pattern 2: Expecting subnormal results

```
// NOTE: Very small results flush to zero, not subnormal
// If you need gradual underflow, use FP32 or a different
implementation
```

15.7.7 Auto-Generated Code

This module is auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/bf16/bf16_multiplier.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/bf16/generate_all.py rtl/math`

Do not edit the generated .sv file manually.

15.8 Related Modules

- **math_bf16_mantissa_mult** - Mantissa multiplication submodule
- **math_bf16_exponent_adder** - Exponent computation submodule

- **math_bf16_fma** - Fused Multiply-Add with FP32 accumulator
- **math_multiplier_dadda_4to2_008** - Underlying 8x8 multiplier

15.9 Testing

From the test suite (`val/math/test_math_bf16_multiplier.py`):

- **Key test scenarios**:
  - Tests the complete BF16 multiplier including:
  - Normal multiplication with RNE rounding
  - Special value handling (zero, infinity, NaN, subnormal)
  - Overflow and underflow detection
  - Invalid operation detection (0 * inf)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

15.10 References

- Google Brain Float (BF16) specification
- IEEE 754-2019 Standard for Floating-Point Arithmetic
- Intel BFloat16 documentation
- NVIDIA TensorFloat documentation

15.11 Navigation

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16 4:2 Compressor

A 4:2 compressor (also called a 5:3 counter) that reduces 5 input bits to 3 output bits, giving you faster column height reduction than traditional 3:2 compressors in multiplier trees.

16.1 Overview

The `math_compressor_4to2` module implements a 4:2 compressor using two cascaded full adders. It accepts 4 primary inputs plus a carry-in, producing a sum, carry, and carry-out. The property you're really buying is that the carry-out (`ow_cout`) is independent of the carry-in (`i_cin`) — that's what enables efficient chaining within multiplier reduction trees.

Key Features: - **5 inputs to 3 outputs** - More efficient column reduction than 3:2 compressors - **Cout independent of Cin** - Enables parallel carry chains - **2 full adder depth** - Predictable timing - **Building block** for Dadda/Wallace trees with 4:2 compressors

16.2 Parameters

None — this is a single-bit cell with nothing to configure.

16.3 Ports

```
module math_compressor_4to2 (
    input  logic i_x1, i_x2, i_x3, i_x4,
    input  logic i_cin,
    output logic ow_sum,
    output logic ow_carry,
    output logic ow_cout
);
```

Port	Direction	Width	Description
i_x1	Input	1	First input bit
i_x2	Input	1	Second input bit
i_x3	Input	1	Third input bit
i_x4	Input	1	Fourth input bit
i_cin	Input	1	Carry input (from previous compressor)
ow_sum	Output	1	Sum output (same column weight)
ow_carry	Output	1	Carry output (weight = column + 1)
ow_cout	Output	1	Carry-out (weight = column + 1, independent of Cin)

16.4 Functional Description

16.4.1 Architecture

The 4:2 compressor is implemented as two cascaded full adders:

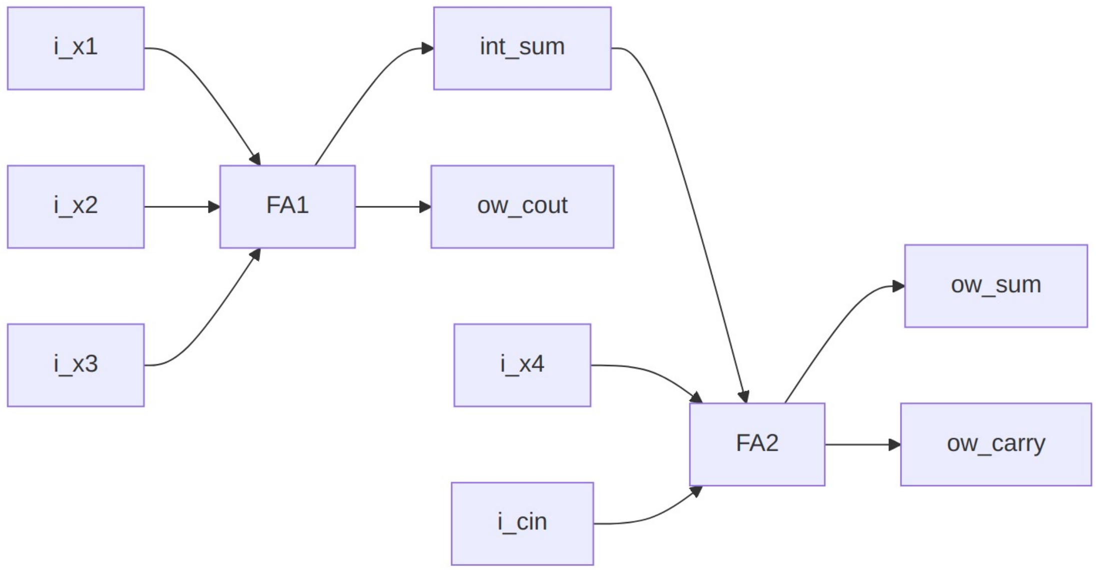

Diagram 5

Stage 1 (FA1): Compresses X1, X2, X3 - Sum goes to Stage 2 - Carry becomes `ow_cout` (independent of i_cin!)

Stage 2 (FA2): Compresses int_sum, X4, Cin - Sum becomes `ow_sum` - Carry becomes `ow_carry`

16.4.2 Truth Table (simplified)

The compressor performs: `i_x1 + i_x2 + i_x3 + i_x4 + i_cin = ow_sum + 2*(ow_carry + ow_cout)`

Sum of inputs (0-5)	ow_cout	ow_carry	ow_sum
0	0	0	0
1	0	0	1
2	0/1	0/1	0
3	0/1	0/1	1
4	1	1	0
5	1	1	1

Note: Multiple input combinations can produce the same sum, with different cout/carry distributions.

16.4.3 Key Property: Cout Independence

The critical property of this implementation is that `ow_cout` depends only on X1, X2, X3 - not on i_cin:

```
// First stage: cout is independent of cin
ow_cout = (i_x1 & i_x2) | (i_x2 & i_x3) | (i_x1 & i_x3);
```

This allows multiple 4:2 compressors to be chained with their cout signals forming independent carry chains, reducing the critical path compared to implementations where all carries depend on cin.

16.5 Timing Characteristics

Metric	Value	Description
Logic Depth	4 gates	2 full adders in series
Cin to Sum	2 XOR gates	Through FA2 only
Cin to Carry	1 XOR + 1 AND-OR	Through FA2 only
X1-X3 to Cout	1 AND-OR	Through FA1 only

Critical Path: X1/X2/X3 -> FA1 sum -> FA2 sum = 4 XOR gate delays

16.5.1 Resource Utilization

Metric	Value
Full Adders	2
Total Gates	~10-12 (technology dependent)
LUTs (FPGA)	~3-4

16.5.2 Comparison with 3:2 Compressor (Full Adder)

Metric	3:2 Compressor	4:2 Compressor
Inputs	3	5
Outputs	2	3
Reduction ratio	3:2	5:3
Gate count	~5	~10-12
Logic depth	2	4
Column reduction	-1 per stage	-2 per stage

Advantage: 4:2 compressors reduce column height faster, resulting in fewer reduction stages for multipliers.

16.6 Usage Examples

16.6.1 Single Compressor

```
logic x1, x2, x3, x4, cin;
logic sum, carry, cout;

math_compressor_4to2 u_comp (
    .i_x1(x1), .i_x2(x2), .i_x3(x3), .i_x4(x4),
    .i_cin(cin),
    .ow_sum(sum),
    .ow_carry(carry),
    .ow_cout(cout)
);
```

16.6.2 Chained Compressors (Multiplier Column)

```
// Reduce 8 bits in a column to 4 using two chained 4:2 compressors
logic [7:0] col_bits;
logic sum1, carry1, cout1;
logic sum2, carry2, cout2;

// First compressor: bits [3:0] + carry from previous column
math_compressor_4to2 u_comp1 (
    .i_x1(col_bits[0]), .i_x2(col_bits[1]),
    .i_x3(col_bits[2]), .i_x4(col_bits[3]),
    .i_cin(cin_from_prev),
    .ow_sum(sum1),
    .ow_carry(carry1),
    .ow_cout(cout1)
);

// Second compressor: bits [7:4], cin tied low.
// cout1 is a NEXT-COLUMN carry (weight 2, independent of cin per the
port
// table): feeding it into this column's cin (weight 1) loses a bit of
// column mass -- with col_bits=8'hFF the chained version totals 7,
not 8.
math_compressor_4to2 u_comp2 (
    .i_x1(col_bits[4]), .i_x2(col_bits[5]),
    .i_x3(col_bits[6]), .i_x4(col_bits[7]),
    .i_cin(1'b0),
    .ow_sum(sum2),
    .ow_carry(carry2),
    .ow_cout(cout2)
```

```
);

// Results: sum1, sum2 stay in this column
// carry1, cout1, carry2, cout2 ALL go to the next column (every
compressor
// output except sum has weight 2)
```

16.6.3 In Dadda Tree Multiplier

```
// Part of Dadda reduction stage
// Column has partial products: pp0, pp1, pp2, pp3, pp4
// Plus carry-in from previous column

math_compressor_4to2 u_dadda_comp (
    .i_x1(pp0), .i_x2(pp1), .i_x3(pp2), .i_x4(pp3),
    .i_cin(cin_prev),
    .ow_sum(reduced_sum),       // Stays in column
    .ow_carry(carry_to_next),   // Goes to column+1
    .ow_cout(cout_to_next)      // Goes to column+1
);

// pp4 passes through to next stage if column height allows
```

16.7 Design Notes

16.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Minimal gate count using two full adders 2. **Wire complexity** - Simple cascaded structure with minimal routing 3. **Logic depth** - Fixed 2 full adder delays

16.7.2 Advantages

- **Faster reduction** - Each compressor removes 2 bits from column height (vs 1 for 3:2)
- **Cin independence** - Cout doesn't depend on Cin, enabling parallel carry computation
- **Predictable timing** - Fixed 2 FA delay
- **Standard building block** - Well-understood for multiplier design

16.7.3 Applications

- **Multiplier trees** - Dadda and Wallace trees with 4:2 compressors
- **Multi-operand adders** - Reducing many operands to 2
- **DSP blocks** - Custom multiply-accumulate units

- **BF16/FP16 arithmetic** - Mantissa multiplication

16.8 Related Modules

- **math_adder_full** - Full adder building block (3:2 compressor)
- **math_adder_half** - Half adder building block
- **math_multiplier_dadda_4to2_008** - 8x8 multiplier using this compressor
- **math_adder_carry_save** - Alternative 3:2 compressor naming

16.9 Testing

No dedicated test wrapper – this block is exercised structurally through every Dadda 4:2 multiplier test (`val/math/test_math_multiplier_dadda_4to2.py` and the FP mantissa multiplier suites). It is also formally proved: `formal/common/math_compressor_4to2/` (prove + cover, SymbiYosys).

16.10 References

- Dadda, L. "Some Schemes for Parallel Multipliers." Alta Frequenza 34, 1965.
- Weinberger, A. "4:2 Carry-Save Adder Module." IBM Technical Disclosure Bulletin, 1981.
- Oklobdzija, V.G. et al. "A Method for Speed Optimized Partial Product Reduction and Generation of Fast Parallel Multipliers Using an Algorithmic Approach." IEEE Trans. Computers, 1996.

16.11 Navigation

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17 FP16 (Half Precision) Modules

IEEE 754 half-precision (FP16) floating-point modules for activation functions, comparison, and format conversion.

17.1 Overview

FP16 (16-bit floating-point) trades the exponent/mantissa split the other way from BF16 — more mantissa, less exponent — so you get higher precision over a smaller range. These modules complement the BF16 library for applications requiring greater precision in the fractional range.

FP16 Format:

```
[15]    = Sign bit
[14:10] = Exponent (5 bits, bias = 15)
[9:0]   = Mantissa (10 bits, implicit leading 1)

Range: ~6.0e-8 to ~65504
Precision: ~0.1% (3-4 decimal digits)
```

Key Differences from BF16: | Property | BF16 | FP16 | |———-|——|——| | Exponent bits | 8 | 5 | | Mantissa bits | 7 | 10 | | Dynamic range | Same as FP32 | Limited (~65504 max) | | Precision | ~1% | ~0.1% |

17.2 Functional Description

17.2.1 Activation Functions

Module	Function	Description
math_fp16_relu	max(0, x)	Rectified Linear Unit
math_fp16_leaky_relu	max(0.01x, x)	Leaky ReLU
math_fp16_gelu	x * Phi(x)	Gaussian Error Linear Unit
math_fp16_sigmoid	1/(1+exp(-x))	Logistic function
math_fp16_tanh	tanh(x)	Hyperbolic tangent
math_fp16_silu	x * sigmoid(x)	SiLU/Swish activation
math_fp16_softmax_8	softmax(x[7:0])	8-input softmax

Every SCALAR activation function in this group uses the same two-port pattern:

```
module math_fp16_{activation} (
    input  logic [15:0] i_a,        // FP16 input
    output logic [15:0] ow_result   // FP16 output
);
```

17.2.2 Comparison and Selection

Module	Function	Description
math_fp16_comparator	a <=> b	Full comparison with flags
math_fp16_max	max(a, b)	Two-input maximum

Module	Function	Description
math_fp16_min	min(a, b)	Two-input minimum
math_fp16_max_tree_8	max(a[7:0])	8-input maximum
math_fp16_min_tree_8	min(a[7:0])	8-input minimum
math_fp16_clamp	clamp(x, lo, hi)	Value clamping

17.2.3 Format Conversions

Module	Conversion	Notes
math_fp16_to_bf16	FP16 -> BF16	Mantissa truncation, exponent adjustment
math_fp16_to_fp32	FP16 -> FP32	Lossless upconversion
math_fp16_to_fp8_e4m3	FP16 -> FP8 E4M3	For ML inference
math_fp16_to_fp8_e5m2	FP16 -> FP8 E5M2	For ML training

FP16 to BF16: - Mantissa truncation: 10 bits -> 7 bits (may lose precision) - Exponent mapping: bias 15 -> bias 127 - Cannot overflow: BF16 range (FP32-class) strictly contains FP16 range, so no FP16 value exceeds BF16 max normal

FP16 to FP32: - Lossless conversion (FP32 can represent all FP16 values) - Simple exponent bias adjustment: 15 -> 127 - Mantissa padding: 10 bits -> 23 bits

17.2.4 Special Values

Value	Encoding	Notes
+0	0x0000	Positive zero
-0	0x8000	Negative zero
+Inf	0x7C00	Positive infinity
-Inf	0xFC00	Negative infinity
NaN	0x7E00	Canonical quiet NaN
Max Normal	0x7BFF	~65504
Min Normal	0x0400	~6.1e-5
Subnormal	Exp=0	Flushed to zero (FTZ)

17.3 Usage Examples

17.3.1 Mixed Precision Pipeline

```
// FP16 computation with FP32 accumulator
logic [15:0] fp16_a, fp16_b, fp16_product;
logic [31:0] fp32_a, fp32_b, fp32_acc;

// Convert to FP32 for accumulation
math_fp16_to_fp32 u_cvt_a
(.i_a(fp16_a), .ow_result(fp32_a), .ow_invalid());
math_fp16_to_fp32 u_cvt_b
(.i_a(fp16_b), .ow_result(fp32_b), .ow_invalid());

// Compute in FP32 for precision
// ... FP32 MAC operation ...

// Apply FP16 activation
math_fp16_relu u_relu (.i_a(layer_output), .ow_result(activated));
```

17.3.2 Comparison Operations

```
logic [15:0] fp16_values [8];
logic [15:0] max_value, min_value;

math_fp16_max_tree_8 u_max (
    .i_data(fp16_values),
    .ow_max(max_value)
);

math_fp16_min_tree_8 u_min (
    .i_data(fp16_values),
    .ow_min(min_value)
);
```

17.4 Design Notes

17.4.1 Auto-Generation

```
# Regenerate all FP16 modules
PYTHONPATH=bin:$PYTHONPATH python3
bin/rtl_generators/ieee754/generate_all.py rtl/math
```

17.5 Related Modules

- **math_bf16_extended** - BF16 equivalent modules
- **math_fp32_modules** - FP32 equivalent modules

- **math_fp8_modules** - FP8 variants
- **math_ieee754_modules** - IEEE 754-2008 compliant arithmetic

17.6 Testing

Covered by 14 test suites:

- `val/math/test_math_fp16_clamp.py`
- `val/math/test_math_fp16_comparator.py`
- `val/math/test_math_fp16_gelu.py`
- `val/math/test_math_fp16_leaky_relu.py`
- `val/math/test_math_fp16_max.py`
- `val/math/test_math_fp16_min.py`
- `val/math/test_math_fp16_relu.py`
- `val/math/test_math_fp16_sigmoid.py`
- … and 6 more (`ls val/math/test_math_*.py`)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

17.7 Navigation

- **← Back to Math Index**
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18 FP32 (Single Precision) Modules

IEEE 754 single-precision (FP32) floating-point modules for activation functions, comparison, and format conversion.

18.1 Overview

FP32 (32-bit floating-point) is the full single-precision IEEE 754 representation. These modules cover activation functions, comparisons, and downconversion to the smaller formats.

FP32 Format:

```
[31]    = Sign bit
[30:23] = Exponent (8 bits, bias = 127)
[22:0]  = Mantissa (23 bits, implicit leading 1)

Range: ~1.2e-38 to ~3.4e38
Precision: ~0.00001% (7 decimal digits)
```

18.2 Functional Description

18.2.1 Activation Functions

Module	Function	Description
math_fp32_relu	max(0, x)	Rectified Linear Unit
math_fp32_leaky_relu	max(0.01x, x)	Leaky ReLU
math_fp32_gelu	x * Phi(x)	Gaussian Error Linear Unit
math_fp32_sigmoid	1/(1+exp(-x))	Logistic function
math_fp32_tanh	tanh(x)	Hyperbolic tangent
math_fp32_silu	x * sigmoid(x)	SiLU/Swish activation
math_fp32_softmax_8	softmax(x[7:0])	8-input softmax

Every SCALAR activation function in this group uses the same two-port pattern:

```
module math_fp32_{activation} (
    input  logic [31:0] i_a,        // FP32 input
    output logic [31:0] ow_result   // FP32 output
);
```

18.2.2 Comparison and Selection

Module	Function	Description
math_fp32_comparator	a <=> b	Full comparison with flags
math_fp32_max	max(a, b)	Two-input maximum
math_fp32_min	min(a, b)	Two-input minimum
math_fp32_max_tree_8	max(a[7:0])	8-input maximum
math_fp32_min_tree_8	min(a[7:0])	8-input minimum
math_fp32_clamp	clamp(x, lo, hi)	Value clamping

18.2.3 Format Conversions (Downconversion)

Module	Conversion	Notes
`math_fp32_to_bf16`	FP32 -> BF16	Truncate mantissa (23->7 bits)
`math_fp32_to_fp16`	FP32 -> FP16	Range check + rounding
`math_fp32_to_fp8_e4m3`	FP32 -> FP8 E4M3	ML inference quantization
`math_fp32_to_fp8_e5m2`	FP32 -> FP8 E5M2	ML training quantization

FP32 to BF16:

```
module math_fp32_to_bf16 (
    input  logic [31:0] i_a,
    output logic [15:0] ow_result,
    output logic        ow_overflow,   // If exp > 254 in BF16
    output logic        ow_underflow,  // If exp < 1 (subnormal/zero)
    output logic        ow_invalid     // NaN input
);
```

- Same exponent range (8 bits), just truncate mantissa
- May lose precision in mantissa (23 -> 7 bits)
- RNE rounding applied

FP32 to FP16: - Exponent range reduction: 8 bits -> 5 bits - May overflow (FP32 value > 65504) - May underflow (FP32 value < FP16 min subnormal) - Mantissa rounding (23 -> 10 bits)

18.2.4 Special Values

Value	Encoding	Notes
+0	0x00000000	Positive zero
-0	0x80000000	Negative zero
+Inf	0x7F800000	Positive infinity
-Inf	0xFF800000	Negative infinity
NaN	0x7FC00000	Canonical quiet NaN
Min Normal	0x00800000	~1.2e-38

18.3 Usage Examples

18.3.1 Training Pipeline (FP32 Master)

```
// FP32 forward pass with activation
logic [31:0] layer_input, layer_output, activated;

// FP32 computation (full precision for training)
// ... matrix multiply in FP32 ...

// FP32 activation
math_fp32_relu u_relu (
    .i_a(layer_output),
    .ow_result(activated)
);

// Convert to BF16 for storage/transfer
logic [15:0] bf16_result;
math_fp32_to_bf16 u_cvt (
    .i_a(activated),
    .ow_result(bf16_result),
    .ow_overflow(overflow_flag),
    .ow_underflow(),
    .ow_invalid()
);
```

18.3.2 Quantization Pipeline

```
// Quantize FP32 model weights to FP8 for inference
logic [31:0] fp32_weight;
logic [7:0]  fp8_weight;

math_fp32_to_fp8_e4m3 u_quantize (
    .i_a(fp32_weight),
    .ow_result(fp8_weight),
    .ow_overflow(quant_overflow),
    .ow_underflow(quant_underflow),
    .ow_invalid()
);
```

18.4 Design Notes

18.4.1 Auto-Generation

```
# Regenerate all FP32 modules
PYTHONPATH=bin:$PYTHONPATH python3
bin/rtl_generators/ieee754/generate_all.py rtl/math
```

18.5 Related Modules

- **math_bf16_extended** - BF16 equivalent modules
- **math_fp16_modules** - FP16 equivalent modules
- **math_fp8_modules** - FP8 variants
- **math_ieee754_modules** - IEEE 754-2008 compliant arithmetic

18.6 Testing

Covered by 14 test suites:

- `val/math/test_math_fp32_clamp.py`
- `val/math/test_math_fp32_comparator.py`
- `val/math/test_math_fp32_gelu.py`
- `val/math/test_math_fp32_leaky_relu.py`
- `val/math/test_math_fp32_max.py`
- `val/math/test_math_fp32_min.py`
- `val/math/test_math_fp32_relu.py`
- `val/math/test_math_fp32_sigmoid.py`
- … and 6 more (`ls val/math/test_math_*.py`)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

18.7 Navigation

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19 FP8 (8-bit Floating-Point) Modules

8-bit floating-point modules in both E4M3 and E5M2 formats for ML inference and training acceleration.

19.1 Overview

FP8 is about extreme memory efficiency for neural network accelerators. Two variants are supported: - **E4M3** - Higher precision, smaller range (inference-optimized) - **E5M2** - Lower precision, larger range (training-friendly)

FP8 E4M3 Format:

```
[7]   = Sign bit
[6:3] = Exponent (4 bits, bias = 7)
[2:0] = Mantissa (3 bits)

Range: ~0.001953 to 448 (no infinity!)
Special: exp=15, mant=111 is NaN (not infinity)
Max normal: exp=15, mant=110 = 448
```

FP8 E5M2 Format:

```
[7]   = Sign bit
[6:2] = Exponent (5 bits, bias = 15)
[1:0] = Mantissa (2 bits)

Range: ~6.1e-5 (2^-14, min normal) to 57344 (has infinity). The smallest
subnormal is 2^-16 ~ 1.5e-5; ~6e-8 is the FP16 subnormal minimum, not E5M2's.
Special: exp=31, mant=0 is Inf; mant!=0 is NaN
```

19.2 Functional Description

19.2.1 Format Comparison

Property	E4M3	E5M2	BF16
Total bits	8	8	16
Exponent bits	4	5	8
Mantissa bits	3	2	7
Max value	448	57344	~3.4e38
Min normal	0.0156	6.1e-5	~1.2e-38
Has infinity	No	Yes	Yes
Use case	Inference	Training	General

19.2.2 E4M3 Core Arithmetic

Module	Function	Description
`math_fp8_e4m3_adder`	a + b	FP8 E4M3 addition
`math_fp8_e4m3_multiplier`	a * b	FP8 E4M3 multiplication
`math_fp8_e4m3_fma`	a*b + c	Fused multiply-add
`math_fp8_e4m3_ma`	Internal	Mantissa multiplication helper

Module	Function	Description
ntissa_mult		
math_fp8_e4m3_exponent_adder	Internal	Exponent computation helper

The multiplier interface:

```
module math_fp8_e4m3_multiplier (
    input  logic [7:0] i_a,
    input  logic [7:0] i_b,
    output logic [7:0] ow_result,
    output logic       ow_overflow,    // Saturates to max normal (448)
    output logic       ow_underflow,   // Flushes to zero (post-round
detection)
    output logic       ow_invalid      // NaN operation
);
```

Note: E4M3 has no infinity - overflow saturates to max normal value (0x7E = 448).

19.2.3 E4M3 Activation Functions

Module	Function	Description
math_fp8_e4m3_relu	max(0, x)	ReLU activation
math_fp8_e4m3_leaky_relu	max(0.01x, x)	Leaky ReLU
math_fp8_e4m3_gelu	x * Phi(x)	GELU activation
math_fp8_e4m3_sigmoid	1/(1+exp(-x))	Sigmoid
math_fp8_e4m3_tanh	tanh(x)	Hyperbolic tangent
math_fp8_e4m3_silu	x * sigmoid(x)	SiLU/Swish
math_fp8_e4m3_softmax_8	softmax(x[7:0])	8-input softmax

19.2.4 E4M3 Comparison and Selection

Module	Function	Description
math_fp8_e4m3_comparator	a <=> b	Full comparison
math_fp8_e4m3_max	max(a, b)	Two-input max
math_fp8_e4m3_min	min(a, b)	Two-input min
math_fp8_e4m3_max_tree_8	max(a[7:0])	8-input max
math_fp8_e4m3_min_tree_8	min(a[7:0])	8-input min

Module	Function	Description
math_fp8_e4m3_clamp	clamp(x, lo, hi)	Value clamping

19.2.5 E4M3 Format Conversions

Module	Conversion	Notes
math_fp8_e4m3_to_bf16	E4M3 -> BF16	Lossless upconversion
math_fp8_e4m3_to_fp16	E4M3 -> FP16	Lossless upconversion
math_fp8_e4m3_to_fp32	E4M3 -> FP32	Lossless upconversion
math_fp8_e4m3_to_fp8_e5m2	E4M3 -> E5M2	May lose precision

19.2.6 E5M2 Core Arithmetic

Module	Function	Description
math_fp8_e5m2_adder	a + b	FP8 E5M2 addition
math_fp8_e5m2_multiplier	a * b	FP8 E5M2 multiplication
math_fp8_e5m2_fma	a*b + c	Fused multiply-add
math_fp8_e5m2_mantissa_mult	Internal	Mantissa multiplication helper
math_fp8_e5m2_exponent_adder	Internal	Exponent computation helper

19.2.7 E5M2 Activation Functions

Module	Function	Description
math_fp8_e5m2_relu	max(0, x)	ReLU activation
math_fp8_e5m2_leaky_relu	max(0.01x, x)	Leaky ReLU
math_fp8_e5m2_gelu	x * Phi(x)	GELU activation
math_fp8_e5m2_sigmoid	1/(1+exp(-x))	Sigmoid
math_fp8_e5m2_tanh	tanh(x)	Hyperbolic tangent
math_fp8_e5m2_silu	x * sigmoid(x)	SiLU/Swish

Module	Function	Description
math_fp8_e5m2_softmax_8	softmax(x[7:0])	8-input softmax

19.2.8 E5M2 Comparison and Selection

Module	Function	Description
math_fp8_e5m2_comparator	a <=> b	Full comparison
math_fp8_e5m2_max	max(a, b)	Two-input max
math_fp8_e5m2_min	min(a, b)	Two-input min
math_fp8_e5m2_max_tree_8	max(a[7:0])	8-input max
math_fp8_e5m2_min_tree_8	min(a[7:0])	8-input min
math_fp8_e5m2_clamp	clamp(x, lo, hi)	Value clamping

19.2.9 E5M2 Format Conversions

Module	Conversion	Notes
math_fp8_e5m2_to_bf16	E5M2 -> BF16	Lossless upconversion
math_fp8_e5m2_to_fp16	E5M2 -> FP16	Lossless upconversion
math_fp8_e5m2_to_fp32	E5M2 -> FP32	Lossless upconversion
math_fp8_e5m2_to_fp8_e4m3	E5M2 -> E4M3	May overflow/lose precision

19.2.10 Special Values

19.2.10.1 E4M3 Special Values

Value	Encoding	Notes
+0	0x00	Positive zero
-0	0x80	Negative zero
Max Normal	0x7E	448 (exp=15, mant=6)
NaN	0x7F	exp=15, mant=7

Value	Encoding	Notes
No Infinity	-	Overflow saturates to 0x7E

19.2.10.2 E5M2 Special Values

Value	Encoding	Notes
+0	0x00	Positive zero
-0	0x80	Negative zero
+Inf	0x7C	exp=31, mant=0
-Inf	0xFC	exp=31, mant=0
NaN	0x7E	exp=31, mant!=0

19.3 Usage Examples

19.3.1 Inference Pipeline (E4M3)

```
// FP8 E4M3 inference - maximum memory efficiency
logic [7:0] weight, activation, product;

math_fp8_e4m3_multiplier u_mult (
    .i_a(weight),
    .i_b(activation),
    .ow_result(product),
    .ow_overflow(overflow),  // Saturates to 448
    .ow_underflow(),
    .ow_invalid()
);

// Apply activation
math_fp8_e4m3_relu u_relu (
    .i_a(product),
    .ow_result(activated)
);
```

19.3.2 Mixed Precision Training

```
// Forward pass in E5M2, upconvert for backward pass
logic [7:0]  fp8_activation;
logic [15:0] bf16_gradient;

// Upconvert for gradient computation
math_fp8_e5m2_to_bf16 u_cvt (
    .i_a(fp8_activation),
    .ow_result(bf16_activation),
```

```
    .ow_invalid()
);

// Compute gradients in BF16
// ... backward pass in BF16 ...
```

19.4 Design Notes

19.4.1 E4M3 Overflow Handling

E4M3 has no infinity encoding. When multiplication would overflow: - Result saturates to max normal (448 = 0x7E) - `ow_overflow` flag asserts - Sign is preserved

```
// E4M3: Large values saturate instead of becoming infinity
// 400 * 2 = 800 -> saturates to 448
```

19.4.2 Subnormal Handling

Both E4M3 and E5M2 use Flush-to-Zero (FTZ): - Subnormal inputs treated as zero - Results that would be subnormal flush to zero - Simplifies hardware, matches AI accelerator conventions

19.4.3 Auto-Generation

```
# Regenerate all FP8 modules
PYTHONPATH=bin:$PYTHONPATH python3
bin/rtl_generators/ieee754/generate_all.py rtl/math
```

Generator files: - `bin/rtl_generators/ieee754/fp8_e4m3_multiplier.py` - `bin/rtl_generators/ieee754/fp8_e5m2_multiplier.py` - `bin/rtl_generators/ieee754/fp_activations.py` - `bin/rtl_generators/ieee754/fp_conversions.py`

19.5 Related Modules

- **math_bf16_extended** - BF16 modules
- **math_fp16_modules** - FP16 modules
- **math_fp32_modules** - FP32 modules
- **math_ieee754_modules** - IEEE 754-2008 arithmetic

19.6 Testing

Covered by 34 test suites:

- `val/math/test_math_fp8_e4m3_adder.py`
- `val/math/test_math_fp8_e4m3_clamp.py`
- `val/math/test_math_fp8_e4m3_comparator.py`
- `val/math/test_math_fp8_e4m3_fma.py`

- `val/math/test_math_fp8_e4m3_gelu.py`
- `val/math/test_math_fp8_e4m3_leaky_relu.py`
- `val/math/test_math_fp8_e4m3_max.py`
- `val/math/test_math_fp8_e4m3_min.py`
- … and 26 more (`ls val/math/test_math_*.py`)

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

19.7 Navigation

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20 IEEE 754-2008 Compliant Modules

Full IEEE 754-2008 compliant floating-point arithmetic modules for FP16 and FP32 formats.

20.1 Overview

These modules implement complete IEEE 754-2008 compliant arithmetic for half-precision (FP16) and single-precision (FP32) floating-point operations. Unlike the simplified BF16/FP8 modules, which use Flush-to-Zero, these handle subnormals properly and give you full IEEE compliance.

Key Features: - **Full IEEE 754-2008 compliance** - Proper subnormal handling, all rounding modes conceptually supported - **Complete special case handling** - Zero, infinity, NaN, subnormal - **Status flags** - Overflow, underflow, invalid - **Pipelined options** - Configurable pipeline stages for timing closure

20.2 Functional Description

20.2.1 Module Summary

20.2.1.1 FP16 (Half Precision) IEEE 754

Module	Operation	Description
`math_ieee754_2008_fp16_adder`	a + b	Full FP16 addition with alignment
`math_ieee754_2008_fp16_multiplier`	a * b	FP16 multiplication
`math_ieee754_2008_fp16_fma`	a*b + c	Fused multiply-add

Module	Operation	Description
math_ieee754_2008_fp16_mantissa_mult	Internal	11x11 mantissa multiply
math_ieee754_2008_fp16_exponent_adder	Internal	Exponent computation

20.2.1.2 FP32 (Single Precision) IEEE 754

Module	Operation	Description
math_ieee754_2008_fp32_adder	a + b	Full FP32 addition with alignment
math_ieee754_2008_fp32_multiplier	a * b	FP32 multiplication
math_ieee754_2008_fp32_fma	a*b + c	Fused multiply-add
math_ieee754_2008_fp32_mantissa_mult	Internal	24x24 mantissa multiply
math_ieee754_2008_fp32_exponent_adder	Internal	Exponent computation

20.2.2 Module Interfaces

20.2.2.1 FP16 Adder

```
module math_ieee754_2008_fp16_adder #(
    parameter int PIPE_STAGE_1 = 0,  // After operand swap
    parameter int PIPE_STAGE_2 = 0,  // After alignment
    parameter int PIPE_STAGE_3 = 0,  // After addition
    parameter int PIPE_STAGE_4 = 0   // After normalization
) (
    input  logic        i_clk,
    input  logic        i_rst_n,
    input  logic [15:0] i_a,
    input  logic [15:0] i_b,
    input  logic        i_valid,
    output logic [15:0] ow_result,
    output logic        ow_valid,
    output logic        ow_overflow,
    output logic        ow_underflow,
    output logic        ow_invalid
);
```

20.2.2.2 FP32 Multiplier

```
module math_ieee754_2008_fp32_multiplier (
    input  logic [31:0] i_a,
    input  logic [31:0] i_b,
    output logic [31:0] ow_result,
    output logic        ow_overflow,
    output logic        ow_underflow,
    output logic        ow_invalid
);
```

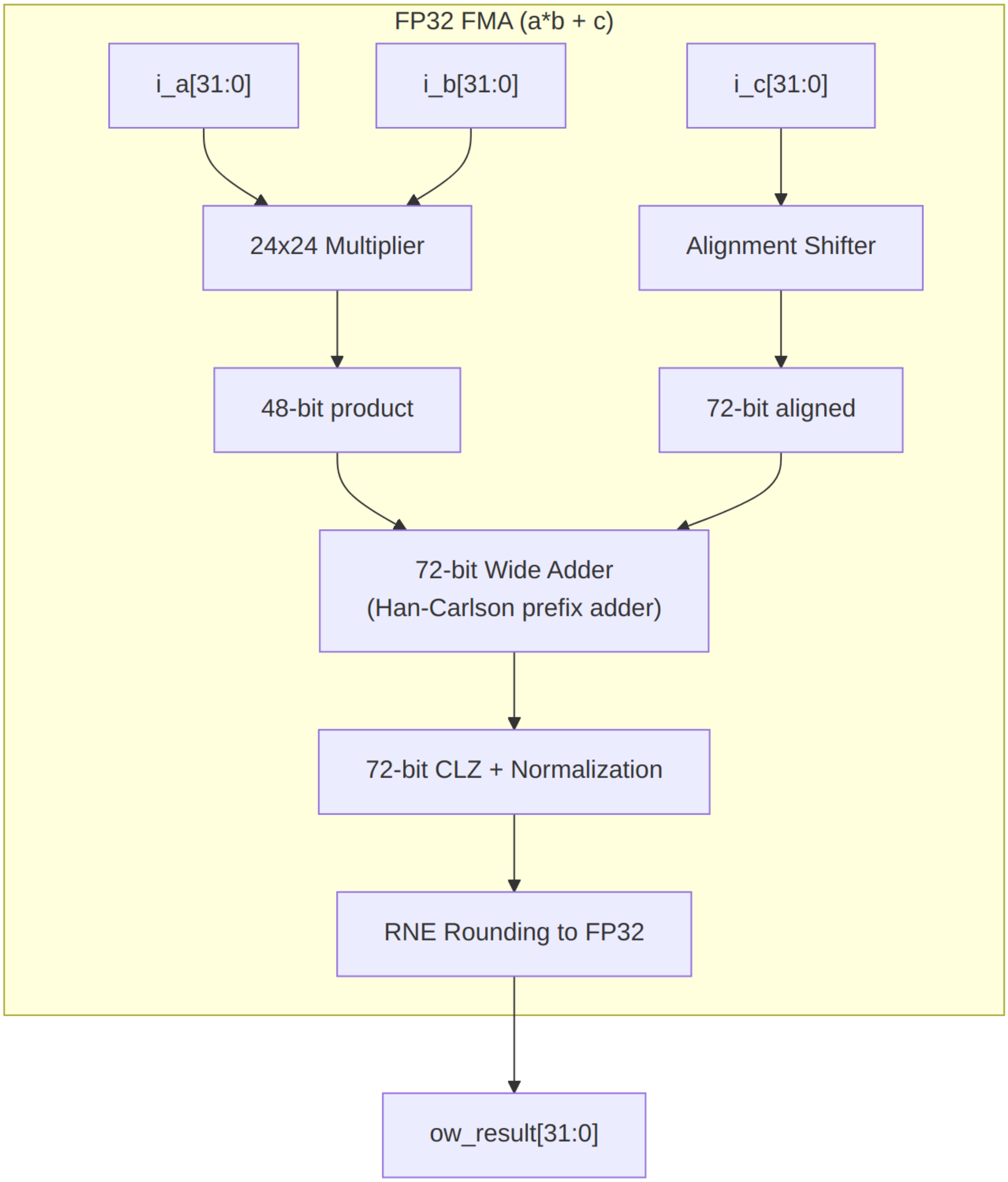
20.2.2.3 FP32 FMA

```
module math_ieee754_2008_fp32_fma (
    input  logic [31:0] i_a,      // Multiplicand
    input  logic [31:0] i_b,      // Multiplier
    input  logic [31:0] i_c,      // Addend
    output logic [31:0] ow_result,
    output logic        ow_overflow,
    output logic        ow_underflow,
    output logic        ow_invalid
);
```

20.2.3 Architecture

20.2.3.1 FP32 Multiplier Pipeline

```
Stage 1: Field extraction + special case detection
         sign_a, exp_a[7:0], mant_a[22:0]
         sign_b, exp_b[7:0], mant_b[22:0]
              |
Stage 2: Sign computation (XOR)
         24x24 Dadda tree multiplication -> 48-bit product
              |
Stage 3: Exponent computation
         exp_sum = exp_a + exp_b - 127 + norm_adjust
              |
Stage 4: Normalization + RNE rounding
         Shift product, apply round-to-nearest-even
              |
Stage 5: Special case priority selection
         NaN > Inf > Overflow > Underflow > Zero > Normal
              |
         Result assembly
```



FP32 FMA Architecture

20.2.4 IEEE 754 Compliance

20.2.4.1 Special Value Handling

Operation	Input	Output
Any	NaN	NaN (propagated)

Operation	Input	Output
x + y	+Inf, -Inf	NaN (invalid)
x * y	0, Inf	NaN (invalid)
x + y	Normal, Normal	Normal/Subnormal
x * y	Normal, Normal	Normal/ Subnormal/Zero

20.2.4.2 Rounding Modes

Currently implements Round-to-Nearest-Even (RNE):

```
// RNE: Round up if guard=1 AND (round=1 OR sticky=1 OR lsb=1)
wire w_round_up = w_guard & (w_round | w_sticky | w_lsb);
```

20.2.4.3 Status Flags

Flag	Condition
`ow_overflow`	Result magnitude exceeds max normal
`ow_underflow`	Rounded result magnitude less than min normal (non-zero); IEEE 754 after-rounding detection, so a rounding carry out of the boundary yields min normal, not a flush (MATH-008)
`ow_invalid`	Invalid operation (0*Inf, Inf-Inf, NaN input)

20.3 Usage Examples

20.3.1 High-Precision Computation

```
// IEEE 754 compliant FP32 FMA for numerical accuracy
logic [31:0] a, b, c, result;

math_ieee754_2008_fp32_fma u_fma (
    .i_a(a),
    .i_b(b),
    .i_c(c),
    .ow_result(result),
    .ow_overflow(overflow),
    .ow_underflow(underflow),
    .ow_invalid(invalid)
);

// Check for exceptions
always_ff @(posedge clk) begin
```

```
    if (invalid)
        $error("Invalid FP operation!");
    else if (overflow)
        $warning("FP overflow - result is infinity");
end
```

20.3.2 Pipelined FP16 Adder

```
// 4-stage pipelined FP16 adder for high frequency
math_ieee754_2008_fp16_adder #(
    .PIPE_STAGE_1(1),  // Pipeline after swap
    .PIPE_STAGE_2(1),  // Pipeline after align
    .PIPE_STAGE_3(1),  // Pipeline after add
    .PIPE_STAGE_4(1)   // Pipeline after normalize
) u_adder (
    .i_clk(clk),
    .i_rst_n(rst_n),
    .i_a(fp16_a),
    .i_b(fp16_b),
    .i_valid(input_valid),
    .ow_result(fp16_sum),
    .ow_valid(output_valid),
    .ow_overflow(),
    .ow_underflow(),
    .ow_invalid()
);
```

20.4 Design Notes

20.4.1 Comparison: IEEE 754 vs Simplified Modules

Feature	IEEE 754 Modules	BF16/FP8 Modules
Subnormals	Full support	Flush to Zero
Infinity	Proper handling	Saturation (FP8 E4M3)
NaN payloads	Preserved	Canonical only
Rounding modes	RNE (extensible)	RNE only
Pipeline options	Configurable	Fixed
Target use	Precision-critical	AI/ML acceleration

20.4.2 Dependencies

These modules use the following building blocks:

Module	Used By	Purpose
math_adder_han_carlson_072	FP32 FMA	72-bit wide addition
math_adder_han_carlson_048	FP32 mult	48-bit product CPA
math_multiplier_dadda_4to2_024	FP32	24x24 mantissa multiply
math_multiplier_dadda_4to2_011	FP16	11x11 mantissa multiply
count_leading_zeros	All	Normalization
shifter_barrel	Adders	Mantissa alignment

20.4.3 Auto-Generation

```
# Regenerate IEEE 754 modules
PYTHONPATH=bin:$PYTHONPATH python3
bin/rtl_generators/ieee754/generate_all.py rtl/math
```

Generator files: - `bin/rtl_generators/ieee754/fp16_adder.py` - `bin/rtl_generators/ieee754/fp16_multiplier.py` - `bin/rtl_generators/ieee754/fp16_fma.py` - `bin/rtl_generators/ieee754/fp32_adder.py` - `bin/rtl_generators/ieee754/fp32_multiplier.py` - `bin/rtl_generators/ieee754/fp32_fma.py`

20.5 Related Modules

- **math_bf16_multiplier** - Simplified BF16 multiply
- **math_bf16_fma** - Simplified BF16 FMA
- **math_adder_han_carlson** - Prefix adder building block
- **math_multiplier_dadda_4to2** - Dadda multiplier

20.6 Testing

Covered by 6 test suites:

- `val/math/test_math_ieee754_2008_fp16_adder.py`
- `val/math/test_math_ieee754_2008_fp16_fma.py`
- `val/math/test_math_ieee754_2008_fp16_multiplier.py`
- `val/math/test_math_ieee754_2008_fp32_adder.py`
- `val/math/test_math_ieee754_2008_fp32_fma.py`
- `val/math/test_math_ieee754_2008_fp32_multiplier.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

20.7 Navigation

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21 math_mod_3_compress (`math_mod_3_compress.sv`)

A combinational 16-bit mod-3 block built from a carry-save compressor tree — no multiplier, no divider, no DSP, just the 2-bit remainder the monbus burst-writer needs.

Location: `rtl/math/` **Status:** Production Ready

21.1 Overview

`math_mod_3_compress` computes `d_in mod 3` for a 16-bit operand and returns the 2-bit remainder (0..2). It's built in the same carry-save-compressor style as `div_by_15_ceil_32compress.sv` — a 3:2 compressor tree feeding a final carry-propagate add and a small fold — but it emits only the remainder, which is exactly what the monbus burst-writer needs to round a beat count **down** to a whole number of 3-beat records: `rounded = X - (X mod 3)`.

There's no `*` or `/` operator anywhere, so it infers no DSP block and no iterative divider — just a shallow tree of carry-save adders and a couple of small adds.

The monbus compressor packs monitor packets into fixed 3-beat records. Working out how many whole records a beat count covers means computing `X - (X mod 3)`, which means you need the remainder `X mod 3`. This module produces that remainder combinationally without a divider, so the burst-writer can round the count down in a single cycle of logic.

- **Combinational, single-cycle:** No clock, no state; pure logic from `d_in` to `rem_out`
- **No multiply / divide:** Avoids inferred DSP and iterative dividers
- **Carry-save tree:** Eight base-4 digits reduced with 3:2 compressors instead of a ripple-add chain
- **Remainder only:** Returns just `d_in mod 3` (0..2), the quantity the monbus record rounding needs
- **Shared style:** Same construction as `div_by_15_ceil_32compress.sv`, reusing `math_adder_carry_save_nbit`

Use Cases:

- Rounding a monbus beat count down to a whole number of 3-beat records `(X - (X mod 3))`
- Any combinational `mod 3` of a 16-bit value where a DSP-free, divider-free implementation is desired
- A worked reference for the base-4 digit-sum residue technique with carry-save reduction

Key Benefit: Produces `d_in mod 3` in one combinational stage with a shallow carry-save tree, no multiplier, divider, or DSP block required.

21.2 Parameters

This module has no parameters. The operand width is fixed at 16 bits and the result at 2 bits.

(Internally, `localparam int BITS = 6` sizes the carry-save datapath so the weight-2 carry left-shifts have headroom.)

21.3 Ports

Port	Direction	Width	Description
`d_in`	Input	16	Operand whose remainder mod 3 is computed
`rem_out`	Output	2	`d_in mod 3`, in the range 0..2

21.4 Functional Description

21.4.1 Base-4 Digit-Sum Method

The trick rests on one identity: `4^k ≡ 1 (mod 3)`. Split `d_in` into 2-bit groups and each group is a base-4 digit with weight `4^k`; since every weight is `1 (mod 3)`, the **sum of the eight digits is congruent to `d_in (mod 3)`**. The problem therefore collapses to summing eight small digits and taking that sum mod 3.

The eight 2-bit groups are zero-extended to the 6-bit carry-save width:

```
assign w_g0 = {{(BITS-2){1'b0}}, d_in[1:0]};
assign w_g1 = {{(BITS-2){1'b0}}, d_in[3:2]};
// ... through ...
assign w_g7 = {{(BITS-2){1'b0}}, d_in[15:14]};
```

21.4.2 3:2 Compressor Tree

Rather than a ripple-add chain, the eight digits are reduced with a tree of `math_adder_carry_save_nbit` 3:2 compressors. Each carry output has weight 2, so wherever a carry feeds the next level it is left-shifted by one (`{carry[BITS-2:0], 1'b0}`); the always-zero top carry bit is dropped, exactly as the div-by-15 reference does:

- **Level 1:** three compressors reduce the eight digits three at a time (the last triple padded with 0)
- **Level 2:** two compressors combine the level-1 sums and shifted carries
- **Level 3:** one compressor, holding `w_carryE2` for the next level
- **Level 4:** one compressor folds in the held carry

A final carry-propagate add collapses the last sum/carry pair into the digit sum (0..24, still `≡ d_in (mod 3)`):

```
assign w_grp_sum = w_sumG4 + {w_carryG4[BITS-2:0], 1'b0};
```

21.4.3 Final Fold and Conditional Subtract

The digit sum `w_grp_sum` tops out at 24, so it is folded once more onto its own base-4 digits (same residue mod 3), giving a value in 0..7, and then a conditional subtract brings it into the final 0..2 range:

```
assign w_fold = {2'b0, w_grp_sum[1:0]} + {2'b0, w_grp_sum[3:2]} +
{3'b0, w_grp_sum[4]};
assign rem_out = 2'((w_fold >= 4'd6) ? (w_fold - 4'd6)
                  : (w_fold >= 4'd3) ? (w_fold - 4'd3)
                                     : w_fold);
```

The two-branch subtract handles `w_fold` values up to 7 (subtract 6, subtract 3, or pass through), yielding the exact remainder 0, 1, or 2.

21.5 Timing Characteristics

Purely combinational. `rem_out` tracks `d_in` with no clock and no state — one cycle of logic through the compressor tree, the final add, and the fold. If you want a pipeline stage, register the result externally.

21.6 Usage Examples

```
// Round a beat count down to a whole number of 3-beat monbus records.
logic [15:0] beat_count;
logic [1:0]  beat_rem;
logic [15:0] rounded_count;

math_mod_3_compress u_mod3 (
```

```
    .d_in    (beat_count),
    .rem_out (beat_rem)
);

// rounded = X - (X mod 3)
assign rounded_count = beat_count - {14'b0, beat_rem};
```

21.7 Design Notes

- **Why remainder only?** The div-by-15 reference computes a quotient; here only the residue is needed, so the quotient machinery is dropped, keeping the logic shallow.
- **Datapath width headroom.** `BITS = 6` (rather than the minimum 5 needed for a 0..24 sum) gives the weight-2 carry left-shifts room so the dropped top carry is always 0, matching the div-by-15 construction.
- **No inferred arithmetic primitives.** Because there is no `*` or `/`, synthesis produces adders and LUTs only, no DSP slices, no multi-cycle divider.

21.8 Related Modules

21.8.1 Used By

- monbus_compressor - Rounds a beat count down to whole 3-beat records when packing monitor packets
- monbus_group_core - Shared filter/FIFO core of the monbus group wrappers

21.8.2 Uses

- math_adder_carry_save - The N-bit carry-save (3:2) compressor instantiated throughout the tree

21.8.3 See Also

- `div_by_15_ceil_32compress.sv` - The reference implementation this module's compressor style follows

21.9 Testing

From the test suite (`val/math/test_math_mod_3_compress.py`):

- **Key test scenarios**:
  - Check rem_out == d_in % 3 across the 16-bit input space.

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

21.10 References

21.10.1 Source Code

- `rtl/math/math_mod_3_compress.sv`
- `rtl/math/math_adder_carry_save.sv` (instantiated submodule)
- `div_by_15_ceil_32compress.sv` (the carry-save-compressor style reference this module was patterned on; the file itself lives outside this repo's tree – historical reference only)

21.10.2 Documentation

- `docs/markdown/rtl-math/index.md`

Last Updated: 2026-07-15

21.11 Navigation

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22 Basic Multiplier Building Blocks

The fundamental multiplication primitives: a basic single-bit multiplier cell and an array-based carry-save multiplier. Simple implementations, suited to educational purposes and as building blocks for more complex multiplier architectures.

22.1 Overview

Two modules live here: - **math_multiplier_basic_cell** - Single-bit multiplier cell with partial product accumulation - **math_multiplier_carry_save** - Array-based N×N multiplier using carry-save addition

These modules demonstrate fundamental multiplication concepts and serve as reference implementations. For production designs, consider using optimized Wallace or Dadda tree multipliers.

Key Features: - **Structural implementation** - Explicit cell-by-cell construction - **Educational value** - Clear demonstration of multiplication algorithm - **Array-based** - Regular structure, easy to understand - **Scalable** - Generic width support via parameter - **Combinational** - Pure logic, no sequential elements

22.2 Parameters

22.2.1 math_multiplier_basic_cell

No parameters (single-bit cell).

22.2.2 math_multiplier_carry_save

Parameter	Type	Default	Description
N	int	4	Bit width (range: 2-32, typical: 4-16)

22.3 Ports

22.3.1 math_multiplier_basic_cell

```
module math_multiplier_basic_cell (
    input  logic i_i,       // Multiplier bit
    input  logic i_j,       // Multiplicand bit
    input  logic i_c,       // Carry input
    input  logic i_p,       // Partial sum input
    output logic ow_c,      // Carry output
    output logic ow_p       // Partial sum output
);
```

Port	Direction	Width	Description
i_i	Input	1	Multiplier bit (row index)
i_j	Input	1	Multiplicand bit (column index)
i_c	Input	1	Carry input (from cell below)
i_p	Input	1	Partial sum input (from cell to the left)
ow_c	Output	1	Carry output (to cell above)
ow_p	Output	1	Partial sum

Port	Direction	Width	Description
			output (to cell on the right)

22.3.2 math_multiplier_carry_save

```
module math_multiplier_carry_save #(
    parameter int N = 4     // Bit width (default: 4)
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

Port	Direction	Width	Description
i_multiplier	Input	N	Multiplier operand (unsigned)
i_multiplicand	Input	N	Multiplicand operand (unsigned)
ow_product	Output	2N	Product result (unsigned)

22.4 Functional Description

22.4.1 Basic Multiplier Cell

The basic cell implements a single position in the multiplication array:

Logic Operations:

```
w_p = i_i & i_j;                                    // Partial product bit
w_sum = i_c ^ i_p ^ w_p;                            // 3-input XOR
w_carry = (i_c & i_p) | (i_c & w_p) | (i_p & w_p); // Majority function

ow_p = w_sum;       // Partial sum output
ow_c = w_carry;     // Carry output
```

Function: - Computes partial product bit: `i_i & i_j` - Adds to carry input and partial sum input using full adder - Outputs new partial sum and carry to adjacent cells

Cell Position in Array:

```
        i_c (carry from below)
         ↓
    i_p → [CELL] → ow_p (to right)
         ↓
        ow_c (carry to above)

Cell computes: (i_i & i_j) + i_c + i_p
```

22.4.2 Carry-Save Array Multiplier

The array multiplier creates an N×N grid of basic cells:

Structure:

```
Multiplicand:  j₃ j₂ j₁ j₀
Multiplier:
  i₀:         [C][C][C][C]
  i₁:         [C][C][C][C]
  i₂:         [C][C][C][C]
  i₃:         [C][C][C][C]

Where each [C] is a math_multiplier_basic_cell

Connections:
- Horizontal: ow_p[i][j] → i_p[i][j+1]
- Vertical:   ow_c[i][j] → i_c[i+1][j]
- Diagonals collect product bits
```

Algorithm: 1. **Generate partial products** - Each cell computes `i_i & i_j` 2. **Accumulate with carry-save** - Each row adds partial products 3. **Collect LSBs** - Left column provides lower product bits 4. **Final addition** - Top row sum and carries added for upper product bits

Implementation:

```
// Create N×N array of cells
genvar i, j;
generate
    for (i = 0; i < N; i++) begin : gen_row
        for (j = 0; j < N; j++) begin : gen_col
            math_multiplier_basic_cell cell (
                .i_i(i_multiplier[i]),
                .i_j(i_multiplicand[j]),
                .i_c(w_cin[i][j]),
                .i_p(w_pin[i][j]),
                .ow_c(w_cout[i][j]),
                .ow_p(w_pout[i][j])
            );
        end
    end
```

```verilog
endgenerate

// Initialize first row
for (j = 0; j < N; j++) begin
    assign w_cin[0][j] = 1'b0;
    assign w_pin[0][j] = 1'b0;
end

// Connect rows (vertical carry propagation)
for (i = 1; i < N; i++) begin
    for (j = 0; j < N; j++) begin
        assign w_cin[i][j] = w_cout[i-1][j];
        assign w_pin[i][j] = (j == N-1) ? 1'b0 : w_pout[i-1][j+1];
    end
end

// Collect LSB of each row (lower product bits)
for (i = 0; i < N; i++) begin
    assign w_product[i] = w_pout[i][0];
end

// Final addition for upper bits
assign final_carries = w_cout[N-1][N-1:0];
assign final_partials = {1'b0, w_pout[N-1][N-1:1]};
assign final_sum = final_carries + final_partials;

assign ow_product = {final_sum, w_product};
```

22.4.3 Multiplication Example: 5 × 6 (4-bit)

```
Multiplicand (6):  0110
Multiplier (5):    0101

Partial Products:
  i₃ (0): 0000   (0 × 0110)
  i₂ (1): 0110   (1 × 0110)
  i₁ (0): 0000   (0 × 0110)
  i₀ (1): 0110   (1 × 0110)

Array accumulation:
       0110
     0000
   0110
 0000
-----------
 0001 1110   = 30 (decimal)
```

```
Product: 5 × 6 = 30 (correct)
```

22.5 Timing Characteristics

Metric	4-bit	8-bit	16-bit
Logic Depth	~8 levels	~16 levels	~32 levels
Typical Delay (ns)	~3.2	~6.4	~12.8
Max Frequency	~310 MHz	~155 MHz	~78 MHz

Logic Depth Breakdown: - Each cell: 2 levels (AND + full adder) - Array depth: 2N levels (carry ripples down rows) - Final addition: Additional adder delay

Critical Path:

```
i_multiplier[N-1] → PP[N-1][0] → row N-1 accumulation → final addition
→ ow_product[2N-1]
```

Scaling: O(N) depth - doubles with doubling of bit width

22.5.1 Resource Utilization

Width	Basic Cells	Final Adder	AND Gates	Estimated LUTs
4-bit	16	4-bit	16	~48
8-bit	64	8-bit	64	~192
16-bit	256	16-bit	256	~768

Area Formula: N^2 basic cells + N-bit final adder ≈ $3N^2$ LUTs

22.5.2 Comparison to Optimized Multipliers

Architecture	8-bit LUTs	8-bit Delay	Best Use Case
Array (Basic Cell)	~192	~6.4 ns	Educational, low speed
Carry-Save Array	**~192**	**~6.4 ns**	**Simple, moderate speed**
Wallace Tree	~120	~3.5 ns	High speed, larger area
Dadda Tree	~108	~3.2 ns	**Production (best**

Architecture	8-bit LUTs	8-bit Delay	Best Use Case
			balance)
Booth Radix-4	~100	~4.5 ns	Signed multiplication

Key Observation: Array multipliers are ~2× slower but simpler than tree multipliers.

22.6 Usage Examples

22.6.1 Basic 4×4 Multiplication

```
logic [3:0] a, b;
logic [7:0] product;

math_multiplier_carry_save #(.N(4)) u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product)
);

// Example: 7 × 9 = 63
initial begin
    a = 4'd7;
    b = 4'd9;
    #1;
    assert(product == 8'd63);
end
```

22.6.2 Generic Width Multiplier

```
module flexible_multiplier #(
    parameter int WIDTH = 8
) (
    input  logic [WIDTH-1:0]   a, b,
    output logic [2*WIDTH-1:0] product
);

    math_multiplier_carry_save #(.N(WIDTH)) u_mult (
        .i_multiplier(a),
        .i_multiplicand(b),
        .ow_product(product)
    );

endmodule

// Instantiate with different widths
```

```
flexible_multiplier #(.WIDTH(8))  u_mult8  (...);
flexible_multiplier #(.WIDTH(16)) u_mult16 (...);
```

22.6.3 Building Custom Array from Cells

```
// Manual 2×2 multiplier using basic cells
module mult_2x2 (
    input  logic [1:0] i_multiplier,
    input  logic [1:0] i_multiplicand,
    output logic [3:0] ow_product
);

    logic c[1:0][1:0];  // Carry signals
    logic p[1:0][1:0];  // Partial sums

    // Row 0, Column 0
    math_multiplier_basic_cell cell_00 (
        .i_i(i_multiplier[0]),
        .i_j(i_multiplicand[0]),
        .i_c(1'b0),
        .i_p(1'b0),
        .ow_c(c[0][0]),
        .ow_p(p[0][0])
    );
    assign ow_product[0] = p[0][0];

    // Row 0, Column 1
    math_multiplier_basic_cell cell_01 (
        .i_i(i_multiplier[0]),
        .i_j(i_multiplicand[1]),
        .i_c(1'b0),
        .i_p(1'b0),
        .ow_c(c[0][1]),
        .ow_p(p[0][1])
    );

    // Row 1, Column 0
    math_multiplier_basic_cell cell_10 (
        .i_i(i_multiplier[1]),
        .i_j(i_multiplicand[0]),
        .i_c(c[0][0]),
        .i_p(p[0][1]),
        .ow_c(c[1][0]),
        .ow_p(p[1][0])
    );
    assign ow_product[1] = p[1][0];

    // Row 1, Column 1
```

```
    math_multiplier_basic_cell cell_11 (
        .i_i(i_multiplier[1]),
        .i_j(i_multiplicand[1]),
        .i_c(c[0][1]),
        .i_p(1'b0),
        .ow_c(c[1][1]),
        .ow_p(p[1][1])
    );

    // Final addition
    assign ow_product[3:2] = c[1][0] + p[1][1] + c[1][1];

endmodule
```

22.6.4 Multiplier with Valid/Ready Handshake

```
module handshake_multiplier #(
    parameter int N = 8
) (
    input  logic              clk,
    input  logic              rst_n,
    input  logic [N-1:0]      i_multiplier,
    input  logic [N-1:0]      i_multiplicand,
    input  logic              i_valid,
    output logic              o_ready,
    output logic [2*N-1:0]    o_product,
    output logic              o_valid
);

    logic [2*N-1:0] product_comb;

    // Combinational multiplier
    math_multiplier_carry_save #(.N(N)) u_mult (
        .i_multiplier(i_multiplier),
        .i_multiplicand(i_multiplicand),
        .ow_product(product_comb)
    );

    // Pipeline register
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            o_product <= '0;
            o_valid   <= 1'b0;
        end else begin
            o_product <= product_comb;
            o_valid   <= i_valid;
        end
    end
```

```
    assign o_ready = 1'b1;  // Always ready (combinational)

endmodule
```

22.7 Design Notes

22.7.1 Advantages

- **Simple structure** - Regular array, easy to understand
- **Parameterizable** - Generic width support via parameter
- **Educational value** - Clearly demonstrates multiplication algorithm
- **Moderate area** - Reasonable gate count for small widths
- **Purely combinational** - No state, easy to integrate

22.7.2 Limitations

- **Slow for large widths** - O(N) depth vs O(log N) for trees
- **Poor scaling** - Area grows as N^2, delay as N
- **Not optimal** - Better alternatives exist (Dadda, Wallace)
- **Unsigned only** - Requires sign handling for signed operands

22.7.3 When to Use Array Multipliers

Appropriate Use Cases: - Educational demonstrations - Low-frequency designs (<100 MHz for 8-bit) - Simple prototypes - Designs where area is more critical than speed - Parameterized width support needed

Prefer Optimized Multipliers When: - High speed required → Wallace or Dadda tree - Area critical → Dadda tree (optimized CSA) - Signed operands → Booth multiplier - Behavioral sufficient → Use * operator (synthesis optimizes)

22.7.4 Basic Cell Design Notes

Verilator Lint Suppression:

```
/* verilator lint_off UNOPTFLAT */
output logic ow_c, ow_p
/* verilator lint_on UNOPTFLAT */
```

Reason: Array structure creates combinational feedback paths that are intentional. The explicit output buffering (`always_comb` block) breaks the path for simulation.

22.7.5 Synthesis Considerations

For FPGA Designs:

```
# Let synthesis tool optimize array structure
set_dont_touch false

# Enable register retiming if pipelining
set_optimize_registers true
```

For ASIC Designs:

```
# Flatten for better optimization
set_flatten true

# May benefit from custom standard cell multipliers
```

Note: Modern synthesis tools often recognize multiplication patterns and replace with optimized implementations.

22.7.6 Pipelining Strategy

For higher frequency, add pipeline stages:

2-Stage Pipeline:

```
// Stage 1: Register inputs
always_ff @(posedge clk) begin
    a_reg <= a;
    b_reg <= b;
end

// Stage 2: Multiply and register output
math_multiplier_carry_save #(.N(8)) u_mult (
    .i_multiplier(a_reg),
    .i_multiplicand(b_reg),
    .ow_product(product_comb)
);

always_ff @(posedge clk) begin
    product_reg <= product_comb;
end
```

Multi-Stage Pipeline (split array into stages):

```
// Register after every K rows
// Example: 16-row array split into 4 stages (4 rows each)
```

22.7.7 Common Pitfalls

Anti-Pattern 1: Using for high-speed designs

```
// WRONG: Array multiplier too slow for 200 MHz with 16-bit
math_multiplier_carry_save #(.N(16)) u_mult (...);
// Critical path ~13ns, can't meet 5ns period!

// RIGHT: Use Dadda tree for high-speed
math_multiplier_dadda_tree_016 u_mult (...);
```

Anti-Pattern 2: Not considering behavioral alternative

```
// MANUAL: Explicit array instantiation
math_multiplier_carry_save #(.N(8)) u_mult (
    .i_multiplier(a), .i_multiplicand(b), .ow_product(p)
);

// SIMPLER: Let synthesis optimize
assign p = a * b;  // Synthesis may produce better result!
```

When to use explicit vs behavioral: - Explicit: Educational, specific structure needed, debugging - Behavioral: Production code, synthesis tool optimization

Anti-Pattern 3: Expecting signed multiplication

```
// WRONG: Signed operands misinterpreted
logic signed [7:0] a = -5, b = 3;
math_multiplier_carry_save #(.N(8)) u_mult (...);
// Result: 753 (unsigned 251 × 3), not -15!

// RIGHT: Convert to unsigned, fix sign
// (See signed multiplication techniques)
```

Anti-Pattern 4: Ignoring width limits

```
// CAREFUL: 32×32 array multiplier is very large and slow
math_multiplier_carry_save #(.N(32)) u_mult (...);
// 1024 cells, ~32 levels deep, ~15ns delay

// CONSIDER: Dadda tree or behavioral for large widths
```

22.7.8 Educational Value

The array multiplier clearly demonstrates the algorithm you'd use on paper:

1. Partial Product Generation:

```
Each row computes: multiplicand × multiplier_bit[i]
Just like manual multiplication on paper!
```

2. Shifted Addition:

```
Each row is shifted left by one position
Carry propagation adds partial products
```

3. Carry-Save Concept:

```
Carries propagate downward (next row)
Sums propagate rightward (same row)
Defers final carry-propagate until end
```

Progression to Optimized Multipliers:

Learning Path: 1. **Array Multiplier** (this document) - Understand basic concept 2. **Wallace Tree** - Learn parallel reduction 3. **Dadda Tree** - Learn optimization techniques 4. **Booth Encoding** - Learn signed multiplication

Concept: Array → parallelize → optimize → specialize

22.8 Related Modules

- **math_multiplier_wallace_tree_*.sv** - Fast tree-based multiplication
- **math_multiplier_dadda_tree_*.sv** - Optimized tree-based multiplication
- **math_adder_carry_save.sv** - 3:2 compressor used in trees
- **math_adder_full.sv** - Full adder primitive
- **math_adder_half.sv** - Half adder primitive

22.9 Testing

No dedicated test wrapper, and – correcting an earlier claim – NO structural coverage either: none of the tree multipliers instantiate `math_multiplier_basic_cell` or the `math_multiplier_carry_save` array (the Dadda trees use `math_adder_carry_save`, a different 1-bit 3:2 CSA module). These are reference implementations. Verification is the formal proofs: `formal/common/math_multiplier_basic_cell/` and `formal/common/math_multiplier_carry_save/` (prove + cover, SymbiYosys, re-run 2026-08-10 against current RTL).

22.10 References

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22.11 Navigation

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23 Dadda Multiplier with 4:2 Compressors

An area-optimized 8x8 unsigned multiplier using Dadda reduction with 4:2 compressors, providing fast parallel multiplication. Reduction runs in four 4:2-compressor stages (39 compressors + 2 full adders in math_multiplier_dadda_4to2_008); counts are from the RTL, not estimated.

23.1 Overview

The `math_multiplier_dadda_4to2_008` module implements an 8-bit unsigned multiplier optimized for BF16 mantissa multiplication. It uses 4:2 compressors instead of 3:2 carry-save adders (CSAs), which reduces column height faster and takes fewer stages.

Key Features: - **8x8 unsigned multiply** - 16-bit product output - **4:2 compressor-based** - Faster height reduction than 3:2 CSA - **Dadda scheduling** - Optimal reduction order for minimum area - **Han-Carlson final CPA** - Fast carry-propagate addition - **Purely combinational** - Single-cycle operation

23.2 Parameters

Parameter	Type	Default	Description
N	int	8	Operand width (fixed at 8 for this variant)

23.3 Ports

```
module math_multiplier_dadda_4to2_008 #(
    parameter int N = 8
) (
    input  logic [N-1:0]   i_multiplier,
    input  logic [N-1:0]   i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

Port	Direction	Width	Description
i_multiplier	Input	8	Multiplier operand

Port	Direction	Width	Description
			(unsigned)
i_multiplicand	Input	8	Multiplicand operand (unsigned)
ow_product	Output	16	Full-precision product (unsigned)

23.4 Functional Description

23.4.1 Three-Stage Pipeline

1. **Partial Product Generation** - 64 AND gates create 8x8 bit matrix
2. **Dadda Reduction** - 4:2 compressors reduce columns to 2 rows
3. **Final CPA** - Han-Carlson adder produces final product

23.4.2 Partial Product Matrix

For 8x8 multiplication, partial products form a diamond pattern:

```
Column:    0  1  2  3  4  5  6  7  8  9 10 11 12 13 14
Height:    1  2  3  4  5  6  7  8  7  6  5  4  3  2  1

           pp00
        pp01 pp10
     pp02 pp11 pp20
  pp03 pp12 pp21 pp30
pp04 pp13 pp22 pp31 pp40
  ...           ...
                 pp77

Maximum column height = 8 (columns 7)
```

23.4.3 Dadda Reduction with 4:2 Compressors

4:2 Compressor Advantage: - Reduces 5 inputs to 3 outputs (vs 3:2 for standard CSA) - Effectively removes 2 bits from column height per compressor - Cout independent of Cin enables parallel carry chains

Dadda Height Sequence:

```
d(j) = {2, 3, 4, 6, 9, 13, 19, 28, ...}
Rule: d(j+1) = floor(1.5 * d(j))
```

Reduction Stages for 8-bit (max height = 8):

```
Start:  height 8
Stage 1: reduce to 6 (columns > 6 get 4:2 compressors)
Stage 2: reduce to 4
Stage 3: reduce to 3
Stage 4: reduce to 2 (final for CPA)
```

23.4.4 Reduction Example (Column 7)

```
Initial height: 8 bits
  pp07, pp16, pp25, pp34, pp43, pp52, pp61, pp70

Stage 1: Reduce 8 -> 2
  TWO 4:2 compressors consume ALL 8 partial products:
    u_c4to2_07_001: (pp07, pp16, pp25, pp34, cin=0) -> sum1, carry1,
cout1
    u_c4to2_07_002: (pp43, pp52, pp61, pp70, cin=0) -> sum2, carry2,
cout2
  Remaining: sum1, sum2 (this column) + carry1, cout1, carry2, cout2
(to col 8)
  Height after: 2, plus what arrives from column 6

Stage 2: Reduce to 4
  Continue with 4:2 and 3:2 compressors as needed

...

Final: 2 rows for CPA input
```

23.4.5 Component Instantiation

```
// Partial products (64 AND gates)
wire w_pp_0_0 = i_multiplier[0] & i_multiplicand[0];
wire w_pp_0_1 = i_multiplier[0] & i_multiplicand[1];
// ... 64 total

// 4:2 Compressors for reduction
math_compressor_4to2 u_c4to2_07_001 (
    .i_x1(w_pp_0_7), .i_x2(w_pp_1_6),
    .i_x3(w_pp_2_5), .i_x4(w_pp_3_4),
    .i_cin(1'b0),
    .ow_sum(w_c4to2_sum_07_001),
    .ow_carry(w_c4to2_carry_07_001),
    .ow_cout(w_c4to2_cout_07_001)
);

// Column 1 has only two partial products: no half adder in this
```

```
design --
// w_pp_0_1 and w_pp_1_0 feed the final CPA directly (see the CPA
section)

// Full adders for 3-input reduction
math_adder_full u_fa_02_000 (
    .i_a(w_pp_0_2), .i_b(w_pp_1_1), .i_c(w_pp_2_0),
    .ow_sum(w_fa_sum_02_000),
    .ow_carry(w_fa_carry_02_000)
);

// Final CPA
math_adder_han_carlson_016 u_final_cpa (
    .i_a(w_cpa_a),
    .i_b(w_cpa_b),
    .i_cin(1'b0),
    .ow_sum(ow_product),
    .ow_cout(w_cpa_cout)
);
```

23.5 Timing Characteristics

Metric	Value
Partial Product Gen	1 gate level (AND)
Dadda Reduction	~4 stages of 4:2 compressors
Final CPA	5-6 levels (Han-Carlson 16-bit)
Total Depth	~13-15 gate levels

23.5.1 Critical Path

```
a middle-column partial product (e.g. pp_3_4, column 7) -> Stage 1 4:2
->
Stage 2 4:2 -> Stage 3 4:2 -> Stage 4 FA/4:2 -> CPA -> ow_product[15]

(pp_7_7 is NOT on the critical path -- it enters at stage 4 directly.
Stage 4 holds 2 full adders + 4 compressors; the design has zero half
adders, matching the resource table.)
```

23.5.2 Resource Utilization

Component	Count
AND gates (partial products)	64
4:2 Compressors	39
Full Adders (3:2)	2

Component	Count
Half Adders	0
Han-Carlson 16-bit CPA	1
Total LUTs (est.)	~120-140

23.5.3 Comparison: 4:2 vs 3:2 Dadda

Metric	3:2 CSA Dadda	4:2 Compressor Dadda
Reduction stages	4 (measured, 8x8)	4 (measured, 8x8)
Total adders/compressors	42 (measured: 35 CSA + 7 half adders)	41 (measured: 39 c4:2 + 2 FA)
Logic depth	~13-15	~13-15
Area (LUTs)	~130	~130

Advantage: at 8x8 the 4:2 variant saves one cell (41 vs 42) at the SAME 4 stages – the win is cell count and regularity, not stage count (an earlier ~25%-fewer-stages claim was measured at 4 vs 4 and removed).

23.6 Usage Examples

23.6.1 Basic 8x8 Multiplication

```
logic [7:0] a, b;
logic [15:0] product;

math_multiplier_dadda_4to2_008 u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product)
);

// Example: 12 * 13 = 156
initial begin
    a = 8'd12;
    b = 8'd13;
    #1;
    assert(product == 16'd156);
end
```

23.6.2 In BF16 Mantissa Multiplier

```
// BF16 mantissa with implied leading 1
wire [7:0] w_mant_a_ext = {i_a_is_normal, i_mant_a};  // 1.mantissa
wire [7:0] w_mant_b_ext = {i_b_is_normal, i_mant_b};  // 1.mantissa

wire [15:0] w_product;

math_multiplier_dadda_4to2_008 u_mant_mult (
    .i_multiplier(w_mant_a_ext),
    .i_multiplicand(w_mant_b_ext),
    .ow_product(w_product)
);

// Product is in format: 1x.xxxxxx_xxxxxxxx or 01.xxxxxx_xxxxxxxx
wire w_needs_norm = w_product[15];  // If MSB set, product >= 2.0
```

23.6.3 With Pipeline Register

```
logic [7:0] a_reg, b_reg;
logic [15:0] product_comb, product_reg;

// Input register
always_ff @(posedge clk) begin
    a_reg <= a;
    b_reg <= b;
end

// Multiplier (combinational)
math_multiplier_dadda_4to2_008 u_mult (
    .i_multiplier(a_reg),
    .i_multiplicand(b_reg),
    .ow_product(product_comb)
);

// Output register
always_ff @(posedge clk) begin
    product_reg <= product_comb;
end
```

23.7 Design Notes

23.7.1 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Dadda scheduling minimizes total compressor count 2. **Wire complexity** - Regular structure from systematic reduction 3.

Regularity - 4:2 compressors give a more uniform column structure (measured stage count matches the 3:2 tree at 8x8)

23.7.2 Advantages

- **Fast** - O(log N) depth for reduction + CPA
- **Area-efficient** - Dadda scheduling minimizes hardware
- **Regular structure** - Easy to synthesize and optimize
- **Full precision** - No truncation of result

23.7.3 Limitations

- **Unsigned only** - Requires sign handling for signed operands
- **Fixed width** - N=8 is hardcoded in this variant
- **Combinational** - May need pipelining for high frequency

23.7.4 Signed Multiplication

For signed multiplication, use absolute values and fix sign:

```
// Convert to unsigned
wire [7:0] a_abs = a[7] ? (~a + 1'b1) : a;
wire [7:0] b_abs = b[7] ? (~b + 1'b1) : b;
wire       sign_result = a[7] ^ b[7];

// Multiply unsigned
math_multiplier_dadda_4to2_008 u_mult (
    .i_multiplier(a_abs),
    .i_multiplicand(b_abs),
    .ow_product(product_unsigned)
);

// Fix sign
wire [15:0] product_signed = sign_result ? (~product_unsigned + 1'b1)
                                          : product_unsigned;
```

23.7.5 Auto-Generated Code

This module is auto-generated by Python scripts: - **Generator:** `bin/rtl_generators/bf16/dadda_4to2_multiplier.py` - **Regenerate:** `PYTHONPATH=bin:$PYTHONPATH python3 bin/rtl_generators/bf16/generate_all.py rtl/math`

Do not edit the generated .sv file manually. Modify the generator script instead.

23.8 Related Modules

- **math_compressor_4to2** - 4:2 compressor building block

- **math_adder_full** - Full adder (3:2 compressor)
- **math_adder_half** - Half adder
- **math_adder_han_carlson_016** - Final CPA
- **math_bf16_mantissa_mult** - BF16 wrapper using this multiplier
- **math_multiplier_dadda_tree_008** - Alternative 3:2 CSA version

23.9 Testing

From the test suite (`val/math/test_math_multiplier_dadda_4to2.py`):

- **Key test scenarios**:
  - This multiplier is used in the BF16 mantissa multiplication (8x8).

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

23.10 References

- Dadda, L. "Some Schemes for Parallel Multipliers." Alta Frequenza 34, 1965.
- Weinberger, A. "4:2 Carry-Save Adder Module." IBM Technical Disclosure Bulletin, 1981.
- Oklobdzija, V.G. "A Method for Speed Optimized Partial Product Reduction." IEEE Trans. Computers, 1996.

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24 Dadda Tree Multipliers

Area-optimized fast parallel multipliers using scheduled partial product reduction with carry-save adders. These modules provide unsigned integer multiplication for 8×8, 16×16, and 32×32 operations. The reduction tree has logarithmic depth and uses the minimum number of compressors for that depth; the final carry-propagate adder is a Brent-Kung parallel-prefix adder, which is also logarithmic depth, so end-to-end delay is O(log N).

24.1 Overview

The Dadda tree multiplier family implements high-speed multiplication using an **optimized schedule** of 3:2 compressors (carry-save adders). Unlike Wallace trees, which compress everything they can as early as they can, Dadda trees defer compression: each stage only reduces columns down to a precomputed target height. This reaches height 2 in the same number of stages as Wallace while instantiating measurably fewer compressors.

Key Features: - **Logarithmic reduction depth** - 4 stages for 8-bit, 6 for 16-bit, 8 for 32-bit - **Minimum compressor count** for that depth - 42 cells for 8×8 versus Wallace's 61 - **Structured reduction** - follows the canonical Dadda target-height sequence - **Fixed-width variants** - generated for 8, 16, and 32-bit operands - **Purely combinational** - single-cycle multiplication - **Self-contained** - includes its own final adder; no external adder required

Architecture: 1. **Partial Product Generation** - AND gates create the N×N matrix 2. **Dadda Reduction Schedule** - scheduled CSA stages reduce every column to height 2 3. **Final Addition** - an on-chip Brent-Kung parallel-prefix carry-propagate adder sums the two surviving rows into `ow_product`

Note on the final adder: these modules are complete multipliers. The reduction tree stops at column height 2, the two remaining rows are packed into the 2N-bit vectors `w_cpa_row0` / `w_cpa_row1`, and those are summed internally by a single `math_adder_brent_kung_{2N}` instance named `u_final_cpa`. Earlier revisions of this module used a ripple carry-propagate adder in that position, and revisions before that collapsed every column to height 1 with no final adder at all; neither is the case now, and no external adder is needed.

The CPA width is the **product** width, not the operand width: the 8-bit multiplier instantiates `math_adder_brent_kung_016`, the 16-bit one `math_adder_brent_kung_032`, and the 32-bit one `math_adder_brent_kung_064`. Carry-in is tied to `1'b0`, and the adder's carry-out is left unread on `w_cpa_carry_unused` - an N x N product is strictly less than 2**(2N), so the top column can never carry out.

24.2 Parameters

Parameter	Type	Default	Description
N	int	8/16/32	Bit width (fixed per variant)

Note: The `N` parameter is present but fixed per module variant. It is not intended for user modification.

24.3 Ports

24.3.1 8-bit Dadda Tree Multiplier

```
module math_multiplier_dadda_tree_008 #(
    parameter int N = 8
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
```

```
    output logic [2*N-1:0] ow_product
);
```

24.3.2 16-bit Dadda Tree Multiplier

```
module math_multiplier_dadda_tree_016 #(
    parameter int N = 16
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

24.3.3 32-bit Dadda Tree Multiplier

```
module math_multiplier_dadda_tree_032 #(
    parameter int N = 32
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

Port	Direction	Width	Description
i_multiplier	Input	N	Multiplier operand (unsigned)
i_multiplicand	Input	N	Multiplicand operand (unsigned)
ow_product	Output	2N	Product result (unsigned)

Signal Types: - **Unsigned only** - All operands and results are unsigned integers - **Full precision** - Output is full 2N-bit product (no truncation)

24.4 Functional Description

24.4.1 Dadda Reduction Algorithm

The Dadda tree uses an **optimized reduction schedule** rather than immediate reduction:

Step 1: Calculate Reduction Schedule

```
Define target heights: d(1) = 2, d(j+1) = 1.5 × d(j)
Sequence: 2, 3, 4, 6, 9, 13, 19, 28, 42, ...
```

Pick the smallest sequence entry that is greater than or equal to the tallest column, then walk the sequence downwards. Each reduction stage takes every column to at or below the next target height.

Step 2: Partial Product Generation

```
For N×N multiplication:
- Generate N² partial products: PP[i][j] = multiplier[i] &
multiplicand[j]
- Arrange in diagonal columns (like manual multiplication)
```

Step 3: Scheduled Reduction

```
For each stage k (from high to low in d-sequence):
    For each column:
        While column height > d(k):
            Use Full Adder (3→2 reduction)
        If column height == d(k) and can compress:
            Use Half Adder (2→2 compression)
        Otherwise:
            Pass through to next stage
```

Step 4: Final Addition

```
Every column is now at height 2. Pack the two surviving rows into two
2N-bit vectors and sum them with a Brent-Kung parallel-prefix
carry-propagate adder to produce the 2N-bit product.
```

Key Difference from Wallace: Dadda **waits** to reduce until a column exceeds the stage target, so it spends the fewest compressors that still reach height 2 in the same number of stages Wallace needs.

24.4.2 Dadda Sequence Example

For 8-bit multiplication:

```
Initial heights (15 columns): [1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3,
2, 1]

Tallest column is 8, so the first target is the sequence entry above
it: 9
is not needed because stage targets are applied downwards from 6.

Stage targets: 6 → 4 → 3 → 2

Stage 1: Reduce to height 6
Stage 2: Reduce to height 4
Stage 3: Reduce to height 3
```

```
Stage 4: Reduce to height 2
  - All 15 columns now at height 2; hand off to the final adder
```

The longer sequences apply to the wider variants:

Variant	Tallest column	Stage targets	Stages
8-bit	8	6 → 4 → 3 → 2	4
16-bit	16	13 → 9 → 6 → 4 → 3 → 2	6
32-bit	32	28 → 19 → 13 → 9 → 6 → 4 → 3 → 2	8

24.4.3 8-bit Implementation Structure

Generated signals are named `w_sum_{column}_{op}` / `w_carry_{column}_{op}`, and instances are named `CSA_{column}_{op}` or `HA__{column}_{op}`, where op is the operation index within that column. Excerpts below are taken verbatim from `rtl/math/math_multiplier_dadda_tree_008.sv`.

```
// Partial Products (64 AND gates for 8×8)
wire w_pp_0_0 = i_multiplier[0] & i_multiplicand[0];
wire w_pp_0_1 = i_multiplier[0] & i_multiplicand[1];
// ... 64 total partial products
wire w_pp_7_7 = i_multiplier[7] & i_multiplicand[7];

// Dadda reduction stage 1: max column height 6
// Column 6 has height 7, so exactly one 2:2 compression brings it to
6.
wire w_sum_06_01, w_carry_06_01;
math_adder_half HA__06_01 (
    .i_a(w_pp_0_6),
    .i_b(w_pp_1_5),
    .ow_sum(w_sum_06_01),
    .ow_carry(w_carry_06_01)
);

// Column 7 has height 8 and needs a 3:2 plus a 2:2 to reach 6.
wire w_sum_07_01, w_carry_07_01;
math_adder_carry_save CSA_07_01 (
    .i_a(w_pp_0_7),
    .i_b(w_pp_1_6),
    .i_c(w_pp_2_5),
    .ow_sum(w_sum_07_01),
    .ow_carry(w_carry_07_01)
);
```

```
wire w_sum_07_02, w_carry_07_02;
math_adder_half HA__07_02 (
    .i_a(w_pp_3_4),
    .i_b(w_pp_4_3),
    .ow_sum(w_sum_07_02),
    .ow_carry(w_carry_07_02)
);

// ... stages 2 and 3 follow the same pattern with targets 4 and 3

// Dadda reduction stage 4: max column height 2
// Later stages consume sums and carries produced by earlier stages.
wire w_sum_04_03, w_carry_04_03;
math_adder_carry_save CSA_04_03 (
    .i_a(w_sum_04_01),
    .i_b(w_carry_03_01),
    .i_c(w_sum_04_02),
    .ow_sum(w_sum_04_03),
    .ow_carry(w_carry_04_03)
);
```

Once every column is at height 2, the two surviving rows are packed into a pair of 16-bit vectors and handed to one Brent-Kung prefix adder. There are no `math_adder_full` or `math_adder_half` instances in this stage at all - every half and full adder in the file belongs to the reduction tree:

```
    // Final addition stage: two reduced rows into a Brent-Kung CPA
    wire [15:0] w_cpa_row0 = {
        1'b0,
        w_pp_7_7,
        // ... one bit per column, taken from the surviving row
        w_pp_2_0,
        w_pp_0_1,
        w_pp_0_0
    };
    wire [15:0] w_cpa_row1 = {
        1'b0,
        w_carry_13_01,
        w_sum_13_01,
        // ... one bit per column, taken from the other surviving row
        w_sum_02_01,
        w_pp_1_0,
        1'b0
    };

    /* verilator lint_off UNUSEDSIGNAL */
    wire w_cpa_carry_unused;
    /* verilator lint_on UNUSEDSIGNAL */
```

```
math_adder_brent_kung_016 #(
    .N(16)
) u_final_cpa (
    .i_a(w_cpa_row0),
    .i_b(w_cpa_row1),
    .i_c(1'b0),
    .ow_sum(ow_product),
    .ow_carry(w_cpa_carry_unused)
);
```

The prefix adder drives `ow_product` directly, so there is no per-bit `assign ow_product[i]` fan-out stage either.

24.4.4 Reduction Comparison: Dadda vs Wallace

Both trees reach column height 2 in the **same number of stages**. The difference is entirely in how many compressors they spend getting there. The following counts are instance counts from the generated RTL.

For 8×8 multiplication:

	Dadda	Wallace
Reduction stages / layers	4	4
Reduction 3:2 compressors	35	36
Reduction half adders	7	25
Total reduction cells	**42**	**61**
Final CPA	`math_adder_brent_kung_016`	`math_adder_brent_kung_016`

The 8×8 figure of 35 full adders and 7 half adders is exactly the textbook Dadda count.

Across all widths:

Width	Stages	Reduction CSAs	Reduction HAs	Final CPA
8-bit	4	35	7	`math_adder_brent_kung_016`
16-bit	6	195	15	`math_adder_brent_kung_032`
32-bit	8	899	31	`math_adder_brent_kung_064`

Result: Dadda reaches height 2 in the same depth as Wallace while spending 19 fewer cells at 8×8.

Wallace used to buy something back for that extra hardware. When both multipliers ended in a ripple CPA, Wallace's eager compression flattened the low-order columns to height 1 early, so its ripple spanned only 11 columns against Dadda's 14 - a shorter serial carry chain that partly offset the larger tree. **That offset no longer exists.** Both multipliers now feed a full-width Brent-Kung prefix adder over all 2N columns, and a prefix adder's depth is logarithmic in its width regardless of how many low-order inputs happen to be zero. The two designs therefore have an *identical* final adder, in both cell count and delay.

What remains is the pure statement of the tradeoff: Wallace spends more compressor cells than Dadda to reach the same reduction depth, and now gets nothing in return. Dadda is the better choice on both axes.

24.4.5 Algorithm Reference: Dadda Sequence

The Dadda reduction sequence is defined as:

```
d(1) = 2           (final stage: 2 rows)
d(j+1) = 1.5 × d(j)  (recursive definition)

Sequence: 2, 3, 4, 6, 9, 13, 19, 28, 42, 63, 94, 141, ...
```

For N×N multiplication: 1. Find maximum column height (N for the middle column) 2. Find the largest sequence entry strictly below max_height - that is the first stage target 3. Work downwards through the sequence until the target is 2

Example (8×8): - Max height = 8 - Largest sequence entry below 8 is 6, so that is the first target - Stage targets: 6 → 4 → 3 → 2 (four stages)

This matches the generated RTL, whose stage comments read `Dadda reduction stage 1: max column height 6` through `stage 4: max column height 2`.

24.5 Timing Characteristics

Metric	8-bit	16-bit	32-bit
Logic Depth	~13-15 levels	~17-20 levels	~22-28 levels
Typical Delay (ns)	~6.5-7.5	~8.5-10.5	~11-14
Max Frequency	~130-150 MHz	~95-115 MHz	~70-90 MHz

Logic Depth Breakdown: - Partial product generation: 1 level (AND gates) - Dadda reduction stages: logarithmic - 4 / 6 / 8 stages for 8 / 16 / 32-bit - Final addition: on-chip **Brent-Kung** parallel-prefix carry-propagate adder, O(log N) levels

Critical Path:

```
a middle-column partial product (the tallest column feeds the most CSA
levels; corner PPs like PP[N-1][N-1] join late and are NOT critical)
→ Stage 1 CSA → ... → Stage K CSA
→ Brent-Kung prefix network (~2·log2(2N) levels) → ow_product[2N-1]
```

Important: both halves of the datapath are logarithmic. The reduction tree is log-depth, and the final adder is a Brent-Kung parallel-prefix adder, which is also log-depth, so the **end-to-end delay of these modules is O(log N)**. There is no serial carry chain anywhere in the design. The 32-bit variant, for example, follows an 8-stage tree with a 64-bit prefix network rather than the 61-deep ripple it used to carry.

24.5.1 Resource Utilization

Instance counts below are exact, taken from the generated RTL. “3:2 cells” counts `math_adder_carry_save` and “half adders” counts `math_adder_half`; every one of them lives in the reduction tree. The final CPA is a single prefix-adder instance and contributes no discrete adder cells, so `math_adder_full` does not appear in these files at all.

Width	3:2 cells (tree)	Half adders (tree)	Final CPA	AND Gates	Total adder cells
8-bit	35	7	1 × `math_adder_brent_kung_016`	64	42
16-bit	195	15	1 × `math_adder_brent_kung_032`	256	210

Width	3:2 cells (tree)	Half adders (tree)	Final CPA	AND Gates	Total adder cells
32-bit	899	31	1 × `math_adder_brent_kung_064`	1024	930

Area Comparison (measured cell counts, Dadda versus Wallace):

Both families now instantiate the same Brent-Kung CPA at the same width, so the adder-cell totals below compare reduction trees on equal terms.

Width	Dadda total cells	Wallace total cells	Dadda saving
8-bit	42	61	31%
16-bit	210	274	23%
32-bit	930	1116	17%

Totals count every instantiated discrete adder cell in the corresponding generated `.sv` file, all of which are in the reduction tree. The shared Brent-Kung CPA is identical in both families and is excluded from the totals; adding it back shifts both columns by the same amount. LUT and gate-level area will differ by technology and synthesis settings; these are structural counts, not synthesis results.

Key Advantage: for the same reduction depth, Dadda instantiates substantially fewer compressor cells than Wallace.

24.6 Usage Examples

24.6.1 Basic 8×8 Multiplication

```
logic [7:0] a, b;
logic [15:0] product;

math_multiplier_dadda_tree_008 u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product)
);

// Example: 12 × 13 = 156
initial begin
    a = 8'd12;
    b = 8'd13;
```

```
    #1;  // Allow combinational delay
    assert(product == 16'd156);
end
```

24.6.2 16×16 Multiplication with Pipeline

```
logic [15:0] a, b;
logic [31:0] product_comb, product_reg;
logic clk, rst_n;

// Dadda tree multiplier (combinational)
math_multiplier_dadda_tree_016 u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product_comb)
);

// Output register for timing closure
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)
        product_reg <= '0;
    else
        product_reg <= product_comb;
end
```

24.6.3 Multiply-Accumulate (MAC) Unit

```
module mac_unit (
    input  logic        clk,
    input  logic        rst_n,
    input  logic [15:0] a, b,
    input  logic        accumulate,  // 1=accumulate, 0=clear
    output logic [31:0] result
);

    logic [31:0] product;
    logic [31:0] accumulator;

    // Dadda tree multiplier
    math_multiplier_dadda_tree_016 u_mult (
        .i_multiplier(a),
        .i_multiplicand(b),
        .ow_product(product)
    );

    // Accumulator
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n)
```

```
            accumulator <= '0;
        else if (accumulate)
            accumulator <= accumulator + product;
        else
            accumulator <= product;  // Clear and load
    end

    assign result = accumulator;

endmodule
```

24.6.4 FIR Filter Tap

```
module fir_tap #(
    parameter int DATA_WIDTH = 16,
    parameter int COEFF_WIDTH = 16
) (
    input  logic                        clk,
    input  logic                        rst_n,
    input  logic [DATA_WIDTH-1:0]       data_in,
    input  logic [COEFF_WIDTH-1:0]      coefficient,
    input  logic [31:0]                 partial_sum_in,
    output logic [31:0]                 partial_sum_out
);

    logic [31:0] product;

    // Multiply coefficient by data
    math_multiplier_dadda_tree_016 u_mult (
        .i_multiplier(coefficient),
        .i_multiplicand(data_in),
        .ow_product(product)
    );

    // Add to partial sum (pipeline stage)
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            partial_sum_out <= '0;
        else
            partial_sum_out <= partial_sum_in + product;
    end

endmodule
```

24.6.5 Parameterized Multiplier Selector

```
module flexible_multiplier #(
    parameter int WIDTH = 8
```

```
) (
    input  logic [WIDTH-1:0]   i_multiplier,
    input  logic [WIDTH-1:0]   i_multiplicand,
    output logic [2*WIDTH-1:0] ow_product
);

    generate
        if (WIDTH == 8) begin : gen_8bit
            math_multiplier_dadda_tree_008 u_mult (
                .i_multiplier(i_multiplier),
                .i_multiplicand(i_multiplicand),
                .ow_product(ow_product)
            );
        end else if (WIDTH == 16) begin : gen_16bit
            math_multiplier_dadda_tree_016 u_mult (
                .i_multiplier(i_multiplier),
                .i_multiplicand(i_multiplicand),
                .ow_product(ow_product)
            );
        end else if (WIDTH == 32) begin : gen_32bit
            math_multiplier_dadda_tree_032 u_mult (
                .i_multiplier(i_multiplier),
                .i_multiplicand(i_multiplicand),
                .ow_product(ow_product)
            );
        end else begin : gen_default
            // Fallback: behavioral multiplication
            assign ow_product = i_multiplier * i_multiplicand;
        end
    endgenerate

endmodule
```

24.7 Design Notes

24.7.1 Advantages

- **Smallest fast multiplier** - 17-31% fewer adder cells than Wallace at the same reduction depth, with an identical final adder
- **Optimized structure** - Mathematically proven reduction schedule
- **Fewer cells to synthesize** - same stage count as Wallace, less hardware in each stage
- **Pure combinational** - Easy to pipeline where needed
- **Scalable** - Algorithm extends to any bit width

24.7.2 Limitations

- **Complex design** - Reduction schedule requires calculation
- **Unsigned only** - Requires sign handling for signed operands
- **Fixed width** - Not parameterizable (instantiate specific variant)
- **Irregular structure** - Not as regular as array multipliers

24.7.3 When to Use Dadda Tree

Appropriate Use Cases: - Production designs requiring fast multiplication - Area-constrained high-speed applications - DSP functions (filters, FFT, correlation) - Unsigned integer arithmetic - **Default choice** for most multiplication needs

Consider Alternatives When: - Operands are signed → Booth multiplier (fewer PPs) - Very low area required → Array multiplier - Variable width needed → Behavioral * operator - Ultra-high frequency → Multi-stage pipelined multiplier

24.7.4 Dadda vs Wallace Decision

Choose Dadda When: - Production code (better area-speed balance) - Area matters - Targeting ASIC (fewer gates = lower cost)

Choose Wallace When: - Educational purposes (simpler to understand) - Existing design uses it (consistency) - Specific timing requirements favor it

In Practice: Dadda is preferred for almost all production designs.

24.7.5 Multiplier Architecture Selection

Requirement	Best Choice	Reasoning
High-speed unsigned	**Dadda Tree**	Log-depth tree with the fewest compressors
Signed multiplication	Booth Radix-4	Fewer partial products
Minimal area	Array Multiplier	Sequential, low gates
Variable width	Behavioral (*)	Synthesis optimizes
Very high frequency	Pipelined Dadda	Split into stages

24.7.6 Signed Multiplication

Dadda trees output **unsigned products only**. For signed multiplication:

Option 1: Sign Extension and Correction

```
logic [N-1:0] a, b;
logic [2*N-1:0] product_unsigned, product_signed;
logic sign_bit;

// Compute absolute values
assign a_abs = a[N-1] ? (~a + 1'b1) : a;
assign b_abs = b[N-1] ? (~b + 1'b1) : b;
assign sign_bit = a[N-1] ^ b[N-1];

// Multiply unsigned
math_multiplier_dadda_tree_008 u_mult (
    .i_multiplier(a_abs),
    .i_multiplicand(b_abs),
    .ow_product(product_unsigned)
);

// Apply sign
assign product_signed = sign_bit ? (~product_unsigned + 1'b1) :
product_unsigned;
```

Option 2: Use Booth Multiplier (More efficient for native signed)

24.7.7 Pipelining for High Frequency

2-Stage Pipeline:

```
// Stage 1: Register inputs
always_ff @(posedge clk) begin
    a_reg <= a;
    b_reg <= b;
end

// Stage 2: Multiply and register output
math_multiplier_dadda_tree_016 u_mult (
    .i_multiplier(a_reg),
    .i_multiplicand(b_reg),
    .ow_product(product_comb)
);

always_ff @(posedge clk) begin
    product_reg <= product_comb;
end
```

4-Stage Pipeline (for maximum frequency): Split Dadda tree into pipeline stages based on reduction schedule.

24.7.8 Common Pitfalls

Anti-Pattern 1: Expecting width parameterization

```
// WRONG: N parameter is fixed
math_multiplier_dadda_tree_008 #(.N(10)) u_mult (...);  // Won't work!

// RIGHT: Use appropriate fixed variant
math_multiplier_dadda_tree_016 u_mult (...);  // Use 16-bit
```

Anti-Pattern 2: Using for signed without conversion

```
// WRONG: Signed inputs treated as unsigned
logic signed [7:0] a = -5, b = 3;
logic signed [15:0] product;
math_multiplier_dadda_tree_008 u_mult (
    .i_multiplier(a),      // Treated as 251 (unsigned)!
    .i_multiplicand(b),    // Treated as 3
    .ow_product(product)   // = 753, not -15!
);

// RIGHT: Convert to unsigned, multiply, fix sign
// (See "Signed Multiplication" section)
```

Anti-Pattern 3: Adding a redundant external adder

```
// WRONG: ow_product is already the final summed product.
// Bolting on another adder both wastes area and computes garbage.
math_multiplier_dadda_tree_008 u_mult (..., .ow_product(p));
assign final_product = p + {carry_vector[14:0], 1'b0};  // No!

// RIGHT: use ow_product directly.
math_multiplier_dadda_tree_008 u_mult
(..., .ow_product(final_product));
```

These modules contain their own final carry-propagate adder. They do not expose separate sum and carry vectors.

Anti-Pattern 4: Ignoring timing at high frequencies

```
// WRONG: 32×32 Dadda at 400 MHz without pipeline
// Critical path ~12-14ns, can't meet 2.5ns period!

// RIGHT: Add pipeline stages
// Target: 1 stage per 5-6ns of delay
```

24.8 Related Modules

- **math_multiplier_wallace_tree_*.sv** - same reduction depth, 17-31% more adder cells, identical final CPA
- **math_adder_brent_kung_016/032/064.sv** - logarithmic-depth parallel-prefix adder used as the final CPA here (8/16/32-bit multipliers use the 016/032/064 widths respectively)
- **math_adder_carry_save.sv** - 3:2 compressor building block; the only compressor these files use
- **math_adder_half.sv** - Half adder primitive, used in the reduction tree only

24.9 Testing

From the test suite (`val/math/test_math_multiplier_dadda.py`):

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

24.10 References

- Dadda, L. "Some Schemes for Parallel Multipliers." Alta Frequenza 34, 1965.
- Wallace, C.S. "A Suggestion for a Fast Multiplier." IEEE Trans. Electronic Computers, 1964.
- Oklobdzija, V.G. "High-Speed VLSI Arithmetic Units: Adders and Multipliers." Springer, 2002.

24.11 Navigation

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25 Wallace Tree Multipliers

25.1 Overview

Fast parallel multipliers built on tree-based partial product reduction with carry-save adders. The family covers unsigned integer multiplication for 8×8, 16×16, and 32×32 operations, and the recipe is the same at every width: AND gates build the N×N partial-product matrix, a tree of 3:2 compressors (carry-save adders) works through it in parallel, and a Brent-Kung parallel-prefix adder sums the two surviving rows. The reduction is built with maximal parallelism—everything that can compress in a given layer does compress, rather than following a schedule. The tree is logarithmic in depth, the final adder is log depth too, so end-to-end delay is O(log N).

Key Features: - **Logarithmic reduction depth** - 4 layers for 8-bit, 6 for 16-bit, 8 for 32-bit - **Maximal parallelism** - every group of 3 in a column compresses in the same layer - **Structural implementation** - explicit full adder and half adder instantiation - **Fixed-width variants** - generated for 8, 16, and 32-bit operands - **Purely combinational** - single-cycle multiplication - **Self-contained** - includes its own final adder; no external adder required

Architecture: 1. **Partial Product Generation** - AND gates create the N×N matrix 2. **Wallace Reduction Layers** - parallel CSA layers reduce every column to height 2 3. **Final Addition** - an on-chip Brent-Kung parallel-prefix carry-propagate adder sums the two surviving rows into `ow_product`

A note on the final adder, because this trips people up: these modules are complete multipliers. The reduction tree stops at column height 2, the two remaining rows are packed into the 2N-bit vectors `w_cpa_row0` / `w_cpa_row1`, and a single `math_adder_brent_kung_{2N}` instance named `u_final_cpa` sums them internally. Earlier revisions of this module used a ripple carry-propagate adder in that position, and revisions before that collapsed every column to height 1 with no final adder at all. Neither is the case now, and no external adder is needed.

The CPA width is the **product** width, not the operand width: the 8-bit multiplier instantiates `math_adder_brent_kung_016`, the 16-bit one `math_adder_brent_kung_032`, and the 32-bit one `math_adder_brent_kung_064`. Carry-in is tied to `1'b0`, and the adder's carry-out goes unread on `w_cpa_carry_unused`—an N x N product is strictly less than 2**(2N), so the top column can never carry out.

Two variants are generated. `math_multiplier_wallace_tree_NNN` builds its reduction tree from `math_adder_full`; `math_multiplier_wallace_tree_csa_NNN` builds it from `math_adder_carry_save`. Both cells are 3:2 compressors, so the two trees come out structurally identical—same instance counts, same topology. The final CPA is the same `math_adder_brent_kung_{2N}` instance in **both** variants. Both are standalone top-level multipliers with the same port list. Pick either.

25.2 Parameters

Parameter	Type	Default	Description
N	int	8/16/32	Bit width (fixed per variant)

Note: `N` is present but fixed per module variant. It's not intended for user modification.

25.3 Ports

25.3.1 Module Declarations

25.3.1.1 8-bit Wallace Tree Multiplier

```
module math_multiplier_wallace_tree_008 #(
    parameter int N = 8
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

25.3.1.2 16-bit Wallace Tree Multiplier

```
module math_multiplier_wallace_tree_016 #(
    parameter int N = 16
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

25.3.1.3 32-bit Wallace Tree Multiplier

```
module math_multiplier_wallace_tree_032 #(
    parameter int N = 32
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

25.3.2 Port List

Port	Direction	Width	Description
i_multiplier	Input	N	Multiplier operand (unsigned)
i_multiplicand	Input	N	Multiplicand operand (unsigned)
ow_product	Output	2N	Product result (unsigned)

Signal Types: - **Unsigned only** - All operands and results are unsigned integers - **Full precision** - Output is full 2N-bit product (no truncation)

25.4 Functional Description

25.4.1 Wallace Tree Algorithm

The Wallace tree plays an aggressive reduction game:

Stage 1: Partial Product Generation

```
For N×N multiplication:
- Generate N² partial products: PP[i][j] = multiplier[i] &
multiplicand[j]
- Arrange in diagonal columns (like manual multiplication)
```

Stage 2: Wallace Reduction Layers

Reduction proceeds in **layers**. Within a single layer, every column is partitioned into groups of 3, and all groups across all columns compress **simultaneously**. Layers repeat until every column is at height 2.

```
While any column has height > 2:            // one iteration = one
layer
    For each column, partition its bits into groups of 3:
        - group of 3 bits -> 3:2 compressor (sum stays, carry goes
left)
        - group of 2 bits -> half adder     (sum stays, carry goes
left)
        - group of 1 bit  -> passes through to the next layer
untouched
```

Because the partitioning is done per layer rather than per opportunity, a column of height 8 becomes ceil(8/3) = 3 sums plus carries arriving from the column to its right, and so on down the layers.

Stage 3: Final Addition

```
Every column is now at height 2. Pack the two surviving rows into two
2N-bit vectors and sum them with a Brent-Kung parallel-prefix
carry-propagate adder to produce the 2N-bit product.
```

Key characteristic: Wallace compresses **everything it can as early as it can**. That's the entire distinction from Dadda, which defers compression using per-stage target heights. Both reach height 2 in the same number of layers; Wallace simply spends more compressors getting there.

25.4.2 8-bit Example Structure

Generated signals are named `w_sum_{column}_{layer}_{op}` / `w_carry_{column}_{layer}_{op}`, and instances are named `FA_{column}_{layer}_{op}` (or `CSA_...` in the `_csa_` variant) and `HA_{column}_{layer}_{op}`. The excerpts below are taken verbatim from `rtl/math/math_multiplier_wallace_tree_008.sv`.

```
// Partial Products (64 AND gates for 8×8)
wire w_pp_0_0 = i_multiplier[0] & i_multiplicand[0];
wire w_pp_0_1 = i_multiplier[0] & i_multiplicand[1];
// ... 64 total partial products
wire w_pp_7_7 = i_multiplier[7] & i_multiplicand[7];

// Wallace reduction layer 1
// Column 1 holds only 2 bits, so it gets a half adder.
wire w_sum_01_1_01, w_carry_01_1_01;
math_adder_half HA_01_1_01 (
    .i_a(w_pp_0_1),
    .i_b(w_pp_1_0),
    .ow_sum(w_sum_01_1_01),
    .ow_carry(w_carry_01_1_01)
);

// Column 2 holds 3 bits: one full group, one 3:2 compressor.
wire w_sum_02_1_01, w_carry_02_1_01;
math_adder_full FA_02_1_01 (
    .i_a(w_pp_0_2),
    .i_b(w_pp_1_1),
    .i_c(w_pp_2_0),
    .ow_sum(w_sum_02_1_01),
    .ow_carry(w_carry_02_1_01)
);

// Column 3 likewise; every one of these fires in the same layer.
wire w_sum_03_1_01, w_carry_03_1_01;
math_adder_full FA_03_1_01 (
    .i_a(w_pp_0_3),
    .i_b(w_pp_1_2),
    .i_c(w_pp_2_1),
    .ow_sum(w_sum_03_1_01),
    .ow_carry(w_carry_03_1_01)
);

// ... layers 2, 3, 4 repeat the same grouping on the surviving rows
```

In the `_csa_` variant the identical structure is emitted with `math_adder_carry_save` standing in for `math_adder_full`:

```
// math_multiplier_wallace_tree_csa_008.sv, same position in layer 1
wire w_sum_02_1_01, w_carry_02_1_01;
math_adder_carry_save CSA_02_1_01 (
    .i_a(w_pp_0_2),
    .i_b(w_pp_1_1),
    .i_c(w_pp_2_0),
    .ow_sum(w_sum_02_1_01),
    .ow_carry(w_carry_02_1_01)
);
```

Once every column is at height 2, the two surviving rows are packed into a pair of 16-bit vectors and handed to one Brent-Kung prefix adder. There are no `math_adder_full` or `math_adder_half` instances in this stage at all—every half and full adder in the file belongs to the reduction tree:

```
    // Final addition stage: two reduced rows into a Brent-Kung CPA
    wire [15:0] w_cpa_row0 = {
        w_carry_14_4_01,
        w_carry_13_4_01,
        // ... one bit per column, taken from the surviving row
        w_sum_02_2_01,
        w_sum_01_1_01,
        w_pp_0_0
    };
    wire [15:0] w_cpa_row1 = {
        w_sum_15_4_01,
        w_sum_14_4_01,
        // ... one bit per column, taken from the other surviving row
        w_sum_05_4_01,
        1'b0,
        1'b0,
        1'b0,
        1'b0,
        1'b0
    };

    /* verilator lint_off UNUSEDSIGNAL */
    wire w_cpa_carry_unused;
    /* verilator lint_on UNUSEDSIGNAL */
    math_adder_brent_kung_016 #(
        .N(16)
    ) u_final_cpa (
        .i_a(w_cpa_row0),
        .i_b(w_cpa_row1),
        .i_c(1'b0),
        .ow_sum(ow_product),
        .ow_carry(w_cpa_carry_unused)
    );
```

Wallace's eager compression has already flattened columns 0-4 to height 1, and that shows up as the five `1'b0` entries at the bottom of `w_cpa_row1`. It no longer buys any delay, though: the prefix adder still spans all 16 columns, and its depth is logarithmic in that width whether or not the low-order inputs are constant zero. The prefix adder drives `ow_product` directly, so there's no per-bit `assign ow_product[i]` fan-out stage either.

25.4.3 Reduction Pattern

Layer counts and instance counts, measured from the generated RTL:

Width	Layers	Reduction 3:2 compressors	Reduction half adders	Final CPA
8-bit	4	36	25	`math_adder_brent_kung_016`
16-bit	6	196	78	`math_adder_brent_kung_032`
32-bit	8	900	216	`math_adder_brent_kung_064`

For 8×8 multiplication: 64 partial products across 15 columns reduce to 2 rows in 4 layers, then a single 16-bit Brent-Kung prefix CPA sums them.

Number of layers: measured, this is **4 layers for 8-bit, 6 for 16-bit, and 8 for 32-bit**—count the `// Wallace reduction layer N` comments in the generated RTL if you want to check.

Don't trust a closed form here. The often-quoted $\log_{1.5}(N)$ is wrong: it describes reduction from N rows down to 1, and this tree stops at 2. Correcting it to $\log_{1.5}(N/2)$ fixes 8-bit and 16-bit but still under-predicts 32-bit—it gives 7 where the generator emits 8:

Layer-count formulas against the generated RTL

N	$\log_{1.5}(N)$	$\log_{1.5}(N/2)$	measured
8	6	4	**4**
16	7	6	**6**
32	9	7	**8**

Both formulas assume every column shrinks by a clean 3:2 every layer. It doesn't. A column whose height isn't a multiple of 3 leaves a remainder that passes through untouched, and carries arriving from the column below can raise a column between layers. Those two effects cost an extra layer by the time N reaches 32. Take the layer count from the RTL, not from a formula.

25.4.4 The _csa_ Variant

math_multiplier_wallace_tree_csa_008/016/032.sv are **standalone top-level multipliers**, not internal sub-components. Each declares the same module interface as the plain variant:

```
module math_multiplier_wallace_tree_csa_008 #(
    parameter int N = 8
) (
    input  logic [  N-1:0] i_multiplier,
    input  logic [  N-1:0] i_multiplicand,
    output logic [2*N-1:0] ow_product
);
```

The only difference is which cell builds the reduction tree:

	Plain variant	_csa_ variant
Reduction tree cell	`math_adder_full`	`math_adder_carry_save`
Final CPA cell	`math_adder_brent_kung_{16,32,64}`	`math_adder_brent_kung_{16,32,64}`
Tree topology	identical	identical
Instance counts	identical	identical

Both cells are 3:2 compressors, so the trees are structurally identical. The plain variant is **not** built out of the _csa_ variant—they're two independent generated modules, and neither instantiates the other. Instantiate whichever one you prefer.

25.5 Timing Characteristics

Metric	8-bit	16-bit	32-bit
Logic Depth	~14-16 levels	~18-22 levels	~24-30 levels

Metric	8-bit	16-bit	32-bit
Typical Delay (ns)	~7-8	~9-11	~12-15
Max Frequency	~125 MHz	~90-110 MHz	~65-80 MHz

Logic Depth Breakdown: - Partial product generation: 1 level (AND gates) - Wallace reduction layers: logarithmic - 4 / 6 / 8 layers for 8 / 16 / 32-bit - Final addition: on-chip **Brent-Kung** parallel-prefix carry-propagate adder, O(log N) levels

Critical Path:

```
i_multiplier[N-1] → PP generation → Layer 1 → Layer 2 → ... → Layer K
→ Brent-Kung prefix network (~2·log2(2N) levels) → ow_product[2N-1]
```

One thing worth stating plainly: both halves of the datapath are logarithmic. The reduction tree is log-depth, and the final adder is a Brent-Kung parallel-prefix adder, which is also log-depth, so the **end-to-end delay of these modules is O(log N)**. There is no serial carry chain anywhere in the design. The 32-bit variant, for example, follows an 8-layer tree with a 64-bit prefix network rather than the 54-deep ripple it used to carry.

Note: Actual timing depends heavily on synthesis optimization and target technology.

25.6 Usage Examples

25.6.1 Basic 8×8 Multiplication

```
logic [7:0] a, b;
logic [15:0] product;

math_multiplier_wallace_tree_008 u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product)
);

// Example: 15 × 17 = 255
initial begin
    a = 8'd15;
    b = 8'd17;
    #1;  // Allow combinational delay
    assert(product == 16'd255);
end
```

25.6.2 16×16 Multiplication with Pipeline Register

```
logic [15:0] a, b;
logic [31:0] product_comb, product_reg;
logic clk, rst_n;

// Wallace tree multiplier (combinational)
math_multiplier_wallace_tree_016 u_mult (
    .i_multiplier(a),
    .i_multiplicand(b),
    .ow_product(product_comb)
);

// Optional output register for pipelining
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)
        product_reg <= '0;
    else
        product_reg <= product_comb;
end
```

25.6.3 32×32 Multiplication for DSP

```
module dsp_multiply (
    input  logic        clk,
    input  logic        rst_n,
    input  logic [31:0] coeff,      // Filter coefficient
    input  logic [31:0] sample,     // Input sample
    input  logic        valid_in,
    output logic [63:0] product,
    output logic        valid_out
);

    logic [63:0] product_comb;
    logic [63:0] product_pipe;
    logic        valid_pipe;

    // Wallace tree multiplier
    math_multiplier_wallace_tree_032 u_mult (
        .i_multiplier(coeff),
        .i_multiplicand(sample),
        .ow_product(product_comb)
    );

    // Pipeline register
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            product_pipe <= '0;
```

```
            product      <= '0;
            valid_pipe   <= 1'b0;
            valid_out    <= 1'b0;
        end else begin
            product_pipe <= product_comb;
            product      <= product_pipe;  // 2 stages, like the valid chain --
                                           // a 1-stage product with a 2-stage
                                           // valid shows the NEXT input's result
            valid_pipe   <= valid_in;
            valid_out    <= valid_pipe;
        end
    end

endmodule
```

25.6.4 Multi-Stage Pipelined Multiplier

```
// Split Wallace tree into pipeline stages for higher frequency
module pipelined_multiplier (
    input  logic        clk,
    input  logic        rst_n,
    input  logic [15:0] a, b,
    output logic [31:0] product
);

    // Stage 1: Partial product generation
    logic [15:0][15:0] pp_stage1;
    genvar i, j;
    generate
        for (i = 0; i < 16; i++) begin : gen_pp
            for (j = 0; j < 16; j++) begin
                always_ff @(posedge clk)
                    pp_stage1[i][j] <= a[i] & b[j];
            end
        end
    endgenerate

    // Stage 2-3: Wallace reduction (using tree, registered)
    // ... intermediate pipeline stages

    // Final stage: Assign product
    // ... final adder and output register

endmodule
```

25.7 Design Notes

25.7.1 Advantages

- **Log-depth end to end** - the tree collapses N rows to 2 in O(log N) layers, versus O(N) for an array multiplier, and the Brent-Kung prefix CPA that follows is O(log N) as well
- **Highly parallel** - Exploits 3:2 compression at all levels
- **No sequential logic** - Pure combinational (easy to pipeline)
- **Unsigned friendly** - Natural fit for unsigned operands
- **Scalable** - Algorithm extends to any bit width

25.7.2 Limitations

- **Large area** - More adder cells than Dadda tree at the same depth (61 versus 42 at 8×8, 1116 versus 930 at 32×32), with no offsetting delay advantage
- **Irregular structure** - Complex synthesis, harder to hand-layout
- **Unsigned only** - Requires additional logic for signed multiplication
- **Fixed width** - Not parameterizable (must instantiate specific variant)
- **Long critical path** - May require pipelining for high-frequency designs

25.7.3 When to Use Wallace Tree

Appropriate Use Cases: - High-speed DSP applications (FIR filters, FFT butterfly) - Single-cycle multiplication requirements - FPGA designs with abundant LUT resources - Unsigned integer multiplication

Consider Alternatives When: - Area is critical → Use Dadda tree (17-31% fewer adder cells at the same depth) - Operands are signed → Use Booth multiplier - Low frequency → Use array multiplier (much smaller) - Variable width needed → Use parameterized array/Booth

25.7.4 Resource Utilization

Instance counts below are exact, taken from the generated RTL. "3:2 cells" counts the reduction-tree compressors (`math_adder_full` in the plain variant, `math_adder_carry_save` in the `_csa_` variant). Every discrete adder cell in these files lives in the reduction tree; the final CPA is a single prefix-adder instance and contributes none.

Width	3:2 cells (tree)	Half adders (tree)	Final CPA	AND Gates	Total adder cells
8-bit	36	25	1 × `math_adder_bre`	64	61

Width	3:2 cells (tree)	Half adders (tree)	Final CPA	AND Gates	Total adder cells
			`nt_kung_016`		
16-bit	196	78	1 × `math_adder_brent_kung_032`	256	274
32-bit	900	216	1 × `math_adder_brent_kung_064`	1024	1116

25.7.5 Comparison to Other Multiplier Architectures

Architecture	Area (relative)	Delay (relative, lower = faster)	Best Use Case
Wallace Tree	**1.2×**	**1.0×**	**High-speed, clearest teaching structure**
Dadda Tree	1.0×	~1.0×	Same depth, fewer compressors
Array Multiplier	0.8×	2.5×	Low-speed, minimal area
Booth (radix-4)	0.9×	1.5×	Signed, reduced partial products

25.7.6 Wallace Tree vs Dadda Tree

For 8×8, both reach column height 2 in **4 layers/stages—the same depth**. What differs is what that depth costs:

	Wallace	Dadda
Layers / stages to height 2	4	4
Reduction	36	35

	Wallace	Dadda
compressors		
Reduction half adders	25	7
Total reduction cells	**61**	**42**
Final CPA	`math_adder_brent_kung_016`	`math_adder_brent_kung_016`

Wallace compresses everything it can as early as it can. Dadda defers, using per-stage target heights to spend the fewest compressors for the same depth. **That is the entire distinction between the two.**

Wallace used to get something back for its extra 19 cells. When both multipliers ended in a ripple CPA, eager compression flattened the low-order columns early, so Wallace's final ripple spanned only 11 columns against Dadda's 14—a shorter serial carry chain that partly paid for the bigger tree. **That offset is gone.** Both now feed a full-width Brent-Kung prefix adder over all 2N columns, and a prefix adder's depth is logarithmic in its width regardless of how many low-order inputs are constant zero. The final adder is now *identical* in the two families, in both cell count and delay, so the extra 19 cells buy nothing.

Measured totals across widths (adder cells in the reduction tree; the shared Brent-Kung CPA is identical in both and excluded):

Width	Wallace total cells	Dadda total cells
8-bit	61	42
16-bit	274	210
32-bit	1116	930

Both use CSA trees; they differ in reduction strategy:

Aspect	Wallace Tree	Dadda Tree
Strategy	Compress everything as early as possible	Defer; compress only down to a per-stage target height
Layers / stages to height 2	4 / 6 / 8 (8/16/32-bit)	4 / 6 / 8 - **the same**
Reduction cells	61	42

Aspect	Wallace Tree	Dadda Tree
(8×8)		
Final CPA (8×8)	`math_adder_brent_kung_016`	`math_adder_brent_kung_016` - **the same**
Total adder cells (8×8)	61	42
Design	Simpler (greedy grouping)	More complex (scheduled targets)

Dadda uses **fewer compressors for the same depth**—not fewer stages. The stage counts are identical.

Recommendation: Use Dadda tree for production designs (fewer cells at equal depth). Wallace remains the clearer teaching example, but with both families now ending in the same Brent-Kung prefix adder, it no longer has a delay advantage to trade against its larger tree.

25.7.7 Area-Speed Tradeoffs

For High-Speed Requirements: - Either tree works—they have the same reduction depth and the same final adder - Consider pipelining for even higher frequency - If you pick Wallace, accept the larger area overhead for no speed gain

For Area-Constrained Designs: - Use Dadda tree instead (see `math_multiplier_dadda_tree.md`) - Consider Booth encoding for signed multiplication - Use sequential multipliers if latency acceptable

25.7.8 Signed Multiplication

Wallace trees output **unsigned products only**. For signed multiplication:

Option 1: Sign-Magnitude Conversion

```
// Convert to unsigned, multiply, then fix sign
logic sign_result;
logic [N-1:0] a_abs, b_abs;
logic [2*N-1:0] product_unsigned;

assign sign_result = a[N-1] ^ b[N-1];
assign a_abs = a[N-1] ? -a : a;
assign b_abs = b[N-1] ? -b : b;

math_multiplier_wallace_tree_008 u_mult (
    .i_multiplier(a_abs),
    .i_multiplicand(b_abs),
    .ow_product(product_unsigned)
);
```

```
assign product_signed = sign_result ? -product_unsigned :
product_unsigned;
```

Option 2: Booth Encoding (More efficient for signed)

```
// Use Booth radix-4 multiplier instead
// (Not covered by Wallace tree modules)
```

25.7.9 Pipelining Strategy

Single-Stage Pipeline:

```
// Register output (adds 1 cycle latency)
always_ff @(posedge clk) begin
    product_reg <= ow_product;
end
```

Multi-Stage Pipeline: Split Wallace tree into stages: 1. **Stage 1:** Partial product generation → register 2. **Stage 2:** First half of reduction tree → register 3. **Stage 3:** Second half of reduction tree → register 4. **Stage 4:** Final addition → output

Benefit: Achieves 2-3× higher frequency at cost of 3-4 cycle latency

25.7.10 Synthesis Considerations

Optimization Directives:

```
# Let synthesis optimize structure
set_dont_touch false

# For timing-critical designs
set_flatten true
set_boundary_optimization true

# If targeting ASIC
set_implementation rtl  # vs gate-level

# If targeting FPGA
# Let tool map to DSP blocks if available
```

FPGA Notes: - Modern FPGAs have dedicated DSP blocks (DSP48 on Xilinx, DSP on Intel) - Synthesis tools may map Wallace tree to DSP block - **Check resource utilization** - may use LUTs instead of DSP if tree doesn't fit

25.7.11 Common Pitfalls

Anti-Pattern 1: Expecting parameterized width

```
// WRONG: Trying to override N parameter
math_multiplier_wallace_tree_008 #(.N(12)) u_mult (...);  // Won't
work!

// RIGHT: Use fixed variant or create custom width
math_multiplier_wallace_tree_016 u_mult (...);  // Use 16-bit variant
```

Anti-Pattern 2: Using for signed multiplication directly

```
// WRONG: Signed operands won't work correctly
logic signed [7:0] a, b;
logic signed [15:0] product;
math_multiplier_wallace_tree_008 u_mult (
    .i_multiplier(a),        // Interpreted as unsigned!
    .i_multiplicand(b),
    .ow_product(product)
);

// RIGHT: Convert to unsigned, then fix sign
// (See "Signed Multiplication" section above)
```

Anti-Pattern 3: Adding a redundant external adder

```
// WRONG: ow_product is already the final summed product.
// Bolting on another adder both wastes area and computes garbage.
math_multiplier_wallace_tree_008 u_mult (..., .ow_product(p));
assign product = p + {carry[14:0], 1'b0};  // No!

// RIGHT: use ow_product directly.
math_multiplier_wallace_tree_008 u_mult (..., .ow_product(product));
```

These modules contain their own final carry-propagate adder. They don't expose separate sum and carry vectors. There's no ambiguity here—every generated variant definitively has a final CPA inside.

Anti-Pattern 4: Not pipelining for high frequency

```
// WRONG: Using 32×32 multiplier at 500 MHz (won't meet timing)
math_multiplier_wallace_tree_032 u_mult (...);  // Critical path too
long!

// RIGHT: Add pipeline stages
always_ff @(posedge clk) begin
    stage1 <= inputs;
    stage2 <= wallace_tree_partial(stage1);
    stage3 <= wallace_tree_final(stage2);
    product <= stage3;
end
```

25.8 Related Modules

- **math_multiplier_dadda_tree_*.sv** - same reduction depth, 17-31% fewer adder cells, identical final CPA
- **math_adder_brent_kung_016/032/064.sv** - logarithmic-depth parallel-prefix adder used as the final CPA here (8/16/32-bit multipliers use the 016/032/064 widths respectively)
- **math_adder_carry_save.sv** - 3:2 compressor, used in the `_csa_` variant's reduction tree
- **math_adder_full.sv** - Full adder primitive, used in the plain variant's reduction tree
- **math_adder_half.sv** - Half adder primitive, used in the reduction tree of both variants

25.9 Testing

Covered by 2 test suites:

- `val/math/test_math_multiplier_wallace.py`
- `val/math/test_math_multiplier_wallace_csa.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

25.10 References

- Wallace, C.S. "A Suggestion for a Fast Multiplier." IEEE Transactions on Electronic Computers, 1964.
- Dadda, L. "Some Schemes for Parallel Multipliers." Alta Frequenza, 1965.
- Oklobdzija, V.G. "High-Speed VLSI Arithmetic Units: Adders and Multipliers." Springer, 2002.

25.11 Navigation

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26 Prefix Cell Gray

26.1 Overview

The `math_prefix_cell_gray` module (the "gray cell") is the reduced-area sibling of the prefix cell—an area-optimized parallel prefix building block that computes only the group generate (G) signal. It earns its keep in reverse tree stages and final carry computation, anywhere the propagate signal isn't needed downstream. The final carry computation needs G, not P, so gray cells save ~33% area in exactly those stages.

Key Features: - **Outputs G only** - Optimized for carry-only computation - **~33% smaller** than black cells (2 gates vs 3) - **Same delay** as black cell for G output - **Used in** the Han-Carlson prefix stages (all six widths); Brent-Kung's reverse tree uses math_adder_brent_kung_gray instead

26.2 Parameters

None. This is a fixed single-bit cell—there's nothing to parameterize.

26.3 Ports

26.3.1 Module Declaration

```
module math_prefix_cell_gray (
    input  logic i_g_hi, i_p_hi,
    input  logic i_g_lo,          // No P needed from lower position
    output logic ow_g             // Only G output (this IS the
carry)
);
```

26.3.2 Port List

Port	Direction	Width	Description
i_g_hi	Input	1	Generate signal from higher bit position
i_p_hi	Input	1	Propagate signal from higher bit position
i_g_lo	Input	1	Generate signal from lower bit position
ow_g	Output	1	Combined group generate (the carry into position i+1)

Note: `i_p_lo` isn't needed here—it would only feed the group propagate computation, and this cell doesn't output one.

26.4 Functional Description

26.4.1 Implementation

```
assign ow_g = i_g_hi | (i_p_hi & i_g_lo);
```

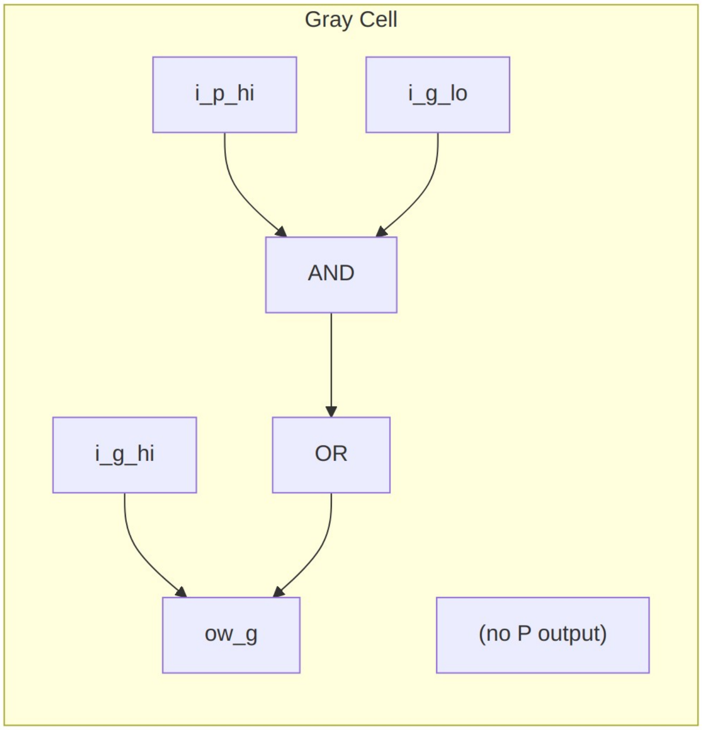
That's the group generate: - **G[i:j]** = G[i:k] OR (P[i:k] AND G[k-1:j])

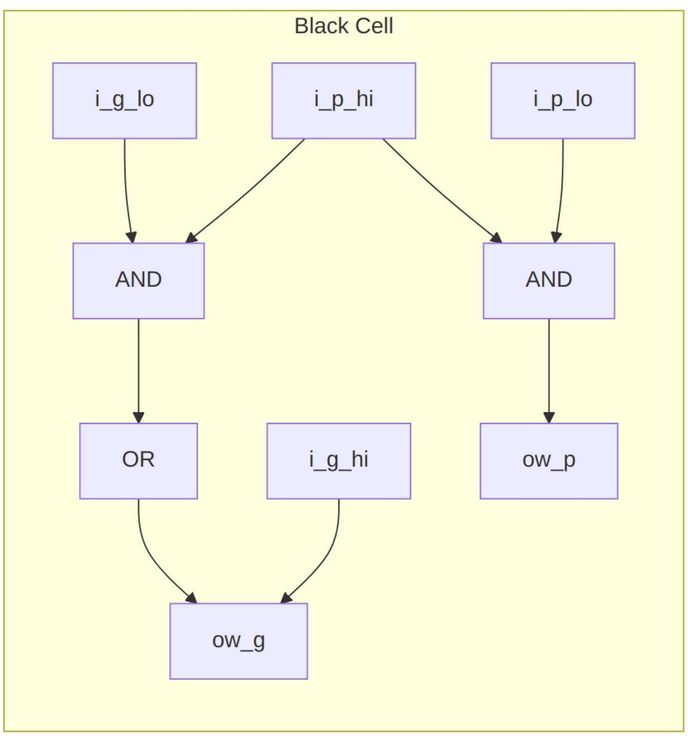
Interpretation: A carry is generated for the combined range [i:j] if: - The high range [i:k] generates a carry, OR - The high range propagates AND the low range generates

26.4.2 Why Gray Cells Don't Need P Output

In parallel prefix adders, the final computation is: - **Sum[i]** = P[i] XOR C[i-1] - **C[i]** = G[i:-1] (group generate from bit i down to carry-in)

The sum computation uses the **original** single-bit propagate P[i], not the group propagate. So once we've computed all the carries (group generates), the group propagates have no consumers left.





Visual Comparison

26.5 Timing Characteristics

Metric	Value	Description
Logic Depth	2 gates	1 AND + 1 OR
Critical Path	AND-OR	i_g_lo -> ow_g
Gate Count	2	1 AND + 1 OR

26.6 Usage Examples

26.6.1 In Han-Carlson Final Stage

```
// Han-Carlson: Final stage fills odd positions using gray cells
// Odd positions get their carry from the even neighbor
generate
    for (i = 0; i < N; i++) begin : gen_final_stage
        if (i % 2 == 1) begin : gen_odd
            // Odd positions: compute G[i:-1] from G[i] and G[i-1:-1]
            math_prefix_cell_gray u_pf_gray (
                .i_g_hi(w_g_prev[i]),   // G[i] (single bit)
                .i_p_hi(w_p_prev[i]),   // P[i] (single bit)
                .i_g_lo(w_g_prev[i-1]), // G[i-1:-1] (group from even
neighbor)
                .ow_g(w_g_final[i])     // G[i:-1] (the carry)
            );
        end else begin : gen_even
            // Even positions: already computed, pass through
            assign w_g_final[i] = w_g_prev[i];
        end
    end
endgenerate
```

26.6.2 In Brent-Kung Reverse Tree

```
// Brent-Kung: the reverse tree uses gray cells to fill intermediate
positions.
// After the forward tree, the complete carries sit at positions 2^k -
1 --
// with 0-based bit indexing that is 1, 3, 7, 15 for a 16-bit adder,
NOT at
// the powers of two. The reverse tree fills the rest -- for 16-bit
that is
// twelve positions ({2,4,5,6,8,9,10,11,12,13,14,16} in
// math_adder_brent_kung_grouppg_016.sv), not just {2,4,8}.

// Example: position 5 combines the span-2 group [4:3] with the
complete
// carry already available at position 3. (Indices: ow_gg[k] is the
```

```
carry
// INTO bit k -- math_adder_brent_kung_sum drives ow_sum[k] = gg[k] ^
p[k+1],
// so ow_gg[5] = G[4:-1], the carry into bit 5, not 6.)
math_prefix_cell_gray u_bk_gray_5 (
    .i_g_hi(w_g_2[5]),  // G[4:3] (span-2 group from forward tree
level 1)
    .i_p_hi(w_p_2[5]),  // P[4:3] (matching group propagate)
    .i_g_lo(w_gg[3]),   // G[2:-1] (complete carry into bit 3, already
resolved)
    .ow_g(w_gg[5])      // G[4:-1] (carry into bit 5)
);
```

This mirrors `gray_block_5_3` in `math_adder_brent_kung_grouppg_008.sv`, which wires `G_5_4`/`P_5_4` against `ow_gg[3]`. Many gray cells in the Brent-Kung reverse tree pair a *group* generate/propagate with an already-complete carry—but the library's Brent-Kung fill level also uses single-bit gray cells: `math_adder_brent_kung_grouppg_016.sv` contains `gray_block_1_0`, `gray_block_2_1`, `gray_block_4_3`, … which take a single-bit `G[i:i]` against the completed carry below. So the single-bit pattern isn't exclusive to Han-Carlson; Brent-Kung uses it in its fill-in stage too.

26.6.3 Computing Final Sum

```
// After all carries computed with gray cells:
generate
    for (i = 0; i < N; i++) begin : gen_sum
        if (i == 0) begin
            assign sum[0] = p_original[0] ^ cin;
        end else begin
            // Sum = original P XOR carry from previous position
            assign sum[i] = p_original[i] ^ g_final[i-1];
        end
    end
endgenerate

// Carry out is the final group generate
assign cout = g_final[N-1];
```

26.7 Design Notes

26.7.1 Resource Utilization

Metric	Value
AND gates	1
OR gates	1
Total gates	2

Metric	Value
LUTs (FPGA)	1

26.7.2 Comparison with Black Cell

Property	Black Cell	Gray Cell
Inputs	4 (Ghi, Phi, Glo, Plo)	3 (Ghi, Phi, Glo)
Outputs	2 (G, P)	1 (G)
Gate count	3	2
Area savings	-	~33%
G output delay	Same	Same

26.7.3 Area Savings in Adder

For an N-bit adder, the mix of gray cells versus black cells sets the total area.

The two 16-bit rows below are counted from this library's own RTL, not estimated. Kogge-Stone is a textbook reference figure; there is no Kogge-Stone adder in this library.

Prefix-cell counts at N=16, counted from the RTL

Architecture	Black Cells	Gray Cells	Total Cells	Prefix Levels
Kogge-Stone N=16 (textbook, not implemented here)	49	0	49	4
Brent-Kung N=16 (`math_adder_brent_kung_grouppg_016.sv`)	11	16	27	6
Han-Carlson N=16 (`math_adder_han_carlson_016.sv`)	24	8	32	5

How these were counted:

- Brent-Kung: `grep -c math_adder_brent_kung_black rtl/math/math_adder_brent_kung_grouppg_016.sv` gives 11, and the matching `_gray` count gives 16. Note that this library uses a gray cell wherever the low operand is already a complete carry, even inside the forward tree, so gray cells outnumber black ones.
- Han-Carlson: elaborating the five generate stages of `math_adder_han_carlson_016.sv` yields 7 + 7 + 6 + 4 = 24 `math_prefix_cell` instances and 8 `math_prefix_cell_gray` instances.
- Kogge-Stone: `N x log2(N) - N + 1 = 16 x 4 - 16 + 1 = 49` black cells.

Han-Carlson uses gray cells only in the final fill-in stage: N/2 cells for an N-bit adder, which is the 8 above.

26.7.4 When to Use Gray Cells

Use gray cells when: - Computing final carries (no further prefix operations needed) - In reverse tree stages (Brent-Kung) - Final fill-in stage (Han-Carlson) - Any position where P is not needed downstream

Use black cells when: - P signal needed for subsequent prefix stages - Building Kogge-Stone (all stages need both P and G) - Forward tree stages in hybrid architectures

26.7.5 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Minimal 2-gate implementation 2. **Wire complexity** - One fewer input than black cell 3. **Logic depth** - Same 2-gate delay as black cell

26.7.6 Architectural Trade-offs

Architecture	Gray Cell Usage	Trade-off
Kogge-Stone	None	Maximum speed, maximum area
Brent-Kung	Reverse tree (~50%)	Minimum area, 2x depth
Han-Carlson	Final stage only (~20%)	Balanced speed/area

26.7.7 Applications

- **Parallel prefix adders** - Final carry computation stages
- **Area-optimized adders** - Brent-Kung and Han-Carlson architectures
- **Multiplier CPAs** - Final addition where area matters

- **Low-power designs** - Fewer gates = lower dynamic power

26.8 Related Modules

- **math_prefix_cell** - Black cell variant (outputs both G and P)
- **math_adder_han_carlson_016** - 16-bit Han-Carlson adder using this cell
- **math_adder_han_carlson_048** - 48-bit Han-Carlson adder using this cell
- **math_adder_brent_kung_gray** - Brent-Kung gray cell (equivalent)

26.9 Testing

No dedicated test wrapper – this block is exercised structurally through the Han-Carlson adder tests (`val/math/test_math_adder_han_carlson.py`) – all six HC widths instantiate this cell (Brent-Kung uses its own math_adder_brent_kung_black/gray cells, NOT this one). It is also formally proved: `formal/common/math_prefix_cell_gray/` (prove + cover, SymbiYosys).

26.10 References

- Brent, R.P., Kung, H.T. "A Regular Layout for Parallel Adders." IEEE Trans. Computers, 1982.
- Han, T., Carlson, D.A. "Fast Area-Efficient VLSI Adders." IEEE Symposium on Computer Arithmetic, 1987.
- Harris, D. "A Taxonomy of Parallel Prefix Networks." Asilomar Conference, 2003.

26.11 Navigation

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27 Prefix Cell (Black Cell)

27.1 Overview

The `math_prefix_cell` module (the "black cell") is the parallel prefix building block that combines generate (G) and propagate (P) signals from adjacent bit groups—the cell at the heart of high-performance adders like Kogge-Stone, Brent-Kung, and Han-Carlson. It takes the generate and propagate signals from a higher bit position and a lower one, and computes the group generate and group propagate for the combined range.

Key Features: - **Outputs both G and P** - Full prefix cell for forward tree stages - **O(1) delay** - Fixed 2-gate delay per stage - **Building block** for all parallel prefix adder architectures - **Enables O(log N) addition** - Parallel carry computation

27.2 Parameters

None. This is a fixed single-bit cell—there's nothing to parameterize.

27.3 Ports

27.3.1 Module Declaration

```
module math_prefix_cell (
    input  logic i_g_hi, i_p_hi,   // From higher bit position
    input  logic i_g_lo, i_p_lo,   // From lower bit position
    output logic ow_g, ow_p
);
```

27.3.2 Port List

Port	Direction	Width	Description
i_g_hi	Input	1	Generate signal from higher bit position [i:k]
i_p_hi	Input	1	Propagate signal from higher bit position [i:k]
i_g_lo	Input	1	Generate signal from lower bit position [k-1:j]
i_p_lo	Input	1	Propagate signal from lower bit position [k-1:j]
ow_g	Output	1	Combined group generate [i:j]
ow_p	Output	1	Combined group propagate [i:j]

27.4 Functional Description

27.4.1 Parallel Prefix Theory

In parallel prefix adders, carries are computed from generate (G) and propagate (P) signals:

Bit-level definitions: - **Generate:** G[i] = A[i] AND B[i] - Carry generated at position i - **Propagate:** P[i] = A[i] XOR B[i] - Carry propagated through position i

Group-level recurrence (the prefix operation): - **G[i:j]** = G[i:k] OR (P[i:k] AND G[k-1:j]) - **P[i:j]** = P[i:k] AND P[k-1:j]

This cell computes those group-level signals by combining two adjacent ranges.

27.4.2 Implementation

```
assign ow_g = i_g_hi | (i_p_hi & i_g_lo);
assign ow_p = i_p_hi & i_p_lo;
```

Interpretation: - **Group Generate:** A carry is generated for the combined range if: - The high range generates a carry, OR - The high range propagates AND the low range generates - **Group Propagate:** A carry propagates through the combined range only if both ranges propagate

27.4.3 Example: Combining Adjacent Pairs

```
Initial (bit-level):
  Position:  3    2    1    0
  P[i]:     P3   P2   P1   P0
  G[i]:     G3   G2   G1   G0

After Stage 1 (prefix cells combining adjacent pairs):
  G[1:0] = G1 | (P1 & G0)     // Does range [1:0] generate?
  P[1:0] = P1 & P0            // Does range [1:0] propagate?

  G[3:2] = G3 | (P3 & G2)     // Does range [3:2] generate?
  P[3:2] = P3 & P2            // Does range [3:2] propagate?

After Stage 2 (prefix cells combining pairs of pairs):
  G[3:0] = G[3:2] | (P[3:2] & G[1:0])  // Does range [3:0] generate?
  P[3:0] = P[3:2] & P[1:0]             // Does range [3:0] propagate?
```

27.5 Timing Characteristics

Metric	Value	Description
Logic Depth	2 gates	1 AND + 1 OR (for G), 1 AND (for P)
Critical Path	AND-OR	i_p_hi/i_g_lo -> ow_g
Gate Count	3	2 AND gates + 1 OR gate

27.6 Usage Examples

27.6.1 In Kogge-Stone Adder (Stage 1)

```
// Combine positions 1 and 0
math_prefix_cell u_pf_10 (
    .i_g_hi(g[1]),   .i_p_hi(p[1]),
    .i_g_lo(g[0]),   .i_p_lo(p[0]),
    .ow_g(g_1_0),    .ow_p(p_1_0)
);

// Combine positions 3 and 2
math_prefix_cell u_pf_32 (
    .i_g_hi(g[3]),   .i_p_hi(p[3]),
    .i_g_lo(g[2]),   .i_p_lo(p[2]),
    .ow_g(g_3_2),    .ow_p(p_3_2)
);
```

27.6.2 In Kogge-Stone Adder (Stage 2)

```
// Combine ranges [3:2] and [1:0]
math_prefix_cell u_pf_30 (
    .i_g_hi(g_3_2),  .i_p_hi(p_3_2),
    .i_g_lo(g_1_0),  .i_p_lo(p_1_0),
    .ow_g(g_3_0),    .ow_p(p_3_0)
);
```

27.6.3 In Han-Carlson Adder (Even Position)

```
// Han-Carlson: Only even positions get prefix cells in intermediate
stages
generate
    for (i = 0; i < N; i++) begin : gen_stage
        if (i % 2 == 0 && i >= 2) begin : gen_even
            math_prefix_cell u_pf (
                .i_g_hi(w_g_prev[i]),   .i_p_hi(w_p_prev[i]),
                .i_g_lo(w_g_prev[i-2]), .i_p_lo(w_p_prev[i-2]),
                .ow_g(w_g_curr[i]),     .ow_p(w_p_curr[i])
            );
        end else begin : gen_pass
            assign w_g_curr[i] = w_g_prev[i];
            assign w_p_curr[i] = w_p_prev[i];
        end
    end
endgenerate
```

27.7 Design Notes

27.7.1 Resource Utilization

Metric	Value
AND gates	2
OR gates	1
Total gates	3
LUTs (FPGA)	1-2

27.7.2 Comparison: Black Cell vs Gray Cell

Property	Black Cell (this)	Gray Cell
Outputs	G and P	G only
Gate count	3	2
Use case	Forward tree	Reverse tree / final stage
Module	`math_prefix_cell`	`math_prefix_cell_gray`

27.7.3 When to Use Black vs Gray Cells

Use Black Cells (this module) when: - In forward tree stages where both G and P are needed downstream - Building Kogge-Stone style networks (all cells are black) - Intermediate stages of Brent-Kung or Han-Carlson

Use Gray Cells when: - Only the carry (G) is needed (final sum computation) - In reverse tree stages of Brent-Kung - Final fill-in stage of Han-Carlson - ~33% area savings when P not needed

27.7.4 Design Optimization Priorities

This module is optimized with the following priorities: 1. **Area** - Minimal 3-gate implementation 2. **Wire complexity** - Simple point-to-point connections 3. **Logic depth** - Fixed 2-gate delay

27.7.5 Prefix Network Architectures

Architecture	Black Cells	Gray Cells	Depth	Wiring
Kogge-Stone	All	None	O(log N)	Maximum
Brent-Kung	Forward tree	Reverse tree	O(2 log N)	Minimum
Han-	Even	Final stage	O(log N + 1)	Medium

Architecture	Black Cells	Gray Cells	Depth	Wiring
Carlson	positions			

27.7.6 Applications

- **High-performance adders** - All parallel prefix adder architectures
- **Multiplier final CPA** - Fast carry-propagate addition
- **Incrementers/Decrementers** - Priority-encoded operations
- **Comparators** - Parallel comparison networks

27.8 Related Modules

- **math_prefix_cell_gray** - Gray cell variant (G output only)
- **math_adder_han_carlson_016** - 16-bit Han-Carlson adder using this cell
- **math_adder_han_carlson_048** - 48-bit Han-Carlson adder using this cell
- **math_adder_brent_kung_black** - Brent-Kung black cell (equivalent)

27.9 Testing

No dedicated test wrapper – this block is exercised structurally through the Han-Carlson adder tests (`val/math/test_math_adder_han_carlson.py`) – all six HC widths instantiate this cell (Brent-Kung uses its own math_adder_brent_kung_black/gray cells, NOT this one). It is also formally proved: `formal/common/math_prefix_cell/` (prove + cover, SymbiYosys).

27.10 References

- Kogge, P.M., Stone, H.S. “A Parallel Algorithm for the Efficient Solution of a General Class of Recurrence Equations.” IEEE Trans. Computers, 1973.
- Brent, R.P., Kung, H.T. “A Regular Layout for Parallel Adders.” IEEE Trans. Computers, 1982.
- Han, T., Carlson, D.A. “Fast Area-Efficient VLSI Adders.” IEEE Symposium on Computer Arithmetic, 1987.
- Harris, D. “A Taxonomy of Parallel Prefix Networks.” Asilomar Conference, 2003.

27.11 Navigation

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28 Binary Subtractors

28.1 Overview

A family of binary subtractor modules—half, full, ripple, and carry lookahead—that compute A - B with dedicated subtraction logic. Modern designs typically subtract with an adder and two's complement instead, but these modules remain useful for educational purposes and specific applications.

The complete subtractor module family: - **math_subtractor_half** - Single-bit half subtractor (2 inputs) - **math_subtractor_full** - Single-bit full subtractor (3 inputs with borrow) - **math_subtractor_full_nbit** - N-bit ripple borrow subtractor - **math_subtractor_ripple_carry** - N-bit ripple borrow subtractor (equivalent) - **math_subtractor_carry_lookahead** - N-bit subtractor with lookahead logic

Modern Alternative: Most designs subtract with an adder via two's complement: `A - B = A + (~B) + 1`. Same result, and it reuses adder hardware you already have.

28.2 Parameters

28.2.1 Single-bit Subtractors

No parameters (fixed single-bit operation).

28.2.2 N-bit Subtractors

Parameter	Type	Default	Description
N	int	4	Bit width (range: 1-64)

28.3 Ports

28.3.1 Half Subtractor

```
module math_subtractor_half (
    input  logic i_a,       // Minuend
    input  logic i_b,       // Subtrahend
    output logic o_d,       // Difference
    output logic o_b        // Borrow output
);
```

Port	Direction	Width	Description
i_a	Input	1	Minuend (number to subtract from)

Port	Direction	Width	Description
i_b	Input	1	Subtrahend (number to subtract)
o_d	Output	1	Difference output (A - B)
o_b	Output	1	Borrow output

28.3.2 Full Subtractor

```
module math_subtractor_full (
    input  logic i_a,        // Minuend
    input  logic i_b,        // Subtrahend
    input  logic i_b_in,     // Borrow input
    output logic ow_d,       // Difference
    output logic ow_b        // Borrow output
);
```

Port	Direction	Width	Description
i_a	Input	1	Minuend
i_b	Input	1	Subtrahend
i_b_in	Input	1	Borrow input (from previous stage)
ow_d	Output	1	Difference output
ow_b	Output	1	Borrow output (to next stage)

28.3.3 N-bit Subtractors

The three N-bit subtractors do **not** share a port list—check which one you're instantiating. `math_subtractor_full_nbit` shares only `i_a`/`i_b` with the other two: its borrow in is `i_b_in` (not `i_borrow_in`) and its outputs are `ow_d`/`ow_b` only (no `ow_difference`/`ow_carry_out` aliases).

28.3.3.1 math_subtractor_full_nbit

```
module math_subtractor_full_nbit #(
    parameter int N = 4
) (
    input  logic [N-1:0] i_a,      // Minuend
    input  logic [N-1:0] i_b,      // Subtrahend
```

```
    input  logic         i_b_in,  // Borrow input
    output logic [N-1:0] ow_d,    // Difference
    output logic         ow_b     // Borrow out
);
```

Port	Direction	Width	Description
i_a	Input	N	Minuend vector
i_b	Input	N	Subtrahend vector
i_b_in	Input	1	Initial borrow input
ow_d	Output	N	Difference vector (A - B - Bin)
ow_b	Output	1	Final borrow out

28.3.3.2 math_subtractor_ripple_carry

```
module math_subtractor_ripple_carry #(
    parameter int N = 4
) (
    input  logic [N-1:0] i_a,           // Minuend
    input  logic [N-1:0] i_b,           // Subtrahend
    input  logic         i_borrow_in,   // Borrow input
    output logic [N-1:0] ow_difference, // Difference
    output logic         ow_carry_out   // Borrow out (carry out)
);
```

Port	Direction	Width	Description
i_a	Input	N	Minuend vector
i_b	Input	N	Subtrahend vector
i_borrow_in	Input	1	Initial borrow input
ow_difference	Output	N	Difference vector (A - B - Bin)

Port	Direction	Width	Description
ow_carry_out	Output	1	Final borrow out (carry out)

28.3.3.3 math_subtractor_carry_lookahead

Same inputs as the ripple variant, but every output is exposed twice under two names (`assign ow_d = ow_difference; assign ow_b = ow_borrow_out;`):

```
module math_subtractor_carry_lookahead #(
    parameter int N = 4
) (
    input  logic [N-1:0] i_a,            // Minuend
    input  logic [N-1:0] i_b,            // Subtrahend
    input  logic         i_borrow_in,    // Borrow input
    output logic [N-1:0] ow_difference,  // Difference
    output logic [N-1:0] ow_d,           // Alias of ow_difference
    output logic         ow_borrow_out,  // Borrow output
    output logic         ow_b            // Alias of ow_borrow_out
);
```

Port	Direction	Width	Description
i_a	Input	N	Minuend vector
i_b	Input	N	Subtrahend vector
i_borrow_in	Input	1	Initial borrow input
ow_difference	Output	N	Difference vector
ow_d	Output	N	Alias of `ow_difference`
ow_borrow_out	Output	1	Final borrow output
ow_b	Output	1	Alias of `ow_borrow_out`

28.4 Functional Description

28.4.1 Half Subtractor

Subtracts one bit from another, with no borrow input:

Logic Equations:

```
Difference = A ⊕ B              // XOR
Borrow = ~A • B                 // Borrow when A=0, B=1
```

Truth Table: | A | B | Diff | Borrow | Meaning | |—|—|——|——–|———| | 0 | 0 | 0 | 0 | 0 - 0 = 0 | | 0 | 1 | 1 | 1 | 0 - 1 = -1 (borrow!) | | 1 | 0 | 1 | 0 | 1 - 0 = 1 | | 1 | 1 | 0 | 0 | 1 - 1 = 0 |

Implementation:

```
assign o_d = i_a ^ i_b;          // XOR for difference
assign o_b = ~i_a & i_b;         // Borrow when A < B
```

28.4.2 Full Subtractor

Subtracts two bits with a borrow input:

Logic Equations:

```
Difference = A ⊕ B ⊕ Bin
Borrow = (~A • B) | (~(A ⊕ B) • Bin)
       = (~A • B) | (~A • Bin) | (B • Bin)  // Alternative form
```

Truth Table: | A | B | Bin | Diff | Bout | A - B - Bin | |—|—|—–|——|——|————-| | 0 | 0 | 0 | 0 | 0 | 0 | | 0 | 0 | 1 | 1 | 1 | -1 | | 0 | 1 | 0 | 1 | 1 | -1 | | 0 | 1 | 1 | 0 | 1 | -2 | | 1 | 0 | 0 | 1 | 0 | 1 | | 1 | 0 | 1 | 0 | 0 | 0 | | 1 | 1 | 0 | 0 | 0 | 0 | | 1 | 1 | 1 | 1 | 1 | -1 |

Implementation:

```
assign ow_d = i_a ^ i_b ^ i_b_in;
assign ow_b = (~i_a & i_b) | (~(i_a ^ i_b) & i_b_in);
```

28.4.3 N-bit Ripple Borrow Subtractor

Chains N full subtractors, borrow propagating from LSB to MSB:

```
Bit 0:   FS(A[0], B[0], Bin)     → D[0], B[0]
Bit 1:   FS(A[1], B[1], B[0])    → D[1], B[1]
...
Bit N-1: FS(A[N-1], B[N-1], B[N-2]) → D[N-1], Bout
```

Timing: O(N) delay (borrow ripples sequentially)

28.4.4 N-bit Carry Lookahead Subtractor

Uses lookahead logic to compute borrows faster (similar to CLA adder):

Timing: O(N) delay (simplified lookahead)

28.5 Timing Characteristics

Module	Logic Levels	Typical Delay (ns)
Half Subtractor	1	~0.2
Full Subtractor	2	~0.4
Ripple (N-bit)	2N	~0.4N
CLA (N-bit)	O(N)	Similar to ripple (simplified)

Note: Subtractors have the same timing as the equivalent adders.

28.6 Usage Examples

28.6.1 Basic Subtraction (Using Subtractor)

```
logic [7:0] a, b, diff;
logic borrow;

math_subtractor_ripple_carry #(.N(8)) u_sub (
    .i_a(a),
    .i_b(b),
    .i_borrow_in(1'b0),       // No initial borrow
    .ow_difference(diff),
    .ow_carry_out(borrow)     // Borrow indicates A < B
);

// Example: 100 - 50 = 50
initial begin
    a = 8'd100;
    b = 8'd50;
    #1;
    assert(diff == 8'd50);
    assert(borrow == 1'b0);  // No borrow
end
```

28.6.2 Detecting Underflow (A < B)

```
logic [7:0] a, b, diff;
logic underflow;

math_subtractor_ripple_carry #(.N(8)) u_sub (
    .i_a(a), .i_b(b), .i_borrow_in(1'b0),
    .ow_difference(diff), .ow_carry_out(underflow)
);
```

```
// Borrow output indicates underflow (A < B)
// Example: 50 - 100 → underflow!
initial begin
    a = 8'd50;
    b = 8'd100;
    #1;
    assert(underflow == 1'b1);  // A < B
    assert(diff == 8'd206);     // Wraps around: 50 - 100 + 256 = 206
end
```

28.6.3 Modern Approach: Subtraction Using Adder

```
// PREFERRED: Use adder with two's complement
// A - B = A + (~B) + 1

logic [7:0] a, b, diff;
logic borrow;
logic carry_out;

math_adder_ripple_carry #(.N(8)) u_add (
    .i_a(a),
    .i_b(~b),          // Invert B (one's complement)
    .i_c(1'b1),        // Carry in = 1 (complete two's complement)
    .ow_sum(diff),
    .ow_carry(carry_out)
);

// Borrow = ~carry_out (inverted carry indicates underflow)
assign borrow = ~carry_out;
```

Advantages of Adder-Based Subtraction: - Reuses existing adder hardware - No separate subtractor needed - Standard practice in modern ALUs - Same timing characteristics

28.6.4 Multi-Precision Subtraction (16-bit via 2×8-bit)

```
logic [7:0] a_low, a_high, b_low, b_high;
logic [7:0] diff_low, diff_high;
logic borrow_low, borrow_high;

// Low byte
math_subtractor_ripple_carry #(.N(8)) u_sub_low (
    .i_a(a_low),
    .i_b(b_low),
    .i_borrow_in(1'b0),
    .ow_difference(diff_low),
    .ow_carry_out(borrow_low)
);
```

```
// High byte (chain borrow from low)
math_subtractor_ripple_carry #(.N(8)) u_sub_high (
    .i_a(a_high),
    .i_b(b_high),
    .i_borrow_in(borrow_low),
    .ow_difference(diff_high),
    .ow_carry_out(borrow_high)
);

logic [15:0] diff_16 = {diff_high, diff_low};
```

28.7 Design Notes

28.7.1 Resource Utilization

Module	LUTs	Description
Half Subtractor	2	1 XOR + 1 AND
Full Subtractor	2	Similar to full adder
Ripple (N-bit)	~2N	N full subtractors

28.7.2 Subtractor vs Adder-Based Subtraction

Aspect	Direct Subtractor	Adder + Two's Complement
Logic	Dedicated subtractor cells	Reused adder + inverter
Area	+100% (separate unit)	+1% (just inverter)
Timing	Same as adder	Same as adder
Modern Practice	Rarely used	Standard approach

28.7.3 Why Adders Are Preferred for Subtraction

Historical: Separate subtractors were common in early designs.

Modern: Almost all designs use adders with two's complement:

Advantages: - Hardware reuse (one adder for both add/subtract) - Smaller area (no duplicate logic) - Standard practice (easier to understand/maintain) - ALU simplicity (single arithmetic unit)

When to Use Direct Subtractors: - Educational/demonstration purposes - Very specific applications requiring direct borrow propagation - Compatibility with existing designs using subtractors

28.7.4 Implementing Subtraction in ALU

```
module simple_alu (
    input  logic [7:0] a, b,
    input  logic sub,  // 1 = subtract, 0 = add
    output logic [7:0] result,
    output logic carry_borrow
);

    logic [7:0] b_mux;
    logic cin_mux;
    logic carry_out;

    // Two's complement: A - B = A + (~B) + 1
    assign b_mux = sub ? ~b : b;
    assign cin_mux = sub;

    math_adder_ripple_carry #(.N(8)) u_add (
        .i_a(a),
        .i_b(b_mux),
        .i_c(cin_mux),
        .ow_sum(result),
        .ow_carry(carry_out)
    );

    // For subtraction, borrow = ~carry
    assign carry_borrow = sub ? ~carry_out : carry_out;

endmodule
```

28.7.5 Common Pitfalls

Anti-Pattern 1: Using subtractors instead of adders

```
// DISCOURAGED: Separate adder and subtractor
math_adder_ripple_carry u_add (...);
math_subtractor_ripple_carry u_sub (...);  // Duplicate hardware!

// PREFERRED: Single adder for both operations
math_adder_ripple_carry u_alu (
    .i_a(a),
    .i_b(sub ? ~b : b),  // Two's complement for subtraction
    .i_c(sub),
    ...
);
```

Anti-Pattern 2: Ignoring borrow for underflow detection

```
// WRONG: Ignoring borrow output
math_subtractor_ripple_carry u_sub (
    .i_a(a), .i_b(b), .i_borrow_in(1'b0),
    .ow_difference(diff),
    .ow_carry_out()  // IGNORED - miss underflow detection!
);

// RIGHT: Check borrow for underflow
logic underflow;
math_subtractor_ripple_carry u_sub (
    .i_a(a), .i_b(b), .i_borrow_in(1'b0),
    .ow_difference(diff),
    .ow_carry_out(underflow)  // Indicates A < B
);
```

Anti-Pattern 3: Incorrect borrow interpretation

```
// WRONG: Borrow means A < B, not A > B
if (borrow) begin
    // A is LESS than B (negative result)
end

// Remember: Borrow = 1 means underflow (A < B)
```

28.8 Related Modules

- **math_adder_*.sv** - Preferred for subtraction via two's complement
- **math_addsub_full_nbit.sv** - Combined add/subtract unit

28.9 Testing

Covered by 4 test suites:

- `val/math/test_math_subtractor_carry_lookahead.py`
- `val/math/test_math_subtractor_full_nbit.py`
- `val/math/test_math_subtractor_half.py`
- `val/math/test_math_subtractor_ripple_carry.py`

Run levels come from the standard grid: `REG_LEVEL=GATE|FUNC|FULL` selects the parameter set, `TEST_LEVEL` the per-test depth. Run the whole area with `make -C val/math run-all-func-parallel`, never bare pytest for suites.

28.10 References

- Mano, M.M. "Digital Design." Prentice Hall, 2002.

- Hennessy, J.L., Patterson, D.A. "Computer Architecture: A Quantitative Approach." Morgan Kaufmann, 2011.

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